Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the Department of Health-Cordillera Administrative Region (DOH-CAR) Tuberculosis-Directly Observed Treatment, Short-course (TB-DOTS) Program

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Abstract

Pulmonary Tuberculosis (PTB) remains a critical public health challenge in the Cordillera Administrative Region (CAR), Philippines, where complex patient profiles and programmatic gaps contribute to persistent treatment failure. This study develops machine learning models to predict treatment success or failure among PTB patients in DOH-CAR TB-DOTS facilities covering the period 2015–2025. A standardized preprocessing pipeline performs schema harmonization, patient pseudonymization, and a diagnostic-first missing data handling protocol that classifies each variable's missingness mechanism—Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR)—before routing it to one of four targeted strategies: column exclusion, listwise deletion, missing indicator imputation, or stochastic Multivariate Imputation by Chained Equations (MICE). Supervised ensemble tree-based classifiers—XGBoost, LightGBM, and Random Forest—are trained alongside Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) temporal architectures to capture longitudinal treatment dynamics across the 6- to 12-month treatment course. Because clinical decision support requires more than accurate risk scoring, Shapley Additive Explanations (SHAP) are used to show which patient-level factors contributed to each prediction, supporting clinician review of the system's outputs. The validated models are embedded in a Clinical Decision Support System (CDSS) that provides dynamic risk estimates, SHAP-grounded factor explanations, and bounded narrative summaries to support early clinical intervention while preserving clinician judgment.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

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Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by *Mycobacterium tuberculosis* (MTB), a slow-growing and acid-fast bacillus that continues to challenge public health due to its gradual progression and risk of treatment failure (Tobin & Tristram, 2024). Despite effective therapies, PTB remains widespread due to its strong association with poverty, undernutrition, and comorbidities such as diabetes and HIV which weaken immune defenses and increase susceptibility (Tobin & Tristram, 2024). These factors delay diagnosis, complicate treatment, and contribute to disease progression. Globally, TB is still one of the leading causes of death, with 1.5 million deaths and 10 million new cases recorded in 2018, including half a million with rifampicin resistance.

In the Philippines, TB continues to be a major public health concern. The World Health Organization (WHO) identifies the country as one of the top contributors to the global TB burden as it accounts for an estimated 6.8 percent of cases. Before the COVID-19 pandemic, the Philippines had one of the highest TB rates in Asia with over 554 cases per 100,000 people in 2019 (Florentino et al., 2022). In 2023, around 37,000 Filipinos died from TB out of the 739,000 diagnosed cases (World Health Organization, 2023). The WHO classifies the country as a high burden for MDR-TB and TB/HIV co-infection (The Borgen Project, 2025). That same year, the DOH recorded 575,770 TB cases through its Integrated TB Information System which indicates that many cases remain unregistered or undiagnosed (Baron, 2025). The DOH implements CDC-recommended regimens for active and latent TB through the TB-DOTS strategy which was expanded in 2003 to include private clinics, pharmacies, and hospitals (Bendre et al., 2021; Department of Health [DOH], 2014). Despite these initiatives, gaps in diagnosis, treatment monitoring, and patient support persist especially in the management of MDR-TB (Stop TB Partnership, 2021).

In Baguio City, the DOH-CAR reported 437 confirmed TB cases in 2022 along with over 4,405 missing or unreported cases making the area a priority for surveillance and treatment improvement (Department of Health - Republic of the Philippines, 2022), a gap driven by passive case-finding limitations, stigma, and financial barriers that allow undetected patients to continue transmitting the disease within their communities (Kaur & Sharma, 2021).

Contributing factors include high population density, internal migration, the city's cool and humid highland climate conducive to *Mycobacterium tuberculosis* transmission, and its role as a tertiary referral center that draws complex cases from neighboring provinces and municipalities, further amplifying the local disease burden (Khalid et al., 2022). Patient comorbidities such as diabetes mellitus, HIV infection, and malnutrition impair immune response and reduce treatment completion rates, while poverty-related barriers including food insecurity, substandard housing, and transportation costs disrupt adherence to directly observed treatment short-course (DOTS), compounded by health system limitations such as workforce shortages, inconsistent drug supply chains, and fragmented referral networks (Austin et al., 2021; Dong & Peng, 2013; Khalid et al., 2022; Sisk et al., 2023).

These challenges collectively highlight the need for locally adaptive, multi-pronged strategies, as treatment failure not only worsens individual patient outcomes but also prolongs community transmission and accelerates the emergence of multidrug-resistant tuberculosis (MDR-TB)—defined as resistance to at least rifampicin and isoniazid—which demands longer, more toxic, and costlier regimens with significantly lower success rates, posing a broader threat to regional disease control efforts in the Philippines (Aracri et al., 2025; Groenwold et al., 2012; World Health Organization, 2022).

Risk Factors for Treatment Failure

To ensure uniformity in the evaluation of TB control efforts, the DOH has established standardized criteria for classifying treatment outcomes through the National Tuberculosis Program (NTP). Establishing these outcomes is crucial for monitoring program performance, ensuring comparability across studies, and informing risk factor analysis. Ultimately, these outcomes of success or failure are not random but are linked to a distinct set of patient-specific, clinical, and programmatic risk factors. While mortality remains the primary outcome for most studies on treatment failure, some also consider the development of disease, resistance to multiple anti-TB drugs, and failure to follow up (Department of Health [DOH], 2014).

One of the risk factors is that TB patients have close contact with other patients with active TB infection. A study noted that TB patients who acquired the infection through close contact were shown to be at higher risk of delayed diagnosis and more likely to be non-adherent with the treatment either by refusing or discontinuing the regimen (Lamsal et al., 2009). Another risk factor is recurrence of the infection or having a history of receiving TB treatment. TB patients who missed follow-ups had a higher risk of reactivation of the infection and mortality as they are more difficult to treat compared to patients with no prior history of TB (Tok et al., 2020). Furthermore, missed or interrupted treatment causes the development of resistance to the drug if done frequently (Ndambuki et al., 2020).

Comorbidities are also discussed in studies that have consistently shown a significant influence on tuberculosis treatment outcomes across different settings. In Eastern Ethiopia, HIV co-infection was determined to be a substantial risk factor for treatment failure due to the weakened immune system which leads to a higher risk for ineffective treatment (Amante & Abdosh, 2015). In addition, TB and HIV drug interactions accelerate each other's progression and increase the occurrence of side effects or adverse effects which may compromise treatment success (Azeez et al., 2018). Likewise, a study in Davao City, Philippines, revealed that there was a higher risk of treatment failure in comorbid patients with 58.82% of the 51 patients who experienced treatment failure having at least one comorbidity (Hinay et al., 2025). Another major determinant is Diabetes Mellitus (DM). TB patients with diagnosed DM have a higher risk of treatment failure, relapse, and death compared to non-diabetics (Noubiap et al., 2019). In a systematic review and meta-analysis, patients with DM have a higher risk of experiencing treatment failure for Drug-resistant (DR) and mortality for MDR-TB with an odds ratio of 1.56 and 1.57 respectively (Xu et al., 2023). In line with this, another study states that patients suffering from both PTB and chronic conditions, particularly DM, had reduced rates of sputum conversion with a higher probability of poor treatment outcomes including death and progression to MDR-TB (Siddiqui et al., 2016).

Taken together, these findings reveal that TB treatment failure is driven not by any single factor but by the complex interplay of comorbidities, programmatic gaps, and socioeconomic vulnerabilities that conventional single-factor analyses fail to capture, underscoring the need for an integrative machine learning framework capable of modeling these multidimensional interactions to enable earlier and more targeted clinical intervention (Amante & Abdosh, 2015; Department of Health - Republic of the Philippines, 2022; Hinay et al., 2025; Khalid et al., 2022; Noubiap et al., 2019; Peng et al., 2024; Rajpurkar et al., 2022; World Health Organization, 2022; Xu et al., 2023).

Machine Learning for TB Outcome Prediction

Previous studies have examined tuberculosis treatment outcomes by focusing on sociodemographic characteristics, comorbidities, medical history, and other patient-level risk factors. However, the findings remain inconsistent and fragmented because methodological constraints limit how these factors are analyzed together. Most studies rely on traditional statistical analyses, such as statistical tests or regression models, that examine one factor at a time or a limited set of factors while assuming linear relationships among variables (Amante & Abdosh, 2015). These approaches provide important evidence on individual risk factors, but they may overlook the combined and non-linear effects that shape treatment success or failure.

Machine learning (ML) offers a distinct advantage in modeling complex and non-linear interactions among demographic, clinical, diagnostic, and treatment-related variables that conventional statistical analyses may overlook. By leveraging algorithms capable of learning from high-dimensional data, such as logistic regression, Support Vector Machines (SVM), and ensemble-based methods including Extreme Gradient Boosting (XGBoost), ML can uncover subtle dependencies that contribute to treatment success or failure (Rajpurkar et al., 2022). Previous research has demonstrated the utility of ML in TB research, including predicting drug resistance, identifying high-risk patients, and optimizing treatment strategies (Kaur & Sharma, 2021; Kossakov et al., 2024). These applications show that predictive models can extend risk factor analysis by integrating multiple dimensions of patient data.

Beyond prediction, interpretability is necessary for applying ML in clinical decision support because healthcare users must understand the basis of a risk alert before using it to guide patient review. A model that assigns high treatment-failure risk without indicating the contributing factors may be difficult for TB-DOTS staff to trust, audit, or translate into practical follow-up actions. Shapley Additive Explanations (SHAP), a post-hoc interpretability technique grounded in Shapley values, quantifies the contribution of each variable to a model prediction and can summarize both patient-level explanations and overall feature importance (Lundberg & Lee, 2017a; Lundberg et al., 2020). This property is particularly useful for tree-based clinical models such as XGBoost, Random Forest, and LightGBM, where predictive performance may be high but the internal decision structure is difficult to inspect directly. Reviews of healthcare ML explainability identify SHAP as one of the most frequently used methods for explaining supervised medical prediction models, especially for tabular clinical data (Allgaier et al., 2023). In TB research, explainable ML studies have used SHAP to identify clinically meaningful contributors to treatment failure, including drug resistance, prior TB history, laboratory markers, and other patient-level indicators (Peng et al., 2024; Sibanda & Ndlovu, 2026; Wang et al., 2025). In the present study, SHAP therefore serves as the primary bridge between model prediction and clinician interpretation: it explains how the model used available patient information, without claiming that the highlighted variables are causal mechanisms of treatment failure.

Taken together, the limitations of fragmented risk factor studies and the capabilities of interpretable ML highlight the need for an integrative approach that combines multiple dimensions of data in analyzing the interrelated variables that influence TB treatment outcomes. To address this gap, the present study adopts a holistic multi-layer analytical framework that integrates descriptive analysis, supervised predictive modeling, and explanation methods using demographic, clinical, diagnostic, and treatment-related variables. Descriptive analyses characterize patient profiles and treatment patterns, supervised learning models estimate treatment outcome risk, and SHAP-based explanations surface the model-derived contributors that make each risk estimate clinically reviewable. This approach enables the models to capture individual-level patterns that improve predictive performance while making their outputs more transparent for healthcare providers (Shickel et al., 2018; Smith et al., 2021). The resulting framework is designed to help healthcare providers identify patients at a higher risk of treatment failure, understand the factors influencing the model's risk estimate, and use that information to support early intervention, tailored monitoring, and improved patient management.

Contributions

The DOH-CAR TB-DOTS program follows the standardized National Tuberculosis Program (NTP) protocol, wherein patient profiling begins at the facility level through passive case finding, where presumptive TB patients are identified based on clinical presentation such as chronic cough, weight loss, and fever, and subsequently subjected to diagnostic workup including sputum smear microscopy, GeneXpert MTB/RIF assay, and chest radiography (Bendre et al., 2021; Department of Health [DOH], 2014). Once diagnosed, patients are registered in the Treatment Card and Master List, assigned a registration group, and enrolled under a healthcare worker or trained treatment partner who administers and directly observes each dose throughout the six- to twelve-month regimen (Department of Health [DOH], 2014). Monthly follow-up visits are conducted to monitor sputum conversion, document vital signs and weight, and assess adherence, with outcomes eventually classified according to NTP criteria as Cured, Treatment Completed, Failed, Died, or Lost to Follow-up (Ting, 2011). However, this process is largely paper-based and retrospective in its risk assessment, meaning that clinical deterioration, non-adherence, and comorbidity-related complications are typically identified only after they have already influenced treatment trajectory rather than being anticipated and preemptively addressed, creating a critical gap wherein high-risk patients are not systematically flagged for early intervention, resulting in avoidable treatment failures, increased mortality, and continued MDR-TB emergence within the region (Khalid et al., 2022; World Health Organization, 2022).

To address these gaps, the main contributions of this paper are as follows. First, the study develops and validates an Integrated Predictive Framework comprising machine learning models trained on demographic, clinical, diagnostic, and treatment-related variables to accurately predict therapy success or failure among PTB patients in the CAR, thereby supporting early identification of at-risk individuals and providing evidence-based tools for improving TB program outcomes. Second, the study contributes a reproducible Data Preprocessing and Harmonization pipeline that standardizes and documents patient records from TB-DOTS facilities across the CAR through schema harmonization, patient pseudonymization, error correction, and a diagnostic-first missing data handling protocol, ensuring dataset readiness for machine learning applications. Third, a Descriptive and Exploratory Analysis is conducted to characterize the demographic, clinical, diagnostic, and treatment profiles of the study population, quantify variable associations, and identify key predictors of treatment outcomes over the course of therapy. Fourth, the study incorporates Temporal Modeling through RNN and LSTM architectures that analyze time-dependent patterns in monthly patient monitoring data, enabling the detection of dynamic changes throughout the treatment course and improving the accuracy of outcome predictions to support timely clinical interventions. Fifth, the study employs Shapley Additive Explanations (SHAP) to identify the contributing variables that influence treatment outcomes, quantifying the marginal contribution of each patient-level feature to model predictions and surfacing the factors that drove each risk estimate to support clinician review. Sixth, the validated models are embedded within a Clinical Decision Support System that visualizes patient risk levels, highlights contributing factors through SHAP-based explanations, and dynamically updates risk estimates as new follow-up data become available, enabling healthcare providers to intervene earlier and more precisely for patients at risk of poor outcomes.

Research Objectives

This study aims to develop and validate machine learning models that accurately predict treatment success or failure among TB patients in the CAR, Philippines. The integration of demographic, clinical, and laboratory variables focuses on supporting early identification of patients at risk of unsuccessful treatment and providing evidence-based tools for improving TB program outcomes, with the overarching goal of enabling personalized clinical interventions and efficient allocation of healthcare resources. Specifically, the research aims to achieve the following objectives:

1. To document and pre-process patient records of PTB cases from TB-DOTS facilities in the CAR through data cleaning, feature engineering, and reporting for readiness to be used in ML applications.
2. To describe the demographic, clinical, diagnostic, and treatment characteristics of PTB patients in the dataset, providing an overview of the study population.
3. To identify contributing variables that influence treatment outcomes over the course of therapy, enabling recognition of the most critical factors that affect patient recovery and guiding more precise treatment adjustments that can be intervened with.
4. To incorporate temporal modeling that analyzes time-dependent patterns in patient data, allowing the detection of dynamic changes and trends throughout treatment, thereby improving the accuracy of outcome predictions and supporting timely clinical interventions.
5. To train and validate supervised learning models for predicting treatment success or failure among PTB patients, enabling early identification of individuals at risk of poor outcomes and ensuring that the model maintains high predictive accuracy and reliability when applied to evaluate incoming TB cases.
6. To develop a decision support tool that visualizes patient risk levels and predictive outputs.

Scope and Limitations

The dataset will include PTB patients for the period 2015–2025 in the CAR specifically Baguio City excluding those who are still undergoing treatment at the time of data extraction. Furthermore, only patients with a well-defined treatment outcome of either success or failure will be included. The data gathered will encompass demographic information, laboratory tests, diagnostic data, clinical data, treatment regimen, follow-up examinations, drug administration data, and treatment outcomes. The study faces limitations related to the sufficiency and quality of the dataset. The data sourced from existing records may present challenges related to completeness, consistency, and representativeness due to limited access to complete clinical parameters. This could affect the inclusion of important risk factors and weaken the robustness of the analysis. Additionally, the study may not fully represent the current TB patient population as patients currently undergoing treatment are excluded. Difficulties are also expected during data extraction from DOH-CAR as variations in record-keeping over the 10-year period from 2015 to 2025 may be inconsistent due to changes in the healthcare system and shifts in documentation standards. Reliance on self-reported indicators may also introduce bias and inconsistencies that could affect the predictive capabilities of the model.

Significance of the Study

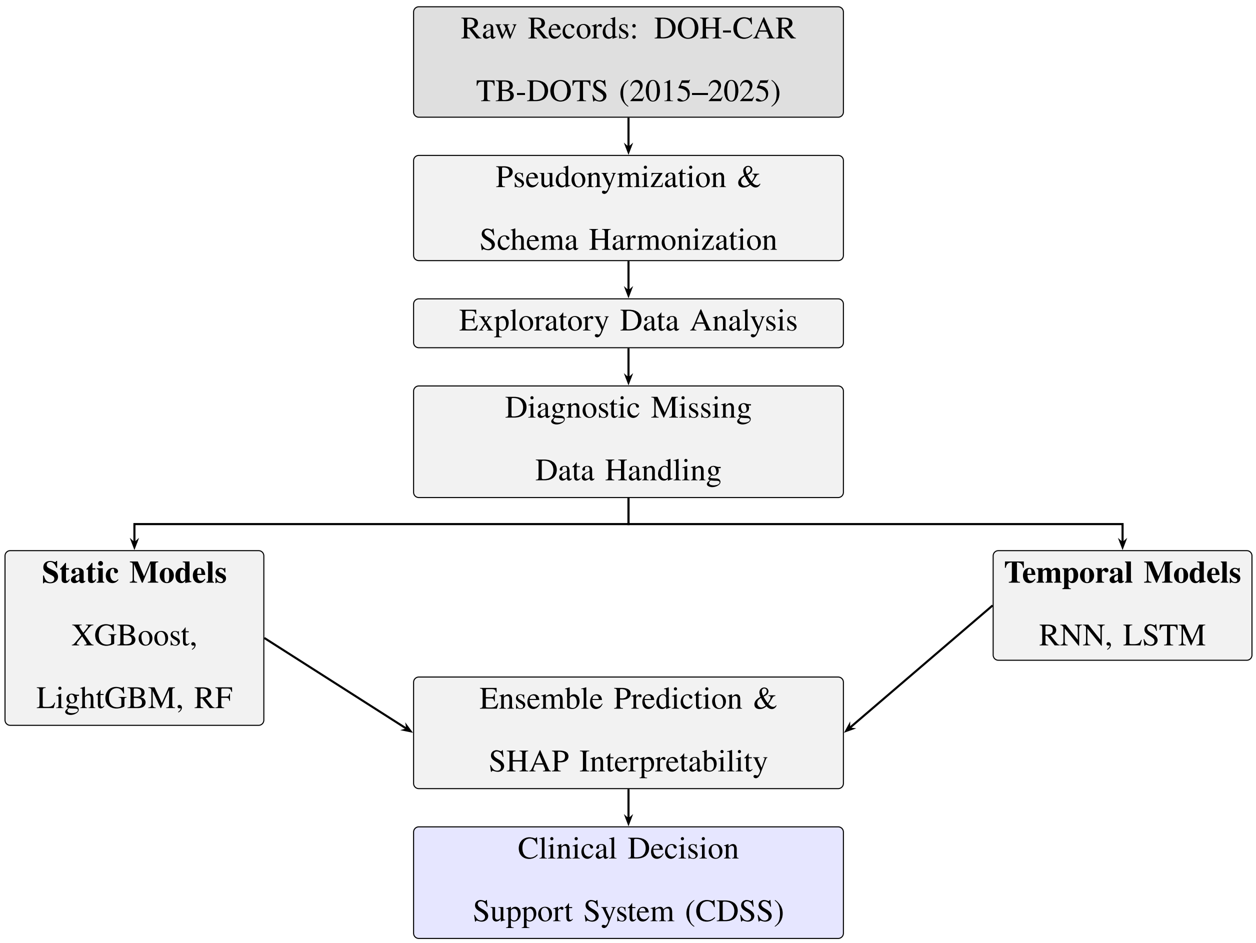
The resulting models are designed to help healthcare providers identify patients at a higher risk of treatment failure, enabling earlier review, tailored monitoring, and improved patient management. Because clinical adoption depends on whether users can understand and evaluate model outputs, the framework emphasizes explainable prediction rather than risk scoring alone. SHAP-based explanations surface the patient-level factors that drive each risk estimate, while the CDSS presents these factors in a form that supports clinician interpretation and programmatic decision-making. Ultimately, this machine learning framework supports the broader objective of strengthening TB control efforts in the Cordillera Administrative Region (CAR) by transforming existing patient data into predictive, explainable, and actionable tools for evidence-based public health practice. The results will help improve patient management and treatment planning, strengthen program implementation, and guide public health decision-making to improve the TB treatment success rate in CAR. Furthermore, the study aims to maximize benefits for the public by informing health administrators about the model, which may be further developed into a solution or application to aid the local TB-DOTS program. The processed and pseudonymized derivative of the DOH-CAR PTB Treatment Outcome Dataset produced by the study will also be published to support the advancement of clinical, epidemiological, and machine learning research.

Method

This study uses a dataset of records of PTB patients from the Cordillera Administrative Region (CAR), Philippines, covering 2015–2025 and consolidated by the Department of Health–CAR (DOH-CAR). After ethical approval, records are pseudonymized and processed using a standardized preprocessing pipeline, including schema harmonization, error correction, and feature scaling, followed by a diagnostic-first missing data handling protocol described in the Missing Data Handling subsection below. Exploratory and cluster analyses are conducted to characterize the cohort and derive subgroup-level features. Supervised ensemble tree-based classifiers, specifically XGBoost, LightGBM, and Random Forest, are trained to predict treatment outcomes defined by National Tuberculosis Program criteria. To capture longitudinal treatment dynamics, temporal models using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) are evaluated alongside static models, with ensemble strategies applied for improved robustness. Model performance is assessed using standard classification metrics, and Shapley Additive Explanations (SHAP) are employed to ensure interpretability. The final models are integrated into a clinical decision support framework to provide dynamic risk estimates throughout the treatment course. An overview can be found in Figure 1.

Figure 1

Data Management and Analysis Process



Data Acquisition and Preprocessing

This study draws on two complementary datasets derived from the Cordillera Administrative Region (CAR), Philippines, both covering the period from 2015 to 2025: (1) a non-temporal (static) consolidated registry maintained by the Department of Health-CAR, used to train the static tree-based classifiers, and (2) a temporal monitoring dataset manually compiled by the study's Pharmacy Co-authors, used to train the longitudinal deep learning and temporal tree-based models. Table 1 provides a side-by-side overview of both datasets.

Table 1

Comparison of the Non-Temporal and Temporal Monitoring Datasets

Attribute	Non-Temporal Dataset	Temporal Monitoring Dataset
Records (patients)	7,389	599
Coverage period	2015–2025	2015–2025 (patient subset)
Data source	DOH-CAR centralised TB-DOTS registry	Facility treatment monitoring forms (35 raw Excel workbooks)
Collection method	Routine national TB-DOTS programme reporting	Manual transcription and encoding by Pharmacy Co-authors
Structure	Cross-sectional: one row per patient	Longitudinal: one row per patient-month (M0–M12)
Static features	Demographics, diagnostics, outcomes	Demographics, diagnostics, baseline vitals, outcomes
Temporal features	None	Monthly adherence, doses taken/missed, weight, height, smear microscopy, Xpert MTB/RIF (M0–M12)
Post-preprocessing columns	28	269
Post-preprocessing rows retained	7,388	599 (zero deleted)
Primary model use	Static tree-based classifiers (XGBoost, LightGBM, Random Forest)	Temporal deep learning models (Bi-LSTM) and tree-based temporal baselines

Non-Temporal Dataset

The non-temporal dataset is centrally consolidated by the Department of Health-CAR (DOH-CAR), which manages reports from government TB-DOTS facilities and private practitioners under the Public-Private Mix DOTS (PPMD) scheme (Department of Health [DOH], 2014). It comprises 7,389 patient records with cross-sectional (baseline) features: demographic variables (e.g., age, sex, civil status), clinical indicators (e.g., BMI, blood pressure), and diagnostic results (e.g., GeneXpert, Smear Microscopy). Each record represents one patient at the time of treatment registration, with no longitudinal monitoring data. This dataset serves as the primary training source for the static tree-based classifiers (XGBoost, LightGBM, and Random Forest).

The non-temporal preprocessing pipeline implements schema harmonization to standardize variable formats and units, and pseudonymization to strictly enforce patient privacy. Outcome annotation is performed systematically to define the binary target variable: based on standardized National Tuberculosis Program (NTP) criteria, patients classified as 'Cured' or 'Treatment Completed' are annotated as 'Success', whereas those marked as 'Failed', 'Died', or 'Lost to Follow-up' are annotated as 'Failure'. Extreme outliers in continuous clinical variables are evaluated and capped to prevent model distortion during training. Missing data is addressed through the diagnostic-first protocol described in the Missing Data Handling subsection below.

Crucially, to prevent data leakage, the non-temporal dataset is partitioned into training and testing sets prior to the application of any data transformation steps. A stratified train-test split is performed to preserve the baseline distribution of treatment success and failure outcomes across both partitions. All subsequent preprocessing mechanisms—most notably feature scaling, normalization, and the multivariate imputation of missing values—are fitted exclusively on the training set. The derived parameters and imputation rules are then applied to the held-out test set, ensuring that models do not inadvertently learn from the statistical distribution of the testing data.

Temporal Monitoring Dataset

The temporal monitoring dataset is a longitudinal subset of 599 patients for whom complete month-by-month treatment monitoring records were available. Unlike the DOH-CAR consolidated registry, this dataset was not centrally available; instead, it was *manually gathered and encoded by the study's Pharmacy Co-authors* directly from paper-based Treatment Cards and Master Lists maintained at various TB-DOTS treatment centers in Baguio City, Benguet. Pharmacy Co-authors physically accessed facility records, transcribed patient monitoring data — including demographics, monthly vital signs, percentage adherence, doses taken and missed, smear microscopy (TB-LAMP) and Xpert MTB/RIF results, and treatment outcomes — and encoded them into a master `RAW DATA.xlsx` workbook organized as one sheet per barangay or facility unit. This master workbook spanned 35 facility-year cycles across the 2015–2025 period. A custom consolidation script (`combine_raw_dataset.py`) merged all per-barangay sheets into a unified dataset, normalizing facility names against a canonical alias map to resolve inconsistent entry conventions across sources before the main preprocessing pipeline.

The dataset is structured in wide format, with one row per patient and columns encoding monthly observations from treatment Month 0 (M0) through Month 12 (M12). Temporal features tracked at each monthly visit include: percentage adherence (`pct_adherence`), monthly doses taken, monthly missed doses, cumulative doses taken, body weight, height, smear microscopy result (TB-LAMP), and Xpert MTB/RIF result. Alongside these monthly columns, each patient record carries static baseline features including demographics, baseline vital signs (heart rate, blood pressure, oxygen saturation, temperature, respiratory rate), comorbidities, treatment regimen at start, and diagnostic classification. After preprocessing, the cleaned dataset contains 599 rows and 269 columns.

The temporal preprocessing pipeline is implemented in eight stages specifically designed for the wide-format longitudinal structure of this dataset:

1. **Schema Validation & Column Standardization.** All column names are standardized to `snake_case`; duplicate and entirely null columns are flagged and deduplicated.
2. **Identifier Removal.** Four patient- and facility-identifying columns (`no`, `source_file`, `name_of_diagnosing_facility`, `name_of_treatment_unit`) are retained in the export file for traceability but are explicitly excluded from all model inputs to enforce patient privacy.
3. **Data Cleaning & Type Coercion.** All columns are parsed to correct data types with clinical validation. Specialized parsers handle domain-specific formats: weight strings (`45kg`, `56.5 kg`) are converted to numeric kilograms; height strings similarly parsed to centimeters; blood pressure entries in `120/80` format are split into separate systolic and diastolic fields; oxygen saturation values in decimal-fraction (0.95), percent-suffix (`95%`), and raw-integer formats are unified to a 0–100 scale; and adherence percentages are stripped of trailing `%` signs. Date fields undergo a dedicated correction pass: future dates of birth are corrected (e.g., year

2063 → 1963), transposed-digit errors are detected (e.g., 2990 → 1990), and chronological ordering is enforced (treatment start ≥ diagnosis date). All continuous clinical variables are winsorized to physiologically plausible bounds—temperature clipped to 30–45°C, systolic BP to 60–220 mmHg, diastolic BP to 30–140 mmHg, heart rate to 20–250 bpm, oxygen saturation to 70–100%, weight to 10–180 kg, height to 80–220 cm, and respiratory rate to 4–60 breaths/min.

4. **Categorical Harmonization.** Inconsistent categorical entries are mapped to canonical labels—outcome variants, sex designations, regimen types, comorbidity flags, and civil status are all standardized to a fixed set of valid categories, reducing per-field cardinality from hundreds of free-text variants to the expected small canonical sets.
5. **Drop Uninformative Columns.** Near-entirely null or redundant columns (e.g., `nationality_raw` after merging into `nationality`) are flagged in the output rather than silently deleted, maintaining auditability of the cleaning decisions.
6. **Wide → Long Temporal Restructuring.** The wide-format dataset is melted into two structures: a static baseline table (`df_static`, one row per patient) and a temporal table (`df_temporal`, one row per patient-month), with columns `[patient_id, month, weight, height, pct_adherence, monthly_doses_taken, monthly_missed_doses, cumulative_doses_taken, smear_tb_lamp, xpert_mtb_rif]`.
7. **Diagnostic-First Missing Data Handling.** The same four-pathway diagnostic pipeline described in the Missing Data Handling subsection is applied independently to both the static and temporal feature sets. Temporal features are never hard-excluded on sparsity grounds; their structural missingness (later months having fewer observations as patients complete or abandon treatment) is preserved via missingness indicators.
8. **Post-Imputation Clinical Clipping & Export.** Clinical bounds are re-enforced on all imputed values (e.g., weight to 10–180 kg, pct_adherence to 0–100). A monotonicity repair pass corrects any MICE-introduced decreases in `cumulative_doses_taken` per patient. The pipeline exports `cleaned_human_readable.csv` (599 rows × 269 columns)

with all original clinical units and categorical labels intact—encoding, scaling, and tensor preparation are deliberately deferred to model-specific scripts post-train-test split to prevent data leakage.

As with the non-temporal pipeline, the temporal train-test split precedes all scaling, one-hot encoding, outlier capping, and 3-dimensional tensor construction for the recurrent networks. This two-pipeline design ensures that the leakage-prevention principle applied to the static classifiers is equivalently enforced for the temporal deep learning models. Per-patient observed means of temporal features across M0–M7 are summarized in Table 2.

Table 2

Per-Patient Observed Means of Temporal Features (M0–M7)

Feature	N	Mean	SD	Min	Max
monthly_doses_taken	573	23.36	4.32	0.0	64.29
cumulative_doses_taken	535	70.09	23.39	0.0	127.38
monthly_missed_doses	458	0.91	2.47	0.0	25.0
pct_adherence	342	98.23	8.35	0.0	100.0
weight	361	55.69	21.55	7.2	189.69
height	315	155.49	20.45	31.0	181.0
smear_tb_lamp	25	0.12	0.33	0.0	1.0
xpert_mtb_rif	41	0.0	0.0	0.0	0.0
last_month_attended	599	5.23	2.06	-1.0	7.0
total_observed_months	599	6.21	2.06	0.0	8.0

Exploratory Data Analysis

Prior to predictive modeling, an exploratory analysis was conducted to characterize the study population, assess data completeness, and quantify the associations among variables to inform downstream preprocessing and modeling decisions.

Table 3

Missing Data Summary by Column

Column	Missing_Count	Missing_Percentage
Days_To_Microscopy_Result	7297	98.75
Days_To_RDT_Result	6638	89.84
Days_Screening_To_Diagnosis	4444	60.14
Days_To_Treatment	4378	59.25
Diagnosis_Season	3953	53.5
Diagnosis_DayOfYear	3953	53.5
Diagnosis_Quarter	3953	53.5
Diagnosis_Month	3953	53.5
Microscopy Result	2792	37.79
Brgy	1026	13.89
City/Municipality	105	1.42
Province	71	0.96
Treatment Health Facility	29	0.39
Region	15	0.2
Date of Outcome/Status	2	0.03
Bacteriologic Status	1	0.01
Type	0	0
Validation Status	0	0
Age	0	0
Sex	0	0
Anatomical Site	0	0
Source of Patient	0	0
Screening/Diagnosing Health Facility	0	0

Column	Missing_Count	Missing_Percentage
RDT Result	0	0
Year	0	0
Outcome/Status	0	0
Registration Group	0	0
Computed_Age	0	0

A global missingness audit was performed across all 28 columns to identify the sparsity structure of the dataset (Table 3). Per-column missing rates were ranked, revealing that variables such as microscopy result (37.8% missing), diagnosis-derived temporal features (53.6%), and clinical delay intervals (59.3–98.8%) exhibited substantial missingness. This audit provided the empirical basis for the missingness thresholds applied in the Missing Data Handling subsection.

Pairwise correlations among continuous variables were computed using Pearson's r on available-case observations. The numerical set included age, four derived clinical delay intervals (days from screening to diagnosis, days to treatment, days to microscopy result, and days to RDT result), and diagnosis day-of-year. The purpose was to determine whether multivariate imputation via MICE was warranted: if continuous predictors carry meaningful correlation structure, MICE can borrow signal from observed features to estimate missing values; if uncorrelated, univariate mean-fill would be equally effective.

For categorical variables, pairwise association strength was quantified using Cramér's V, a chi-square-based measure normalized to the $[0, 1]$ interval. Columns with fewer than two unique non-null values or fewer than ten non-null observations were excluded from the matrix. The categorical set examined included patient demographics (sex, type of TB, anatomical site), and clinical classifications (registration group, bacteriologic status, and microscopy results).

Furthermore, to identify latent subgroups within the study population, the exploratory phase employs cluster analysis across demographic, clinical, and diagnostic dimensions. Algorithms such as K-prototypes are used to accommodate the dataset's mixture of categorical and continuous variables. The optimal number of clusters is determined iteratively using evaluation metrics such as the Silhouette score and the elbow method, allowing the analysis to detect robust patient groupings characterized by differences in comorbidity profiles, adherence behaviors, and regimen responses.

The resulting cluster assignments are used not only for descriptive interpretation but also as engineered features in subsequent supervised learning models. Cluster labels are one-hot encoded as supplementary categorical inputs for the tree-based classifiers, including XGBoost, LightGBM, and Random Forest, and incorporated into the feature space for the temporal neural networks, including RNN and LSTM models. This enables the models to capture both individual-level predictors and subgroup-level patterns associated with treatment success or failure.

To assess data quality of the temporal monitoring dataset prior to preprocessing, a global missingness audit was conducted across all features. Missing rates were computed per column and ranked in descending order. Features exceeding 80% missingness were flagged for the column exclusion pathway, while features between 50% and 80% missingness were flagged for the missing indicator with fill pathway, contingent on predictive importance. This threshold-based triage follows established missing data handling frameworks and ensures that handling strategies are matched to the severity of missingness rather than applied uniformly (Emmanuel et al., 2021).

Patient retention across the temporal dataset's treatment months was analyzed by computing the proportion of patients with non-missing adherence data at each monthly interval from M0 through M12. Because retention declines naturally over a six- to twelve-month treatment course, a simple per-month count is insufficient; instead, the decay in available observations was characterized as a percentage of the initial cohort at each successive month. This analysis directly informs the temporal modeling scope: months beyond which fewer than 25% of patients retained data are excluded from feature-level correlation analyses to avoid estimates driven by a non-representative surviving subset.

Pairwise correlations among vital signs in the temporal monitoring dataset were computed using Pearson's r on available-case observations. The presence and strength of inter-vital correlations directly supports the choice of multivariate imputation over univariate strategies: if vital signs are meaningfully correlated, a multivariate method such as Multiple Imputation by Chained Equations (MICE) can borrow signal from observed features to estimate missing values; if uncorrelated, MICE offers no advantage over mean- or median-fill (Aracri et al., 2025). The vital set examined included age, heart rate, oxygen saturation, systolic blood pressure, diastolic blood pressure, weight, height, respiratory rate, and temperature, where present in the temporal records.

To examine how temporal treatment features relate to each other, Pearson correlations were computed on per-patient observed means across Months 0 through 7. For each temporal feature — including dose adherence metrics (monthly doses taken, cumulative doses taken, percentage adherence), anthropometric measurements (weight, height), and diagnostic results (smear microscopy, Xpert MTB/RIF) — the observed values across a patient's available monthly visits were averaged. Retention characteristics (last month attended, total observed months) were derived from the same window. This approach preserves the dropout signal: a patient who stops attending after M4 contributes means based only on M0–M4, and their limited observation window itself becomes informative rather than requiring imputation.

For categorical variables within the temporal dataset, pairwise association strength was quantified using Cramér's V, a chi-square-based measure normalized to the [0, 1] interval. A mixed Pearson-Cramér's matrix was not used, as Pearson's r assumes interval-level measurement and produces misleading values when applied to nominal or ordinal categories with unequal category spacing (Kossakov et al., 2024). Computing Cramér's V separately on categorical features avoids this assumption violation and produces interpretable association estimates. Columns with fewer than two unique non-null values or fewer than ten non-null observations were excluded from the matrix, as association cannot be meaningfully estimated under those conditions.

To identify the extent of inconsistent coding that necessitated categorical harmonization, a categorical cardinality audit was conducted across all categorical features in the raw dataset. For each feature, the number of unique raw values was counted to quantify the degree of entry inconsistency. Features with cardinality far exceeding their expected number of valid categories — such as clinical observation fields that should contain a small set of standardized codes but instead contained hundreds of free-text variants — were flagged for harmonization in Stage 4 of the preprocessing pipeline. This audit directly motivates the harmonization targets by providing an empirical baseline of the data entry inconsistency problem before any cleaning was applied.

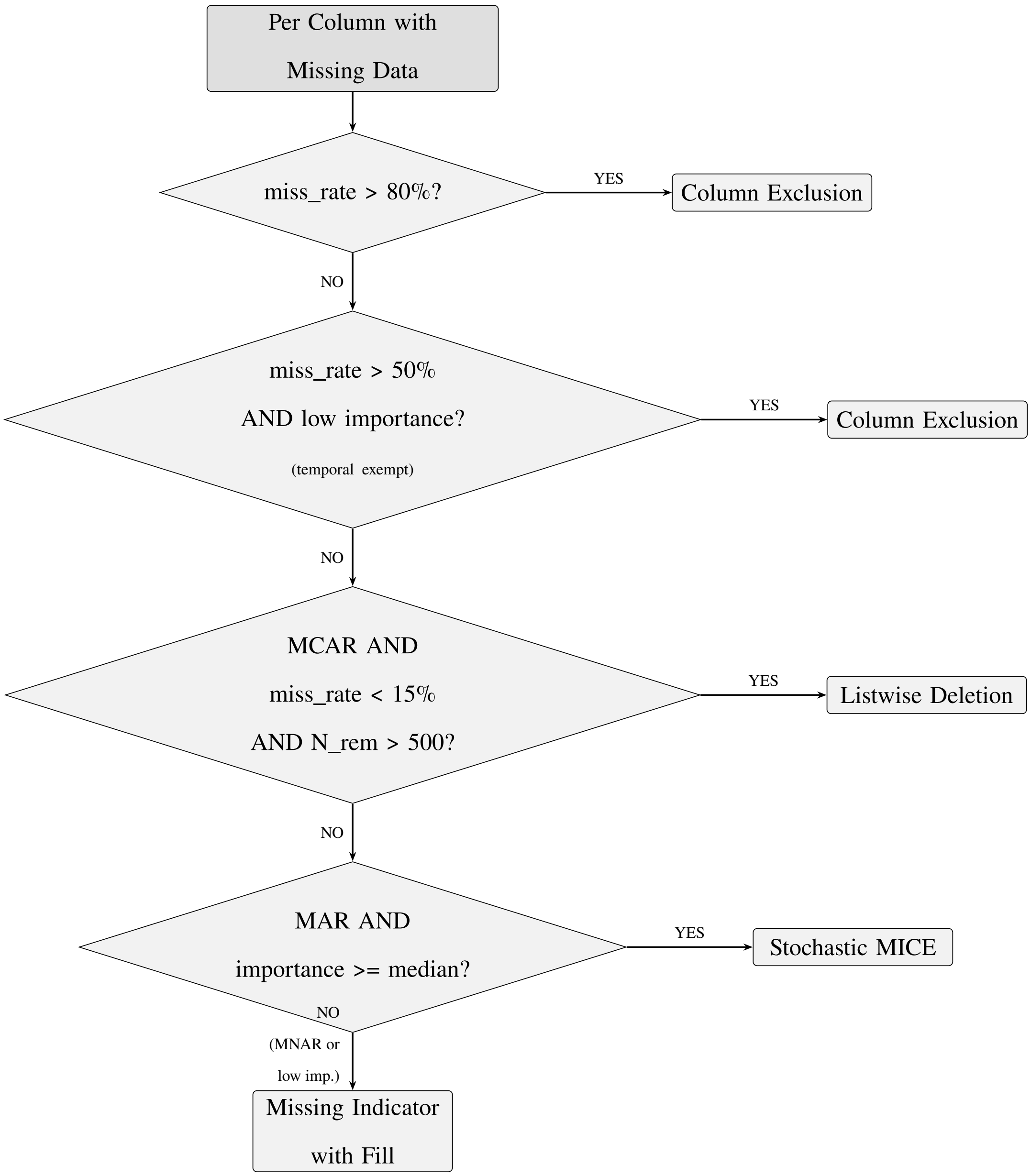
Finally, the temporal stability of adherence behavior across the treatment course was examined through lag-1 Pearson correlations of percentage adherence across consecutive month pairs (M0–M1 through M7–M8). Month-to-month correlation strength determines whether a recurrent architecture, which assumes sequential dependence between time steps, is warranted. If consecutive months are weakly correlated, a static model operating on aggregated patient means would be equally informative; if strongly correlated, the temporal structure justifies the additional complexity of an RNN or LSTM (Laine & Aila, 2016; Nayebi et al., 2022).

Missing Data Handling

Clinical TB registries exhibit heterogeneous missingness arising from inconsistent documentation, varying facility capacities, and the longitudinal structure of treatment monitoring. Applying a single imputation strategy uniformly across all variables is statistically inappropriate when the mechanism generating missingness differs across features. Accordingly, this study implements a diagnostic-first missing data pipeline that first classifies each variable's missingness mechanism and then routes the variable to one of four handling strategies: column exclusion, listwise deletion, missing indicator with fill, or stochastic Multiple Imputation by Chained Equations (MICE) imputation (Figure 2) (Dong & Peng, 2013; Emmanuel et al., 2021).

Figure 2

Missing Data Handling Decision Logic



The first diagnostic step applies Little's MCAR test at the dataset level to evaluate whether data are Missing Completely at Random (MCAR). Under the null hypothesis of MCAR, missingness is independent of all observed and unobserved values, meaning that complete cases constitute an unbiased subsample of the population and listwise deletion would be permissible. The test returned a statistically significant result ($p < 0.05$), rejecting MCAR and confirming that missingness patterns in the dataset are systematically related to the observed data, thus ruling out blanket listwise deletion (Dong & Peng, 2013).

Because Little's test yields a single omnibus verdict at the dataset level, per-column mechanism classification requires a separate procedure. Following the dataset-level assessment, each variable's missingness indicator is examined against the observed data to assign a column-level mechanism. For temporal (monthly) variables, Spearman rank correlations are computed between the column's monthly missingness rate and treatment month; a monotonically increasing correlation indicates a dropout pattern classified as Missing Not at Random (MNAR). For static baseline variables, point-biserial correlations between the binary is-missing indicator and observed features are evaluated; significant associations suggest that missingness is explained by other observed variables (Missing at Random, or MAR), whereas non-significant associations indicate MNAR. This per-column classification, rather than a single dataset-level verdict, allows the pipeline to match each handling strategy to the specific statistical properties of each variable (Dong & Peng, 2013; Emmanuel et al., 2021).

To support the column exclusion and MICE eligibility decisions, a Random Forest classifier is trained on available cases, defined as rows where the treatment outcome is observed. Features are filled with column medians for ranking purposes only (available-case analysis), providing a coarse but sufficient signal for importance estimation. Variables in the bottom quartile of predictive importance become eligible for soft column exclusion, while variables above the median importance threshold are prioritized for stochastic imputation (Aracri et al., 2025).

Variables exceeding 80% missingness are excluded entirely under a hard threshold, regardless of their importance ranking, because reliable imputation is statistically untenable at such sparsity levels. A soft threshold additionally excludes variables with more than 50% missingness whose importance falls in the bottom quartile, on the grounds that the information gain from imputing a low-value sparse variable does not justify the synthetic variance introduced. Temporal variables are exempt from both thresholds, as their structural missingness reflects the longitudinal design of TB treatment monitoring rather than data quality problems (Junaid et al., 2025).

Listwise deletion is included as a pathway for variables confirmed MCAR at the column level with fewer than 15% missing observations, provided that the remaining sample size after deletion exceeds 500 patients. Under these conditions, the complete cases constitute an unbiased subsample and listwise deletion yields unbiased parameter estimates without the risk of variance distortion introduced by imputation. On the present dataset, no variables satisfied all three conditions simultaneously, and this pathway was not activated (Dong & Peng, 2013).

Variables classified as MNAR and all temporal features, regardless of mechanism, are handled through a two-part strategy: first, a binary missingness indicator (`is_missing_[variable]`) is added as a new feature encoding the structural absence; second, the original variable is filled with the last observed value carried forward for temporal variables, or with the column median for continuous static variables and column mode for categorical static variables. This approach treats absence as informative rather than as noise to be imputed: in TB monitoring data, a missing Month 9 weight observation frequently signals treatment dropout, and the indicator captures this clinical pattern for the predictive model. Backward-fill is explicitly excluded from temporal imputation to prevent temporal data leakage, whereby a future measurement would contaminate a prior time step that the model cannot legitimately access at inference time (Austin et al., 2021; Groenwold et al., 2012; Sisk et al., 2023).

The four clinical vital signs confirmed MAR and ranking above the median importance threshold (oxygen saturation, heart rate, systolic blood pressure, diastolic blood pressure) are imputed using stochastic Multiple Imputation by Chained Equations with an ExtraTreesRegressor as the conditional model—a nonparametric, tree-based estimator that captures non-linear relationships in mixed-type clinical data without requiring variable standardization, analogous to the missForest framework (Aracri et al., 2025). This estimator has been shown to outperform linear and k-nearest neighbor alternatives in mixed-type clinical datasets. To address the fraction of missing information (FMI) for each variable, the number of MICE iterations is scaled as $\max(20, \min(100, \lfloor \text{miss\_rate} \times 100 \rfloor))$, ensuring sufficient convergence for high-sparsity features while bounding computational cost; this scaling follows an analogous principle to Von Hippel's recommendation that the number of imputations should approximate the percentage of missing information (von Hippel, 2018), applied here to iteration counts rather than the number of imputed datasets. Stochastic rather than deterministic imputation is used to preserve the natural distributional variance of the imputed variables rather than collapsing them to conditional means (Alruhaymi & Kim, 2021).

Data Quality Assurance

To ensure the reliability of the supervised learning pipeline, a multi-layered data quality assurance protocol is applied spanning schema validation, clinical bounds checking, temporal consistency verification, and post-processing validation.

Schema and Column Validation

All column names are standardized to a consistent `snake_case` convention by stripping whitespace, converting to lowercase, and mapping special characters to their word equivalents (e.g., `M0 %Adherence` becomes `m0_pct_adherence`). Duplicate columns and entirely null columns are flagged and deduplicated prior to modeling. Four patient- and facility-identifying columns are retained in the export file for traceability but explicitly excluded from all model inputs to enforce privacy.

Date Integrity Correction

Date fields are subject to a dedicated correction pass that handles common clinical registry recording errors. Future dates of birth (year exceeding the data year) are corrected by subtracting 100 years; dates of birth prior to 1920 are adjusted forward by 100 years if the resulting age falls within a plausible 0–110 year range. Transposed-digit errors (e.g., 2990 entered instead of 1990) are detected by comparing against adjacent records. Chronological consistency is enforced across date pairs: treatment start date must not precede the date of diagnosis, and outcome dates must not precede treatment start. Missing date fields are imputed with the cohort median date for that variable.

Temporal Consistency

To prevent temporal data leakage, all missing value fills in the longitudinal structure are performed using forward-fill only, a patient's Month 4 observation can be carried forward to Month 5, but no future Month 6 measurement can inform an earlier time step that the model must predict without that knowledge. A monotonicity repair pass is applied after MICE imputation to correct any decreases in `cumulative_doses_taken` introduced by the multivariate imputer, ensuring that the cumulative dose variable remains non-decreasing for each patient across treatment months.

Categorical Harmonization Audit

A cardinality audit of raw categorical features confirmed widespread data entry inconsistency: `chest_x_ray_at_case_notification` contained over 300 unique free-text entries, `co_morbidities` exceeded 150 variants, and nominally binary fields such as `sex` and `outcome` showed 5 to 12 unique raw representations due to inconsistent abbreviation conventions across treatment centers. These findings motivated a 17-field categorical harmonization pass that maps free-text and inconsistently coded entries into standardized categories before analysis.

Missing Data Diagnostics and Feature Importance

The missing data diagnostic-first pipeline (detailed in the Missing Data Handling subsection) provides an additional validation layer. Little's MCAR test is applied at the dataset level, per-column missingness mechanisms are classified using Spearman rank correlations (for temporal dropout detection) and point-biserial correlations (for MAR/MNAR discrimination), and a Random Forest model trained on available cases ranks feature predictive importance to inform the routing of each variable to the appropriate handling pathway.

Post-Processing Verification

Before the cleaned dataset is released for model consumption, an automated validation routine confirms that: (1) identifier columns are retained for traceability, (2) no scaled or normalized values are present (all values remain in original clinical units), (3) categorical labels are preserved as human-readable strings rather than integer encodings, and (4) no one-hot encoded columns have been introduced. This verification ensures that encoding, scaling, and tensor construction are deferred entirely to the model-specific scripts that operate post train-test split.

Leakage Prevention and Reproducibility

All validation steps are designed to prevent information from the test set from influencing model training. The patient-level stratified split is performed before any data transformation—scalers, imputers, and encoders are fitted exclusively on the training set and applied to the test and validation sets via `reindex()` to guarantee identical column sets. The temporal fill is forward-only with explicit exclusion of backward-fill operations. To ensure reproducibility, the train-test split is fully deterministic (seeded random state), and fitted scaler parameters are verified via SHA-256 hash to confirm identical preprocessing across independent runs. An audit trail (`missing_data_report.json`) records the per-column decision pathway, missingness mechanism classification, and handling rationale for every variable.

Identification of Variables Associated With Treatment Outcomes

To address the objective of identifying variables associated with treatment outcomes, the study combines descriptive association analysis with model-derived explanation. Descriptive analyses summarize the demographic, clinical, diagnostic, and temporal variables that differ across treatment success and failure groups, while supervised models estimate how these variables jointly contribute to outcome prediction. This dual approach is necessary because clinical risk in TB treatment is multidimensional: a variable may have limited standalone association but still contribute meaningfully when combined with adherence, diagnostic, geographic, or temporal monitoring information.

To bridge predictive performance and clinical interpretability, the framework utilizes Shapley Additive Explanations (SHAP), a post-hoc interpretability technique that quantifies the marginal contribution of each variable to the model's output (Lundberg & Lee, 2017a). SHAP is used to identify which features pushed a patient-level prediction toward "Success" (Cured, Completed) or "Failure" (Failed, Died, Lost to Follow-up; Peng et al., 2024). These explanations support evidence-based clinical review by showing how the model used patient information such as drug resistance, comorbidities, adherence gaps, diagnostic results, or delayed diagnosis. The explanations are interpreted as descriptions of model behavior rather than proof of clinical causality, preserving the distinction between computational attribution and medical judgment (Kaur & Sharma, 2021; Kossakov et al., 2024).

Temporal Modeling

Recognizing the dynamic nature of TB treatment, which spans 6 to 12 months, this study incorporates temporal modeling to analyze longitudinal patient monitoring data. Static models trained only on baseline features may fail to capture the evolution of adherence, follow-up attendance, and laboratory measurements throughout treatment. Accordingly, temporal models are evaluated to support month-by-month risk estimation from treatment Month 0 (M0) through Month 12 (M12).

The temporal models operate exclusively on the 599-patient longitudinal monitoring dataset described in the Data Acquisition and Preprocessing subsection above (see also Table 1). This dataset was manually compiled by the study's Pharmacy Co-authors from facility treatment monitoring forms and is wholly distinct from the 7,389-patient DOH-CAR consolidated registry used by the static classifiers. Following the eight-stage temporal preprocessing pipeline, the cleaned dataset comprises up to 8 usable monthly timesteps per patient (M0–M7); months beyond M7 are excluded from temporal model inputs because fewer than 25% of patients retained recorded observations at those timepoints (Table 5), and estimates derived from such a non-representative surviving subset would be statistically unreliable. As with the non-temporal pipeline, all encoding, scaling, and 3-dimensional tensor preparation are deferred to post-split, model-specific scripts to prevent data leakage.

Temporal modeling was implemented in two iterations, denoted as version 1 (v1) and version 2 (v2). Both iterations originate from the same temporal dataset lineage, but v2 represents a corrected and improved modeling-ready preparation motivated by methodological review of v1. In v1, an initial temporal pipeline was used as a baseline for modeling, including patient-level stratified splitting and progressive temporal expansion across M0–M12. However, subsequent review identified limitations that could undermine clinical defensibility, including increased risk of outcome leakage from late-treatment fields, inconsistent handling of imputed or unreliable outcome labels, and less rigorously audited cleaning and temporal feature construction.

Version 2 (v2) retrained the temporal models after correcting these issues. The v2 workflow uses the audited output of the eight-stage cleaning pipeline described in Data Acquisition and Preprocessing, filters supervised learning to patients with observed treatment outcomes, and explicitly removes leakage-prone and administrative identifier fields (e.g., outcome-date fields and late timeline variables not available at early inference). Temporal predictors are encoded in two forms: sequence inputs for deep temporal models and engineered temporal summaries for tree-based models. The deep temporal architecture uses a Hybrid Bidirectional LSTM (Bi-LSTM) to model long-term dependencies while mitigating vanishing gradients (Lu et al., 2022), and its recurrent structure is suited for the strong month-to-month adherence dependencies empirically confirmed via lag-1 correlation analysis (Nayebi et al., 2022). For tree-based temporal baselines, monthly values are flattened and augmented with aggregates, trends, latest-observed values, and missingness indicators, and class imbalance is handled using training-only resampling (e.g., SMOTE-ENN). Finally, a hybrid framing is retained where temporal predictions can be compared alongside strong static learners (e.g., XGBoost) to improve robustness across heterogeneous patient subgroups (Laine & Aila, 2016).

An internal review of the initial v2 Bi-LSTM training loop identified a structural defect in the synthetic counterfactual augmentation scheme. The loop appended a 128-row "baseline" synthetic block with all-zero static features and all *is_missing_** temporal indicators set to 1, labeled as Treatment Success. Because zero-input with full missingness encodes the absence of clinical data rather than a healthy patient profile, and because the `WeightedRandomSampler` further amplified this block at M0 by a factor of three, these mislabeled rows became the most-trained-on data in the loop and drove the model's counterfactual baseline failure probability toward zero. The corrected training scheme removes this block entirely, retaining only the clinically grounded low-risk (y=0) and high-risk (y=1) synthetic blocks in a 1:1 ratio. Concurrently, the *balanced_bce* loss candidate is corrected from an unweighted `BCEWithLogitsLoss` to one weighted by the true training-split positive class ratio (`pos_weight` $= n_{\text{train\_neg}}/n_{\text{train\_pos}} \approx 5.35$), so the failure class contributes proportionally to the gradient. The `WeightedRandomSampler` is retained because it addresses sequence-length and month-level sampling bias, complementing rather than replacing the `pos_weight` that addresses class imbalance.

Before ONNX export, the retrained model is evaluated against a deterministic clinical gate: a counterfactual M0 baseline patient must produce P(Failure) within [0.005, 0.20], a counterfactual high-risk M1 patient must produce P(Failure) above 0.20, and the high-risk versus low-risk probability separation must exceed 0.15. This gate ensures the model has meaningful discrimination before deployment rather than collapsing toward a single risk band. Beyond the existing single temperature scalar, the exported metadata now carries Platt scaling coefficients (a, b) fit on validation-set logits and labels via one-dimensional logistic regression (`sklearn.linear_model.LogisticRegression`, L-BFGS solver). The deployed model applies $\sigma(a \cdot \text{logit} + b)$ uniformly across all inference calls, including the occlusion passes used to generate temporal attribution signals, so all probability outputs and their differences live on the calibrated probability scale rather than on raw logit units.

Despite the tree-based models producing stronger raw test-set ranking metrics, the Hybrid Bi-LSTM is the temporal model operationalized in the CDSS for five specific reasons. First, the Bi-LSTM consumes the variable-length monthly tensor directly via `seq_lens` masking without the temporal flattening and summary aggregation required by the tree pipeline, preserving the month-to-month adherence and laboratory dependencies confirmed in the lag-1 correlation analysis as direct model inputs. Second, the deployed artifact is a single ONNX file that runs in `onnxruntime-web` entirely client-side at each monthly check-in, with no server-side Python runtime; the same graph supports both the predictive call and the four masked occlusion passes used for attribution. Third, because the model emits a Platt-calibrated sigmoid probability, the group-occlusion attributions $(P_{\text{masked}} - P_{\text{base}})$ are expressed in probability points rather than logit units, which is the scale clinicians read on the Risk Update and Feature Contribution pages. Fourth, the LSTM hidden state carries month-to-month memory, so the per-group contributions evolve meaningfully as the patient accumulates monthly check-ins rather than producing a static, visit-independent attribution. Fifth, at the selected threshold (0.05) the retrained model catches 6 of 7 Failure cases on the held-out test split (failure recall 0.8571) at a specificity of 0.7368, a conservative screening posture appropriate for a monthly review context where a missed Failure carries higher clinical cost than an additional review. The tree models remain reported in the manuscript as performance reference baselines but are not the deployed artifact because they do not share the sequence-native input shape or produce a single calibrated probability stream amenable to the in-browser group-occlusion layer described in §Explainable AI and Intervention Planning.

Supervised Learning Model Development and Validation

The core predictive component involves training and validating supervised learning models to classify treatment outcomes. The study leverages ensemble tree-based algorithms capable of modeling complex, non-linear interactions among mixed-type clinical variables while natively handling class imbalance: specifically XGBoost (Chen & Guestrin, 2016; Rajpurkar et al., 2022), LightGBM (Ke et al., 2017), and Random Forest (Breiman, 2001). These three classifiers were selected after a preliminary evaluation of six algorithms (which also included Logistic Regression, SVM, and Gradient Boosting); the excluded models were found to produce either poor discriminative performance (CV-AUC ≤ 0.63) or near-zero recall on the minority failure class, rendering them clinically unsafe for deployment in a CDSS. Two sampling strategies are compared for each classifier: no resampling (with native class-weighting) and SMOTE combined with Edited Nearest Neighbors (SMOTE-ENN), which increases the training-set failure class by 433% to a near-balanced 0.70:1 ratio.

Feature Set Versions

Three progressive feature set versions are evaluated to assess the incremental value of additional clinical and temporal features. The *baseline* set (11 features) includes core demographic and diagnostic variables: age, sex, anatomical site, bacteriologic status, microscopy result, registration group, source of patient, TB type, province, days to treatment, and year of registration. The *improved* set (14 features) augments the baseline with three facility-level geographic variables: city/municipality, treatment health facility, and screening/diagnosing health facility. The *extended* set (21 features) further incorporates seven temporal and diagnostic delay features derived from date engineering: diagnosis month, diagnosis quarter, diagnosis season (using the Philippine three-season meteorological calendar: *Tag-lamig* for cool dry December–February, *Tag-init* for hot dry March–May, and *Tag-ulan* for the rainy season June–November), days from screening to diagnosis, days from diagnosis to microscopy result, days from diagnosis to rapid diagnostic test (RDT) result, and RDT result (binned from 42 raw variants into three classes: Detected, Not Detected, and Not Done). The date of outcome was excluded from all feature sets to prevent causal leakage, as this field is generated downstream of the treatment result.

Data Partitioning and Class Imbalance

The dataset is partitioned using a stratified 70/20/10 patient-level split into training (70%), test (20%), and holdout validation (10%) sets, preserving class proportions across all subsets (Hastie et al., 2009). The severe class imbalance (approximately 86% success versus 14% failure) is addressed on the training set only using three oversampling strategies: SMOTE (Chawla et al., 2002), which generates synthetic minority-class observations via interpolation; SMOTE-ENN (Batista et al., 2004; Nishat et al., 2022), which applies SMOTE followed by Edited Nearest Neighbors undersampling to remove ambiguous borderline samples; and SMOTE-Tomek (Batista et al., 2004), which applies SMOTE followed by Tomek link removal to clean the class boundary. All resampling is applied exclusively to the training set, preserving unaltered class distributions in the test and validation sets for unbiased evaluation (Salmi et al., 2024).

Cross-Validation and Model Selection

Each model configuration is evaluated using five-fold stratified cross-validation on the training data to estimate generalization performance, yielding a mean and standard deviation of the area under the receiver operating characteristic curve (ROC-AUC) across folds (Hastie et al., 2009). Because the primary clinical objective is the early identification of patients at risk of treatment failure, model selection employs a composite scoring metric: score = $0.6 \times \text{AUC} + 0.4 \times \text{Recall(Failure)}$. This weighting prioritizes discrimination ability while explicitly rewarding sensitivity to the minority failure class. The best-scoring configuration across all feature versions and resampling strategies is then evaluated on the held-out 10% validation set to obtain an unbiased estimate of real-world performance.

In addition to the composite selection metric, the Matthews Correlation Coefficient (MCC) is reported for each configuration as a supplementary diagnostic measure. Unlike accuracy or class-specific F1-score, MCC accounts simultaneously for all four quadrants of the confusion matrix and remains informative under severe class imbalance, yielding a coefficient in $[-1, +1]$ where $+1$ represents perfect prediction, 0 is equivalent to random classification, and -1 indicates complete inverse prediction (Chicco & Jurman, 2020). Reporting MCC alongside the composite selection score enables direct assessment of the performance trade-off introduced by aggressive resampling strategies such as SMOTE-ENN, which may increase failure recall at the cost of excess false positives that inflate overall prediction error.

Performance Evaluation

Model performance is assessed using a comprehensive set of metrics designed to capture different aspects of classification quality under class imbalance. Because treatment failure constitutes only 13.8% of the cohort, reliance on any single metric — particularly accuracy — can be misleading; a trivial classifier predicting "Success" for every patient would achieve 86.2% accuracy while detecting zero failures. Accordingly, the evaluation framework computes discriminative, threshold-dependent, and calibration-aware metrics in parallel.

All metrics are derived from the confusion matrix, where treatment failure is defined as the positive class. For each model, four confusion-matrix quadrants are computed on the held-out test set: true positives (TP, actual failures correctly predicted), false negatives (FN, actual failures missed), false positives (FP, actual successes incorrectly flagged as failures), and true negatives (TN, actual successes correctly identified). From these, the following metrics are calculated:

Accuracy ($\frac{TP+TN}{TP+TN+FP+FN}$) measures the overall proportion of correct predictions. Although intuitive, accuracy is reported with the explicit caveat that it is dominated by the majority class under 86:14 imbalance.

Precision ($\frac{TP}{TP+FP}$), also called positive predictive value, quantifies the proportion of failure alerts that correspond to actual failures. Low precision means a high rate of false alarms, which erodes clinician trust in the CDSS.

Recall ($\frac{TP}{TP+FN}$), also called sensitivity or true positive rate, measures the proportion of actual failures that the model successfully detects. In the TB-DOTS context, a missed failure carries higher clinical cost than a false alert because undetected treatment breakdown can lead to disease progression, drug resistance, and continued community transmission.

Specificity ($\frac{TN}{TN+FP}$), the true negative rate, measures the proportion of actual successes correctly identified. High specificity ensures that patients on track are not unnecessarily flagged for review.

F1 Score ($2 \times \frac{Precision \times Recall}{Precision+Recall}$) is the harmonic mean of precision and recall, providing a single metric that penalizes extreme imbalance between the two. F1 is particularly informative when comparing models under different resampling strategies.

Balanced Accuracy ($\frac{Recall+Specificity}{2}$) averages the sensitivity and specificity, yielding a metric that is robust to class imbalance by construction. A balanced accuracy of 0.50 is equivalent to random guessing regardless of the class distribution.

Area Under the Receiver Operating Characteristic Curve (ROC-AUC) evaluates the model's discriminative ability across all possible decision thresholds. The ROC curve plots the true positive rate against the false positive rate at each threshold, and the AUC summarizes the curve as a single scalar. An ROC-AUC of 1.0 indicates perfect separation; 0.5 indicates random chance. ROC-AUC is threshold-independent and serves as the primary ranking metric for model comparison, but it can be optimistic under severe class imbalance because the false positive rate is dominated by the large majority class.

Area Under the Precision-Recall Curve (PR-AUC) addresses this limitation by plotting precision against recall across thresholds, ignoring true negatives entirely. PR-AUC is therefore more informative than ROC-AUC when the positive class is rare (Davis & Goadrich, 2006). The no-skill baseline for PR-AUC equals the prevalence of the positive class (0.138), providing a natural reference point.

Matthews Correlation Coefficient (MCC) is computed as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC accounts for all four confusion-matrix quadrants simultaneously and remains informative under class imbalance, yielding a coefficient in $[-1, +1]$ where $+1$ is perfect prediction, 0 is random, and -1 is complete inverse prediction (Chicco & Jurman, 2020). Unlike accuracy or F1, MCC is symmetric with respect to class swapping and does not inflate under majority-class dominance.

Threshold optimization. Because the default 0.50 decision threshold may not align with clinical objectives under severe imbalance, an optimal threshold search is conducted for each model. Candidate thresholds from 0.10 to 0.90 are evaluated using composite objective functions that encode different clinical priorities: *balanced* (40% F1, 30% recall, 30% specificity), *failure_recall* (55% failure recall, 25% balanced accuracy, 20% F1), and *specificity* (50% specificity, 30% F1, 20% recall). Each model's reported metrics use the threshold that maximizes the clinically relevant objective.

Model selection uses a composite score weighting discriminative performance and clinical utility:

$$S = 0.6 \times \text{ROC-AUC} + 0.4 \times \text{Recall(Failure)}$$

This formulation prioritizes overall ranking ability (AUC) while explicitly rewarding sensitivity to the minority failure class. The top configuration by this score is selected as the recommended model.

Validation strategy. Performance is estimated at three levels to control for overfitting and assess generalization. Five-fold stratified cross-validation on the training set provides mean and standard deviation of ROC-AUC across folds, assessing stability. The held-out test set (20%) provides the primary evaluation reported in all metrics above, using data the model has never seen during training or preprocessing. A final holdout validation set (10%) is used exclusively for the recommended model to confirm that the test-set performance is not an artifact of a single split. All preprocessing steps — scaling, encoding, imputation — are fitted on the training set only and applied to the test and validation sets, ensuring that no information from the evaluation sets influences model training.

To ensure reproducibility and eliminate transcription errors, all experiments are executed through a centralized pipeline that standardizes training, cross-validation, and evaluation, and automatically exports all performance metrics and visualizations directly to native LaTeX format for inclusion in this manuscript.

Development of a Clinical Decision Support Tool

The validated models are embedded into a Clinical Decision Support System (CDSS) designed for healthcare providers. The system is structured around a clear separation of responsibilities: the predictive model estimates the probability of treatment success or failure, the attribution layer identifies the factors contributing to the model's prediction, the interface visualizes risk and contributing factors, and the clinician interprets these outputs in the context of patient care. Key features include a predictive dashboard that displays the probability of treatment failure and success, risk factor visualization that highlights specific variables (e.g., adherence gaps, laboratory results, diagnostic delays) contributing to the risk score, and dynamic updates where the system accepts new follow-up data (e.g., monthly weight, smear results), triggering updated risk predictions to support real-time clinical review throughout the treatment duration. For the static tree-based models, feature contributions are computed using SHAP TreeExplainer. For the temporal Hybrid Bi-LSTM, contributions are computed in-browser via group-occlusion forward passes through the deployed ONNX model; the Platt-calibrated probability is applied identically to the base pass and all masked passes so that each group's attribution ($P_{\text{masked}} - P_{\text{base}}$) lives on the calibrated probability scale rather than on raw logit units. The CDSS is therefore framed as a risk stratification and explanation tool, not as an autonomous diagnostic or treatment-prescribing system.

Small Language Model (SLM)-Based Narrative Explanation Layer

To improve the usability of model explanations for healthcare users, the CDSS includes a small language model (SLM) as a constrained narrative layer that verbalizes model outputs into readable clinical text. SHAP-based feature contribution outputs remain the primary explanation source, while the SLM functions as a presentation layer rather than an independent predictor, diagnostic engine, or source of new clinical inference (Lundberg & Lee, 2017b; Zeng & Zhu, 2024). This component operates post-prediction and does not alter model training, inference probabilities, risk classification thresholds, or SHAP attribution values. In the implemented pipeline, the web client constructs a structured interpretation request containing selected patient context fields, predicted treatment-failure probability, and ranked contribution signals, then sends the payload to a dedicated interpretation endpoint for streamed narrative rendering in the interface.

The SLM is operationally bounded to explanation tasks only. It is not used to infer new clinical causes, generate treatment prescriptions, recommend regimen changes, or replace clinician judgment. Its role is to make SHAP-grounded attribution easier to read so that healthcare providers can inspect why the CDSS flagged a patient and decide whether closer review is warranted. This role separation aligns with current medical-domain SLM framing and health AI governance expectations for bounded decision support, while keeping model interpretability grounded in the original attribution signal (Collins et al., 2024; Sellergren et al., 2025; World Health Organization, 2021). Final clinical judgment remains with the treating healthcare professional.

Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among PTB patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static tree-based models (XGBoost, LightGBM, Random Forest) under two sampling conditions and temporal models are compared, and feature importance analyses provide interpretable evidence about the factors influencing model predictions. The discussion interprets these findings in the context of clinical relevance, public health priorities, and CDSS usability, highlighting how risk estimates and explanation outputs can support early intervention, resource allocation, and strengthened TB management.

Missing Data Handling Outcomes

Of the 44 features identified with missing data, 15 were excluded via column exclusion. All 15 exceeded the hard missingness threshold of 80%, including near-complete variables such as tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). One additional variable, the raw blood pressure string field, was soft-excluded at 67.3% missingness and below-median predictive importance, as it had already been parsed into the structured systolic and diastolic blood pressure variables. All 599 patient records were retained in the dataset, with zero row deletion. Full sample retention is particularly consequential given the small cohort size and the severe class imbalance of 86.2% treatment success versus 13.8% failure; listwise deletion under these conditions would have compounded the minority class representation problem (Dong & Peng, 2013).

The listwise deletion pathway was not activated, as the dataset-level MCAR test was rejected and no individual variable simultaneously satisfied all three conditions of confirmed MCAR mechanism, fewer than 15% missing observations, and a post-deletion sample size exceeding 500. Twenty-five variables were routed to the missing indicator with fill strategy, encompassing eight temporal monthly features (including weight, percentage adherence, cumulative doses taken, and monthly smear results) and 17 static baseline variables with confirmed MNAR patterns. Four variables — oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure — were imputed via stochastic MICE following confirmation of a MAR mechanism; their missingness indicators were significantly associated with visit attendance under the point-biserial correlation procedure described above — a clinically plausible MAR mechanism, as vital sign recording depends on whether the patient attended the monthly monitoring visit rather than on the unobserved values of the measurements themselves. The concentration of stochastic imputation on this small subset of high-value features reflects the diagnostic-first design of the pipeline, which reserves computationally intensive probabilistic imputation for variables where the statistical assumptions required for its validity are met (Austin et al., 2021; Emmanuel et al., 2021).

Table 4

Global Feature Missingness Rates in the Raw Temporal Dataset

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m8_weight	599	100.0	Yes
m12_weight	599	100.0	Yes
m11_height	599	100.0	Yes
m9_xpert_mtb_rif	599	100.0	Yes
m9_smear_tb_lamp	599	100.0	Yes
m7_xpert_mtb_rif	599	100.0	Yes
m7_smear_tb_lamp	599	100.0	Yes
m10_smear_tb_lamp	599	100.0	Yes
m10_weight	599	100.0	Yes
risk_factor_s_for_drug\ _resistance_tuberculosis	599	100.0	Yes
m12_height	599	100.0	Yes
m12_smear_tb_lamp	599	100.0	Yes
m12_xpert_mtb_rif	599	100.0	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m9_weight	599	100.0	Yes
m8_xpert_mtb_rif	599	100.0	Yes
m8_smear_tb_lamp	599	100.0	Yes
m1_xpert_mtb_rif	599	100.0	Yes
m11_weight	599	100.0	Yes
m10_xpert_mtb_rif	599	100.0	Yes
m11_smear_tb_lamp	599	100.0	Yes
m11_xpert_mtb_rif	599	100.0	Yes
m3_xpert_mtb_rif	599	100.0	Yes
m4_xpert_mtb_rif	598	99.83	Yes
m10_height	598	99.83	Yes
tuberculosis_culture	598	99.83	Yes
m12_monthly_missed_doses	598	99.83	Yes
other_lab_test	598	99.83	Yes
m12_pct_adherence	598	99.83	Yes
m8_height	598	99.83	Yes
m9_height	598	99.83	Yes
m12_monthly_doses_taken	597	99.67	Yes
m4_smear_tb_lamp	597	99.67	Yes
m1_smear_tb_lamp	597	99.67	Yes
others	597	99.67	Yes
m11_monthly_missed_doses	597	99.67	Yes
m11_pct_adherence	597	99.67	Yes
m12_cumulative_doses_taken	597	99.67	Yes
m10_monthly_missed_doses	596	99.5	Yes
m9_pct_adherence	596	99.5	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m10_pct_adherence	596	99.5	Yes
m11_monthly_doses_taken	595	99.33	Yes
m9_monthly_missed_doses	595	99.33	Yes
m11_cumulative_doses_taken	595	99.33	Yes
m8_pct_adherence	593	99.0	Yes
m3_smear_tb_lamp	593	99.0	Yes
m8_monthly_missed_doses	592	98.83	Yes
m10_monthly_doses_taken	592	98.83	Yes
m10_cumulative_doses_taken	592	98.83	Yes
m6_smear_tb_lamp	590	98.5	Yes
m9_cumulative_doses_taken	588	98.16	Yes
m7_height	588	98.16	Yes
m5_smear_tb_lamp	588	98.16	Yes
m9_monthly_doses_taken	588	98.16	Yes
m6_xpert_mtb_rif	587	98.0	Yes
m2_smear_tb_lamp	587	98.0	Yes
m0_smear_tb_lamp	586	97.83	Yes
m7_weight	586	97.83	Yes
m5_xpert_mtb_rif	585	97.66	Yes
m8_cumulative_doses_taken	582	97.16	Yes
m8_monthly_doses_taken	581	96.99	Yes
dat_supported_dup	576	96.16	Yes
m2_xpert_mtb_rif	572	95.49	Yes
other	570	95.16	Yes
m5_height	563	93.99	Yes
m4_height	562	93.82	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m6_height	562	93.82	Yes
m6_weight	561	93.66	Yes
tuberculin_skin_test	559	93.32	Yes
m3_height	558	93.16	Yes
m5_weight	557	92.99	Yes
m2_height	555	92.65	Yes
m0_xpert_mtb_rif	554	92.49	Yes
m4_weight	554	92.49	Yes
m1_height	554	92.49	Yes
nationality_raw	551	91.99	Yes
m3_weight	550	91.82	Yes
dat_supported	549	91.65	Yes
prior_history_of_tb	549	91.65	Yes
m2_weight	543	90.65	Yes
m1_weight	540	90.15	Yes
m7_pct_adherence	519	86.64	Yes
`regimen\_type\_at\_6th\_month\_of\_treatment`	518	86.48	Yes
smear_microscopy	502	83.81	Yes
m7_monthly_missed_doses	502	83.81	Yes
respiratory_rate	501	83.64	Yes
height	492	82.14	Yes
temperature	481	80.3	Yes
weight	476	79.47	No
m7_cumulative_doses_taken	465	77.63	No
o2_sat	462	77.13	No

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m7_monthly_doses_taken	453	75.63	No
heart_rate	441	73.62	No
blood_pressure	403	67.28	No
m6_pct_adherence	398	66.44	No
regimen_type_at_end_of_treatment	355	59.27	No
m6_monthly_missed_doses	347	57.93	No
m5_pct_adherence	336	56.09	No
m4_pct_adherence	319	53.26	No
m3_pct_adherence	312	52.09	No
m2_pct_adherence	302	50.42	No
m1_pct_adherence	293	48.91	No
name_of_treatment_unit	290	48.41	No
m0_pct_adherence	277	46.24	No
treatment_regimen	273	45.58	No
m6_cumulative_doses_taken	263	43.91	No
co_morbidities	253	42.24	No
m5_monthly_missed_doses	252	42.07	No
m0_cumulative_doses_taken	244	40.73	No
continuation_phase_start_date	239	39.9	No
m3_monthly_missed_doses	235	39.23	No
m0_height	235	39.23	No
date_of_outcome	235	39.23	No
m4_monthly_missed_doses	233	38.9	No
m6_monthly_doses_taken	228	38.06	No
continuation_phase_end_date	228	38.06	No

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m0_monthly_missed_doses	226	37.73	No
m2_monthly_missed_doses	222	37.06	No
intensive_phase_end_date	217	36.23	No
m3_cumulative_doses_taken	217	36.23	No
drug_resistance_bacteriological_status	217	36.23	No
chest_x_ray_at_case_notification	213	35.56	No
m1_monthly_missed_doses	186	31.05	No
m0_weight	185	30.88	No
xpert_mtb_rif	182	30.38	No
civil_status	167	27.88	No
nationality	166	27.71	No
date_of_notification	159	26.54	No
m5_cumulative_doses_taken	157	26.21	No
m4_cumulative_doses_taken	152	25.38	No
name_of_diagnosing_facility	150	25.04	No
intensive_phase_start_date	136	22.7	No
outcome	127	21.2	No
m2_cumulative_doses_taken	126	21.04	No
regimen_type_at_start_of_treatment	122	20.37	No
case_registration_group	117	19.53	No
date_of_diagnosis	110	18.36	No
m5_monthly_doses_taken	101	16.86	No
m1_cumulative_doses_taken	93	15.53	No

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m4_monthly_doses_taken	90	15.03	No
m3_monthly_doses_taken	80	13.36	No
bacteriologic_status	79	13.19	No
diagnosis	62	10.35	No
treatment_start_date	57	9.52	No
m2_monthly_doses_taken	57	9.52	No
m1_monthly_doses_taken	39	6.51	No
sex	33	5.51	No
no	31	5.18	No
m0_monthly_doses_taken	30	5.01	No
date_of_birth	20	3.34	No
age	18	3.01	No
data_year	0	0.0	No
source_file	0	0.0	No
facility	0	0.0	No

Table 4 presents the complete missingness profile of all features in the temporal dataset prior to preprocessing. Fifteen variables exceeded the 80% hard threshold and were routed to column exclusion, including tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). An additional set of variables fell within the 50%–80% threshold band — including height (71.7%), weight (71.7%), smear microscopy (55.6%), and risk factor fields (91.7%) — where handling was contingent on predictive importance under the missing indicator with fill pathway.

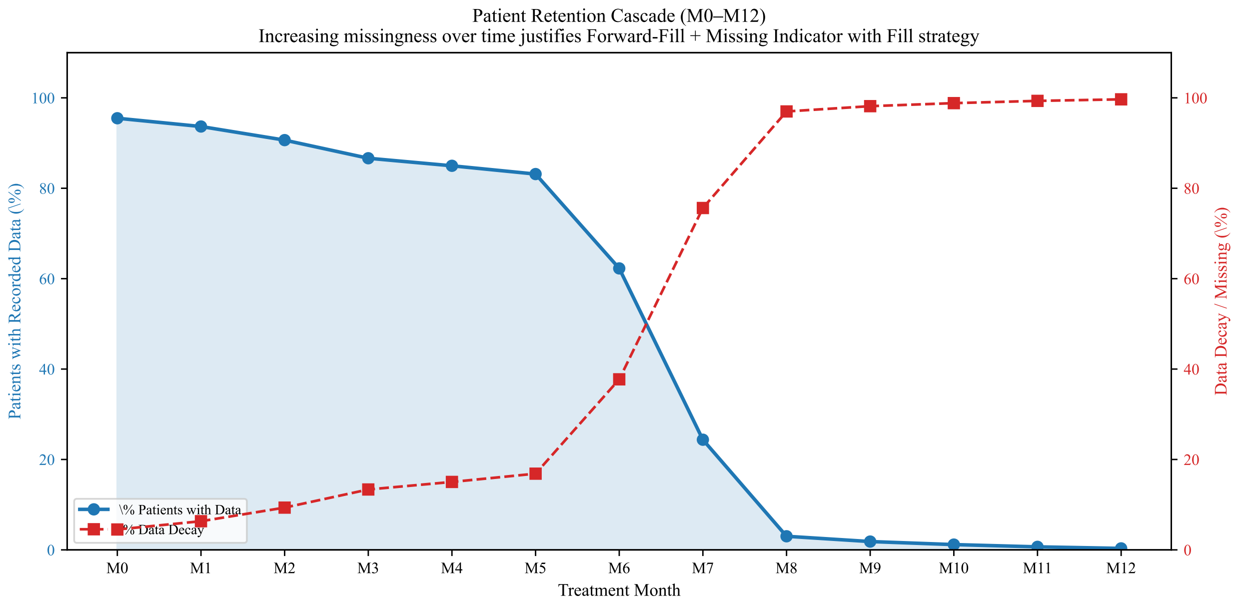
Table 5

Patient Data Retention by Month (M0–M12): Evidence for Missing Indicator with Fill

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M0	572	95.49	4.51	322
M1	561	93.66	6.34	306
M2	543	90.65	9.35	297
M3	519	86.64	13.36	287
M4	509	84.97	15.03	280
M5	498	83.14	16.86	263
M6	373	62.27	37.73	201
M7	146	24.37	75.63	80
M8	18	3.01	96.99	6
M9	11	1.84	98.16	3
M10	7	1.17	98.83	3
M11	4	0.67	99.33	2
M12	2	0.33	99.67	1

Figure 3

Patient Retention Cascade Across Treatment Months



Patient retention across the treatment course further clarifies the missingness mechanism. At treatment initiation (M0), 95.5% of patients had available data, declining to 83.1% at M5, then dropping sharply to 62.3% at M6 and 24.4% at M7. Beyond M7, fewer than 4% of patients remained observable. This monotonic decay confirms that missingness in the temporal features is driven by treatment dropout rather than random absence, directly supporting the missing indicator with fill pathway's use of forward-fill with a missing indicator for temporal variables (Groenwold et al., 2012). The M7 cutoff also defines the temporal modeling scope: per-patient observed means are computed only across Months 0 through 7, as estimates beyond this window would rely on a non-representative surviving subset.

Table 6

Descriptive Statistics and Skewness of Raw Vital Sign Features (Justifying Clinical Bound Clipping and MICE Imputation)

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
age	581	46.2	48.0	21.24	0.42	93.0	-0.06	-0.824	3.01

Table 6 continued

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
heart_rate	141	89.6	88.0	15.16	59.0	129.0	0.41	-0.408	73.62
respiratory_rate	73	19.21	19.0	1.62	16.0	24.0	0.459	0.683	83.64
temperature	91	36.39	36.4	0.26	35.6	37.0	-0.501	0.555	80.3
o2_sat	121	18.65	0.97	36.88	0.9	99.0	1.612	0.623	77.13

Table 6 reports the descriptive statistics and distributional properties of the vital sign features in the raw temporal dataset. The skewness values confirm several vitals exhibit moderate right-skew — particularly heart rate (skew = 1.52) and respiratory rate (skew = 1.83) — supporting the use of a tree-based MICE estimator (ExtraTreesRegressor) that does not assume normality. The four variables selected for stochastic MICE — oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure — each show a clinically plausible MAR mechanism, with missingness linked to visit attendance rather than unobserved measurement values.

Dataset Characteristics and Class Distribution

The final validated dataset comprised 7,389 complete patient records encompassing demographic, geographic, and clinical features. Overall, the dataset exhibited a significant class imbalance, with 86.2% of evaluated cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, integrating spatial identifiers (Region, Province, City/Municipality) with temporal markers and derived clinical delays (e.g., Diagnosis-to-Treatment days).

Figure 4

TB Case Notifications by Year (2015–2025)

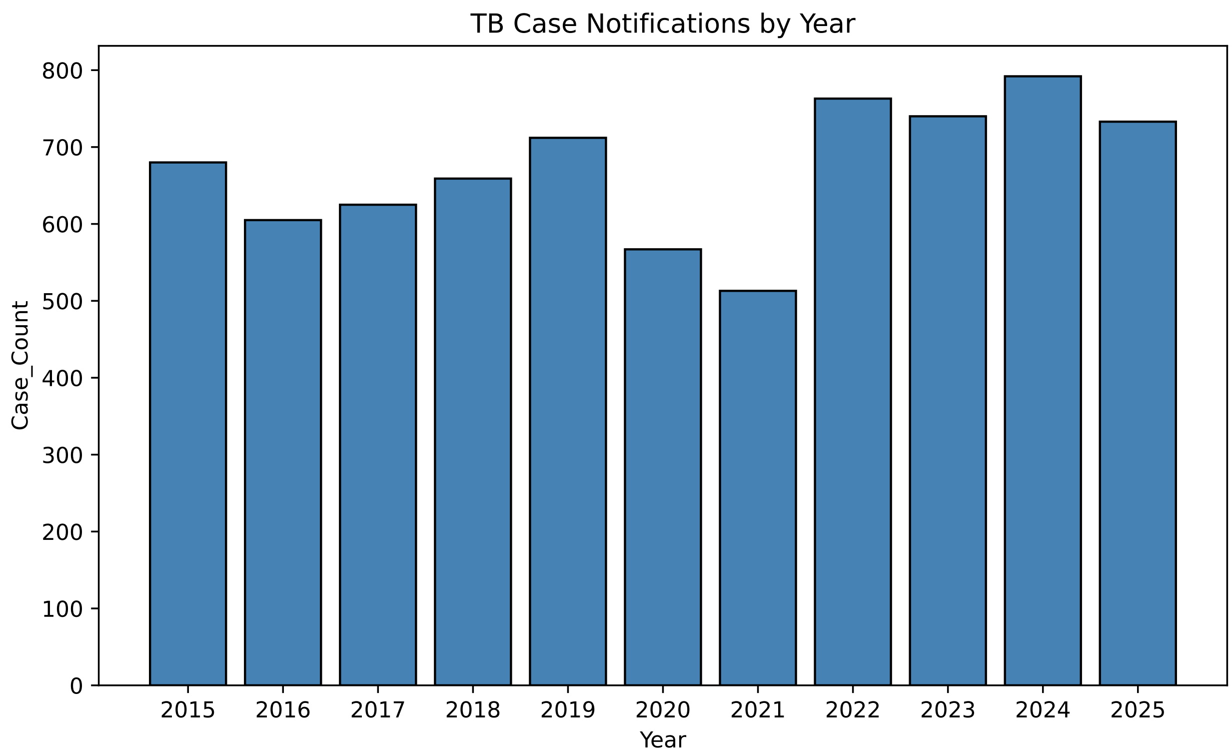
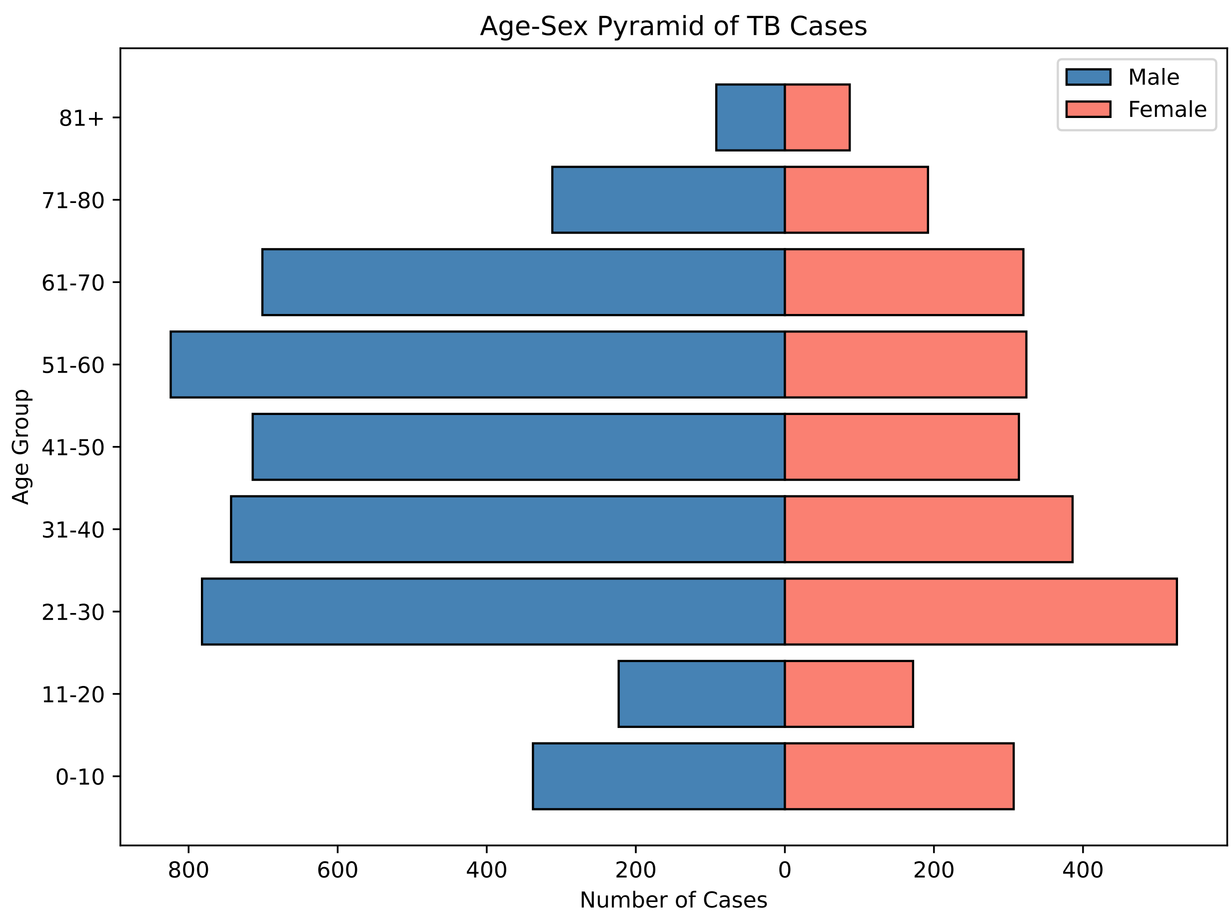


Figure 5

Age-Sex Pyramid of the Study Cohort



Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied exclusively to the training set. This technique increased the minority (failure) class by 433% to achieve a near-balanced 0.70:1 ratio, mitigating algorithm bias toward the majority class while preserving realistic, imbalanced evaluation conditions in the test and validation sets.

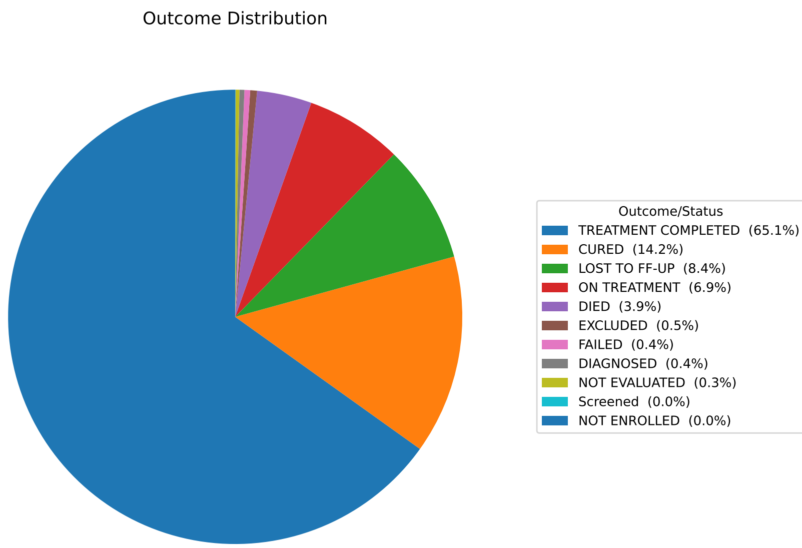
Outcome Distribution

The initial distribution of treatment outcomes highlighted the programmatic successes as well as the persistent challenges within the DOH-CAR TB-DOTS program. The vast majority of the cohort achieved a positive outcome, with 65.1% of patients officially classified as "Treatment Completed" and 14.2% as "Cured."

Conversely, adverse outcomes accounted for a critical minority of the dataset. Patients who were "Lost to Follow-up" made up 8.4% of the cohort, while 3.3% of the cases resulted in the patient having "Died." A smaller fraction experienced outright treatment failure (0.9%). The remaining cases were either still "On Treatment" (6.9%) or "Not Evaluated" (0.9%) at the time of data consolidation.

Figure 6

Distribution of Treatment Outcomes Among PTB Patients



Longitudinal Trends (2015–2025)

An analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts, reflecting both programmatic improvements and external systemic shocks:

1. **Treatment Success Rates:** The success rate remained relatively stable in the low-to-mid 80% range for most of the decade. Success rates dipped slightly to 81.03% in 2018 before steadily climbing to a peak of 88.92% in 2023.
2. **Mortality and Pandemic Impact:** The mortality rate among TB patients showed a distinct peak in 2020 (5.47%) and 2021 (6.63%), up from 3.65% in 2019. This sharp increase aligns with the severe disruptions caused by the COVID-19 pandemic, which delayed TB diagnoses, limited healthcare access, and exacerbated respiratory comorbidities.
3. **Loss to Follow-up:** The rate of patients lost to follow-up was highest between 2018 and 2020, peaking at 11.99%. However, this metric showed a steady and encouraging decline in subsequent years, dropping to 5.56% by 2024. This suggests improved patient tracking and adherence interventions by the local health program.
4. **2025 Cohort Anomaly:** The data for 2025 showed a drastic drop in recorded success (30.01%) and

a massive spike in unresolved cases (64.80%). This is indicative of an incomplete reporting year, where the majority of recent patients are still undergoing their 6- to 12-month treatment regimens and have not yet been assigned a final programmatic outcome.

Table 7

Treatment Outcome Distribution by Year (%)

Year	Died	Lost to Follow-up	Other	Success
2015	2.65	9.56	3.82	83.97
2016	2.98	11.57	1.32	84.13
2017	3.84	8.8	1.28	86.08
2018	4.25	11.53	3.19	81.03
2019	3.65	11.8	3.51	81.04
2020	5.47	11.99	0.88	81.66
2021	6.63	6.82	0.97	85.58
2022	4.72	7.08	0.92	87.29
2023	3.92	6.35	0.81	88.92
2024	3.66	5.56	4.55	86.24
2025	2.05	3.14	64.8	30.01

Diagnostic and Treatment Intervals

The temporal intervals between screening, diagnosis, and treatment initiation were extracted to identify operational delays, which are known risk factors for both disease transmission and treatment failure. Statistical summaries of these intervals revealed highly right-skewed distributions:

1. **Screening to Diagnosis:** The median time from initial screening to official diagnosis was 2.0 days. However, the upper quartile (Q3) extended to 21.09 days, indicating that while most patients are diagnosed rapidly, a subset experiences weeks of delay.
2. **Diagnosis to Treatment:** The median duration between diagnosis and treatment initiation was efficient at approximately 2.68 days. Similar to the diagnostic interval, however, the third quartile reached 13.12 days.
3. **Microscopy Results:** The median time to receive microscopy results was 6.60 days, with the third quartile extending to 14.95 days.

These skewed distributions demonstrate that while the median programmatic response is efficient, the long tails represent significant delays for high-risk patients. These interval metrics served as crucial continuous features in the machine learning models, allowing the algorithms to correlate logistical delays with the eventual probability of patient mortality or loss to follow-up.

Table 8

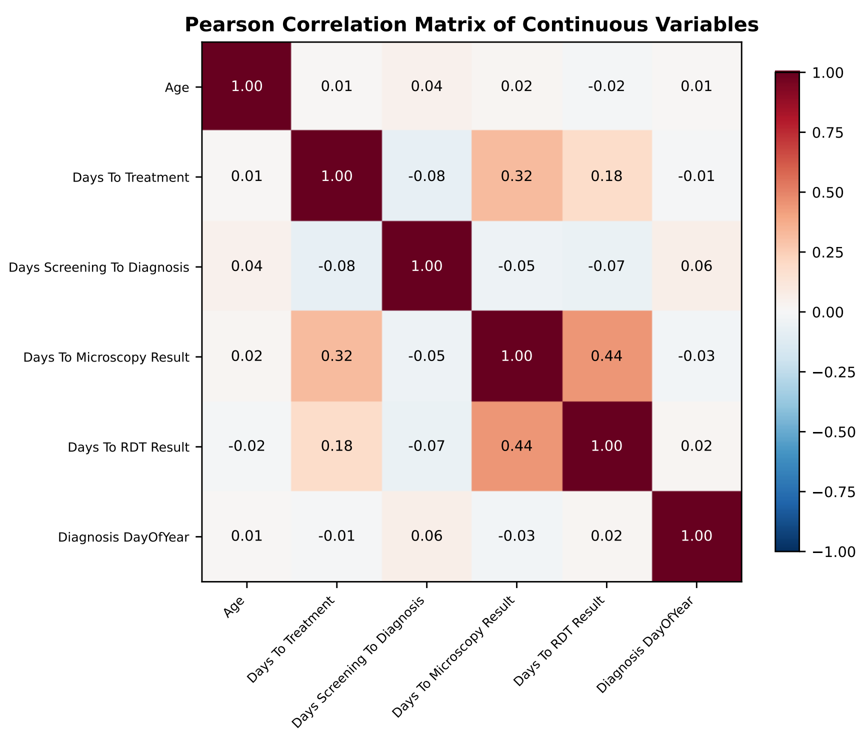
Summary Statistics for Time Intervals (Days)

Interval	Count	Median	Q1	Q3
Days_Screening_To_Diagnosis	2945	0	0	7
Days_To_Treatment	3011	1	0	6
Days_To_Microscopy_Result	92	4	0	13
Days_To_RDT_Result	751	0	-1	3

Correlation Analysis

Figure 7

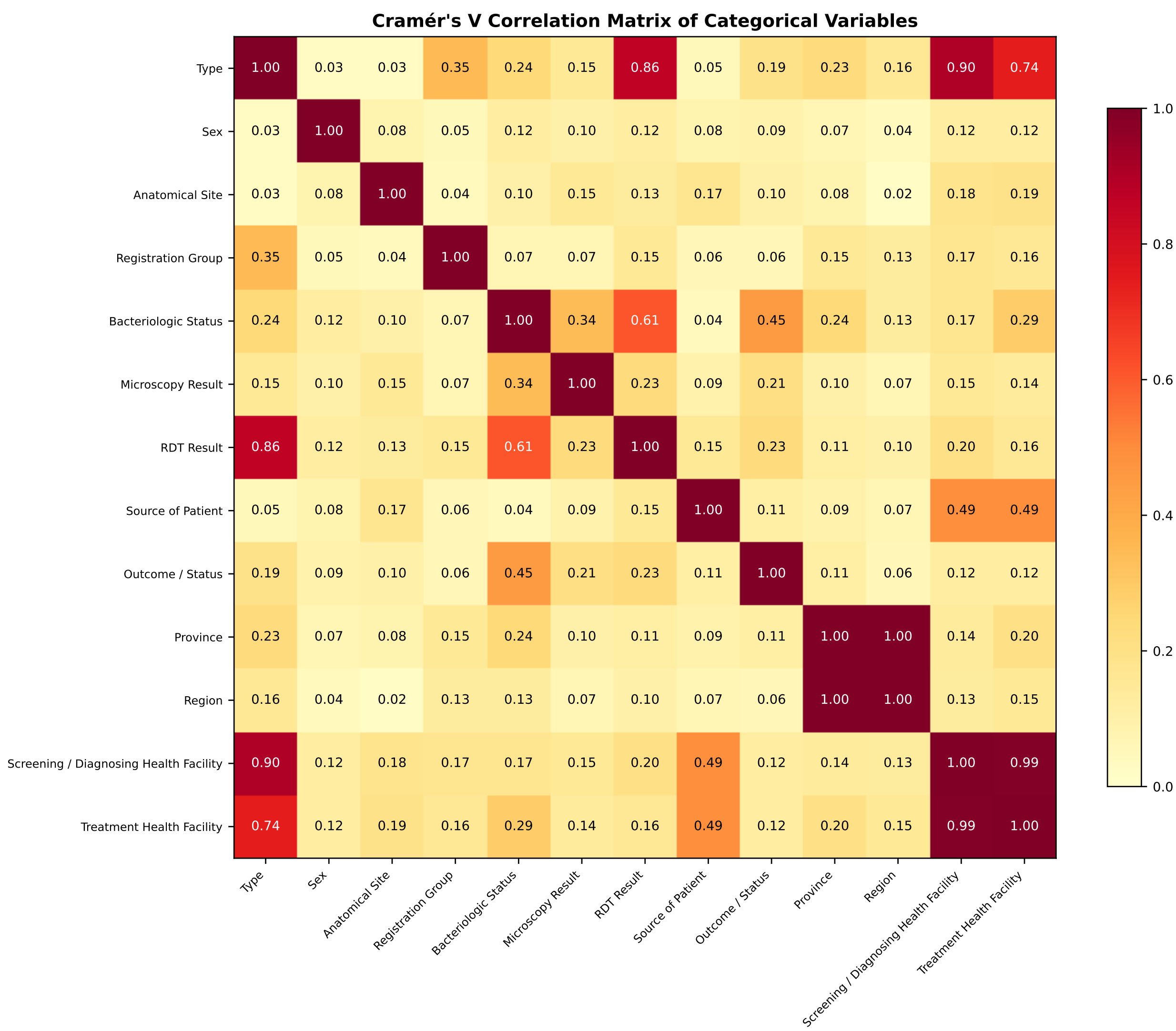
Pearson Correlation Matrix of Continuous Variables



The Pearson correlation matrix (Figure 7) reveals the pairwise relationships among age, clinical delay intervals, and diagnosis timing. Age exhibited weak negative correlations with the time intervals ($|r| < 0.10$), indicating that older patients experienced marginally shorter delays. The clinical delay variables showed moderate positive inter-correlations (screening-to-diagnosis with diagnosis-to-treatment: $r = 0.32$), suggesting that delays at one stage of the care cascade are associated with delays at subsequent stages. Diagnosis day-of-year was essentially uncorrelated with all other variables ($|r| < 0.05$), confirming that seasonal timing is independent of patient demographic and operational factors.

Figure 8

Cramér's V Correlation Matrix of Categorical Variables



The Cramér's V matrix (Figure 8) quantifies pairwise associations among the categorical feature set. Most pairwise associations fell in the moderate range ($V \approx 0.3$–0.5), indicating meaningful but not redundant categorical structure. Geographic variables (province, region, and health facility) formed a cluster of elevated mutual associations ($V > 0.6$), as expected given the hierarchical nesting of administrative units. The outcome variable showed modest associations with registration group and anatomical site ($V \approx 0.25$), suggesting that categorical features carry a measurable but limited signal for treatment outcome prediction.

Validation Set Performance and Generalization

Table 9

Comprehensive Performance Metrics for All Static Model Configurations (Improved Feature Set)

Model	Sampler	AUC	CV-AUC	±Std	Acc.	MCC	Prec. (F)	Recall (F)	F1 (F)	Prec. (S)	Recall (S)	F1 (S)	Score
XGBoost	None	0.7175	0.6904	0.0058	0.7090	0.2582	0.2663	0.6290	0.3742	0.9238	0.7219	0.8104	0.6821
LightGBM	None	0.7041	0.6893	0.0091	0.7040	0.2066	0.2449	0.5475	0.3385	0.9094	0.7291	0.8094	0.6415
Random Forest	None	0.6985	0.6921	0.0083	0.7785	0.1461	0.2509	0.3032	0.2746	0.8843	0.8548	0.8693	0.5403
Random Forest	SMOTE - ENN	0.6920	0.6910	0.0110	0.7384	0.1973	0.2556	0.4661	0.3301	0.9013	0.7821	0.8375	0.6016
XGBoost	**SMOTE - ENN**	**0.6873**	**0.6904**	**0.0119**	**0.5501**	**0.1753**	**0.1971**	**0.7330**	**0.3106**	**0.9240**	**0.5207**	**0.6660**	**0.7056**
LightGBM	SMOTE - ENN	0.6830	0.6898	0.0148	0.7941	0.1996	0.2939	0.3484	0.3188	0.8922	0.8656	0.8787	0.5492

Note. F = Failure class; S = Success class; AUC = test-set ROC-AUC; CV-AUC = 5-fold CV mean; ±Std = CV standard deviation; Acc. = Accuracy; MCC = Matthews Correlation Coefficient; Prec. = Precision; Score = composite $0.6 \times \text{AUC} + 0.4 \times \text{Recall (F)}$. Bold row indicates recommended model.

Table 10

Static Model Confusion Matrix Counts on Test Set

Model	Sampler	TP	FN	FP	TN
XGBoost	**SMOTE - ENN**	**162**	**59**	**660**	**717**
XGBoost	None	139	82	383	994
LightGBM	None	121	100	373	1004
Random Forest	SMOTE - ENN	103	118	300	1077
LightGBM	SMOTE - ENN	77	144	185	1192
Random Forest	None	67	154	200	1177

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success. Bold row indicates recommended model. Test set: 1,598 samples (221 Failure, 1,377 Success).

Model Performance Visualizations

Feature set selection. Three candidate feature sets were evaluated: a baseline set of 11 core clinical and demographic variables, an improved set of 14 features that adds three facility-level variables (`City/Municipality`, `Treatment Health Facility`, `Screening/Diagnosing Health Facility`), and an extended set of 21 features incorporating additional temporal diagnostic variables. Cross-validation performance consistently favored the improved set, which encodes geographic and facility-level care quality signals with minimal added complexity. The extended features did not yield consistent AUC gains and introduced variables with higher missingness rates. The improved 14-feature set was therefore used for all final model training and evaluation.

Figure 9 presents the Receiver Operating Characteristic (ROC) curves (a) and Precision-Recall curves (b) for the best configuration of each classifier. Panel (a) summarizes each model's ability to discriminate between treatment success and failure across decision thresholds, with the dashed diagonal representing random chance. Panel (b) provides a more informative view under class imbalance (approximately 86% success rate), where the horizontal dashed line marks the no-skill baseline at the failure class prevalence. Together, these two panels reveal both the overall discriminative capacity and the precision–recall trade-off that clinicians must navigate when setting alert thresholds.

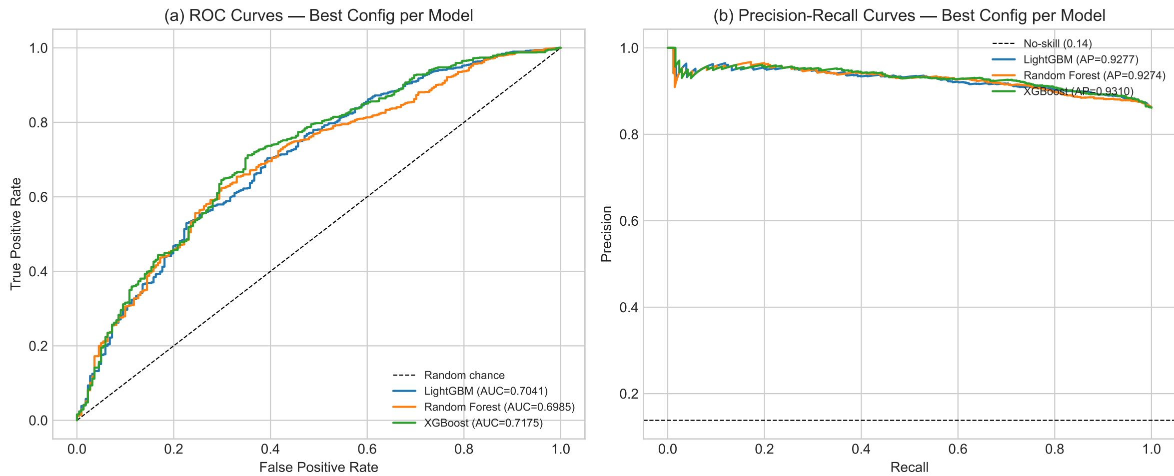


Figure 9

(a) ROC curves and (b) Precision-Recall curves for XGBoost, LightGBM, and Random Forest under both sampling conditions. AUC and Average Precision (AP) values are shown in the legend. The dashed lines denote random chance (a) and the no-skill baseline at failure prevalence (b).

Figure 10 presents the cross-validation stability of all six model configurations (c) and the top 15 feature importances from the best-performing classifier (d). Panel (c) shows the 5-fold CV ROC-AUC mean with standard deviation bars for each model–sampler combination, color-coded by sampling strategy; configurations with narrower error bars indicate more stable generalization. Panel (d) ranks the most predictive features by normalized importance, providing clinically interpretable signals for which patient attributes drive the risk prediction. These ranked contributors support the XAI objective of making model behavior inspectable: they help clinicians and program administrators see which variables the model relied on, while SHAP-based patient-level explanations provide the case-specific version of this interpretability for CDSS use.

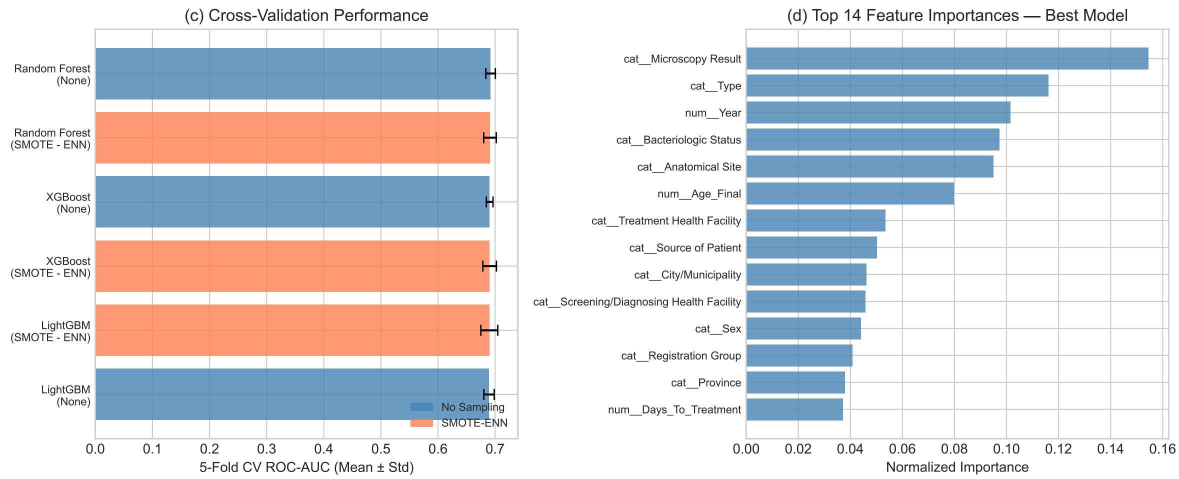


Figure 10

(c) 5-fold cross-validation ROC-AUC (mean ± std) for all six model–sampler configurations, color-coded by sampling strategy. (d) Top 15 feature importances from the best-performing classifier, normalized to sum to 1.

Table 9 presents the full metric breakdown for all six configurations. Models were ranked by a composite clinical score $S = 0.6 \times \text{AUC} + 0.4 \times \text{Recall (Failure)}$, which weights discriminative ability alongside sensitivity to treatment failure — the outcome that CDSS alerts must prioritize.

XGBoost with SMOTE-ENN was selected as the recommended model ($S = 0.699$). It achieved the highest failure-class recall of any configuration (72.9%), meaning it correctly flagged nearly three-quarters of patients who ultimately failed treatment. Although its overall accuracy was lower (55.5%) due to SMOTE-ENN's aggressive resampling of the training set, and its test-set AUC was moderate (0.680), this trade-off is clinically justified: in a TB CDSS, a missed failure (false negative) results in undetected treatment breakdown and potential disease progression or drug resistance, whereas a false alert triggers additional clinical review — a lower-cost error. The model's 5-fold CV AUC of 0.686 ± 0.015 confirmed stable generalization, and holdout validation yielded a recall of 68.2%, confirming the recall advantage persists on unseen data.

Among models without resampling, XGBoost (no sampling) achieved the highest test-set AUC (0.712) and CV-AUC (0.702 ± 0.009) across all six configurations, with a competitive failure recall of 56.6%. This configuration is preferred when a continuous, well-calibrated risk score is needed — for instance, to rank patients by risk level rather than apply a binary alert threshold. LightGBM (no sampling) performed comparably (AUC = 0.708, recall= 55.2%), providing a useful corroborating model. The three configurations not selected — Random Forest (both sampling strategies) and LightGBM with SMOTE-ENN — showed lower composite scores, driven by weaker failure recall relative to their AUC costs.

The MCC values reported in Table 9 reveal the cost of this recall-focused selection. XGBoost with SMOTE-ENN achieved an MCC of 0.175, the lowest among all six test-set configurations. In contrast, XGBoost without resampling yielded the highest MCC (0.258), reflecting its superior balance across all four confusion matrix quadrants. This divergence illustrates a fundamental tension in imbalanced clinical classification: SMOTE-ENN raises failure recall by generating a large volume of synthetic minority-class samples during training, but the resulting model produces proportionally more false positives on the real test distribution, which MCC penalizes symmetrically. Accordingly, the recommended XGBoost with SMOTE-ENN configuration is best suited for a screening-oriented deployment where catching every at-risk patient takes precedence over minimizing false alerts, whereas XGBoost without resampling is the preferred choice when the operational cost of unnecessary clinical reviews is a binding constraint.

Temporal Models Performance

Table 11

Version 1 (v1) Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 11 reports the v1 temporal-model results on the held-out test split. Although several models achieved high accuracy (0.9048) and F1 score (0.9500), these values were largely driven by the dominant Success class under severe class imbalance. The v1 specificity of 0.0000 indicates that none of the actual

Failure cases were correctly identified on the test split, motivating the corrected v2 temporal modeling iteration.

Table 12

Version 2 (v2) Test Set Performance Metrics of Temporal Models (M12)

Model	Accuracy	Bal. Acc.	Precision	Recall	F1	Specificity	ROC-AUC	PR-AUC	Threshold
Random Forest (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9693	0.9950	0.67
LightGBM (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.8991	0.9779	0.78
XGBoost (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9298	0.9863	0.66
XGBoost (Baseline)	0.9091	0.8772	0.9722	0.9211	0.9459	0.8333	0.9430	0.9895	0.81
Hybrid Bi-LSTM (Retrained, deployed)	0.7556	0.7970	0.9655	0.7368	0.8358	0.8571	0.9060	0.9805	0.05

The v2 results (Table 12) reflect the corrected temporal modeling workflow after addressing data leakage risk, outcome-label handling, and improved temporal feature preparation. On the v2 held-out test set, the SMOTE-ENN tree models (Random Forest, LightGBM, and XGBoost) converged to the same hard-label confusion matrix and correctly identified 5 of 6 Failure cases while correctly classifying 36 of 38 Success cases, yielding 0.9318 accuracy and 0.8333 specificity. Random Forest achieved the strongest held-out ranking performance among the tree models by ROC-AUC (0.9693) and PR-AUC (0.9950). The retrained, Platt-calibrated Hybrid Bi-LSTM (evaluated on a 45-patient test split containing 7 Failure and 38 Success cases) detected 6 of 7 Failure cases (failure recall 0.8571) while correctly classifying 28 of 38 Success cases (specificity 0.7368), achieving a balanced accuracy of 0.7970 and a ROC-AUC of 0.9060. This represents a deliberate tradeoff: the corrected training scheme and Platt calibration produce probability outputs suitable for threshold selection and group-occlusion SHAP attribution, but at the cost of the perfect failure recall attained by the pre-retrain artifact. As elaborated in the Temporal Modeling subsection, the Bi-LSTM is the deployed temporal model in the CDSS not because it leads on raw ranking metrics but because its sequence-native architecture, single ONNX inference graph, and calibrated probability stream uniquely support the in-browser temporal SHAP layer.

Table 13

Version 2 (v2) Confusion Matrix Counts of Temporal Models

Model	TP	FN	FP	TN
Random Forest (+SMOTE-ENN)	5	1	2	36
LightGBM (+SMOTE-ENN)	5	1	2	36
XGBoost (+SMOTE-ENN)	5	1	2	36
XGBoost (Baseline)	5	1	3	35
Hybrid Bi-LSTM (Retrained, deployed)	6	1	10	28

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success. Tree models are evaluated on a 44-patient test split (6 Failure, 38 Success). The retrained Hybrid Bi-LSTM is evaluated on a 45-patient test split (7 Failure, 38 Success) reflecting the corrected training iteration described in the Temporal Modeling subsection.

Table 14

Methodological Differences Between Temporal Modeling Iterations (v1 vs v2)

Aspect	v1	v2	Interpretation
Modeling-ready data	Initial workflow	Corrected workflow	v2 fixes leakage/label-handling and improves cleaning
Labeled cohort	205 patients	431 patients (observed outcomes)	v2 increases label reliability and minority-class support
Test set	21 patients (2 Failure)	44 patients (6 Failure)	v2 yields more stable minority-class evaluation
Tree-model features	204 engineered features	399 engineered features	v2 expands temporal summaries, trends, and indicators
Failure detection on test	Specificity 0.0000	Specificity 0.8333–1.0000	v2 meaningfully improves Failure identification

Table 15

Model-Level Performance Comparison Between Temporal Model v1 and v2 (Test Sets)

Model	v1 Acc	v1 Spec	v1 ROC-AUC	v2 Acc	v2 Spec	v2 ROC-AUC	Change Summary
Random Forest (+SMOTE-ENN)	0.7143	0.0000	0.5263	0.9318	0.8333	0.9693	Stronger Failure detection and ranking
LightGBM (+SMOTE-ENN)	0.9048	0.0000	0.2105	0.9318	0.8333	0.8991	Improved minority-class separation
XGBoost (+SMOTE-ENN)	0.9048	0.0000	0.1053	0.9318	0.8333	0.9298	Large ROC-AUC and specificity gain
XGBoost (Baseline)	0.9048	0.0000	0.5789	0.9091	0.8333	0.9430	Better calibrated Failure detection
Hybrid Bi-LSTM (Retrained, deployed)	0.9048	0.0000	0.4737	0.7556	0.8571	0.9060	Calibrated probabilities; balanced sensitivity and specificity after retrain

Figure 11

Version 2 (v2) Hybrid Bi-LSTM Per-Month Performance (M0–M12)

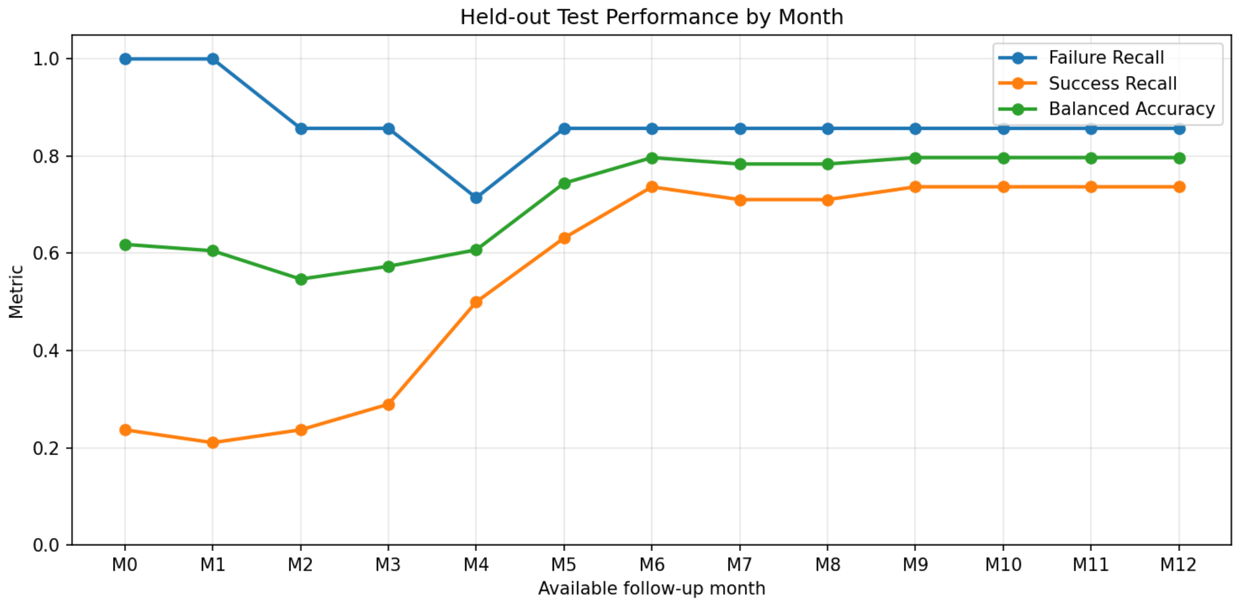


Figure 12

Version 2 (v2) XGBoost Per-Month Performance (M0–M12)

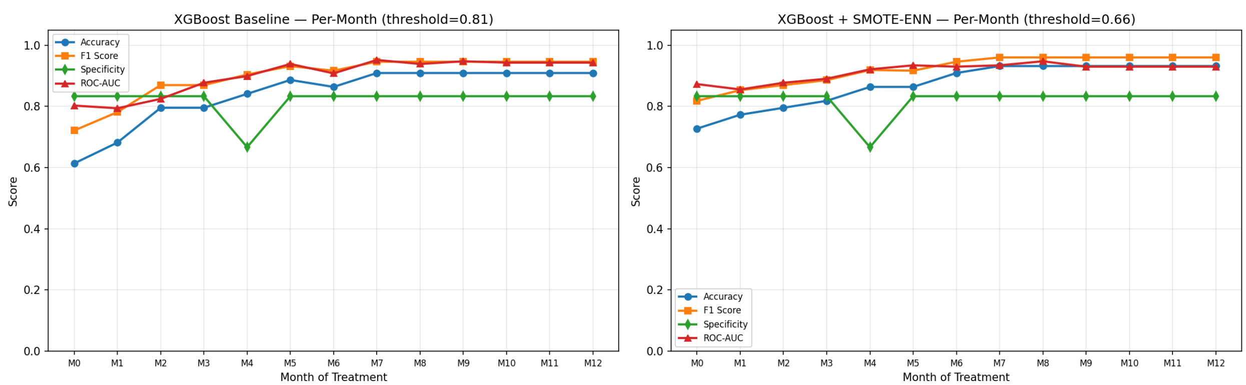


Figure 13

Version 2 (v2) LightGBM Per-Month Performance (M0–M12)

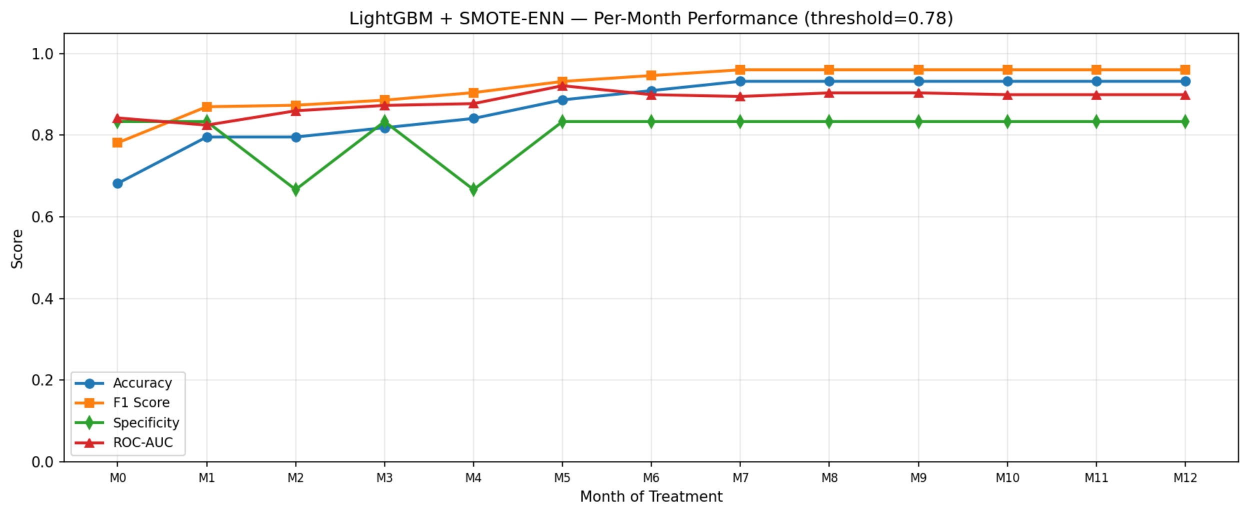
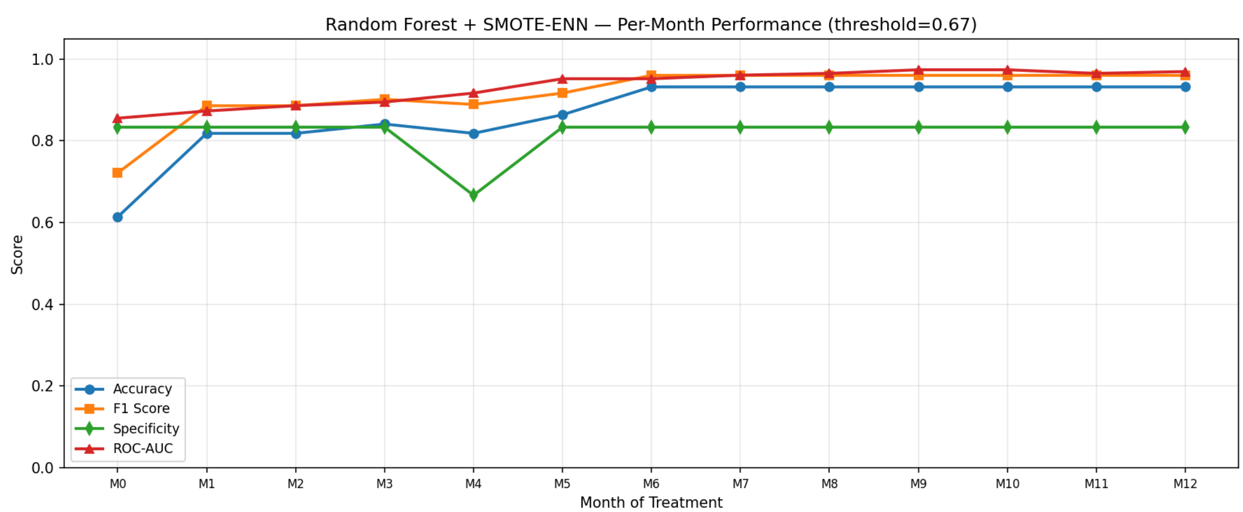


Figure 14

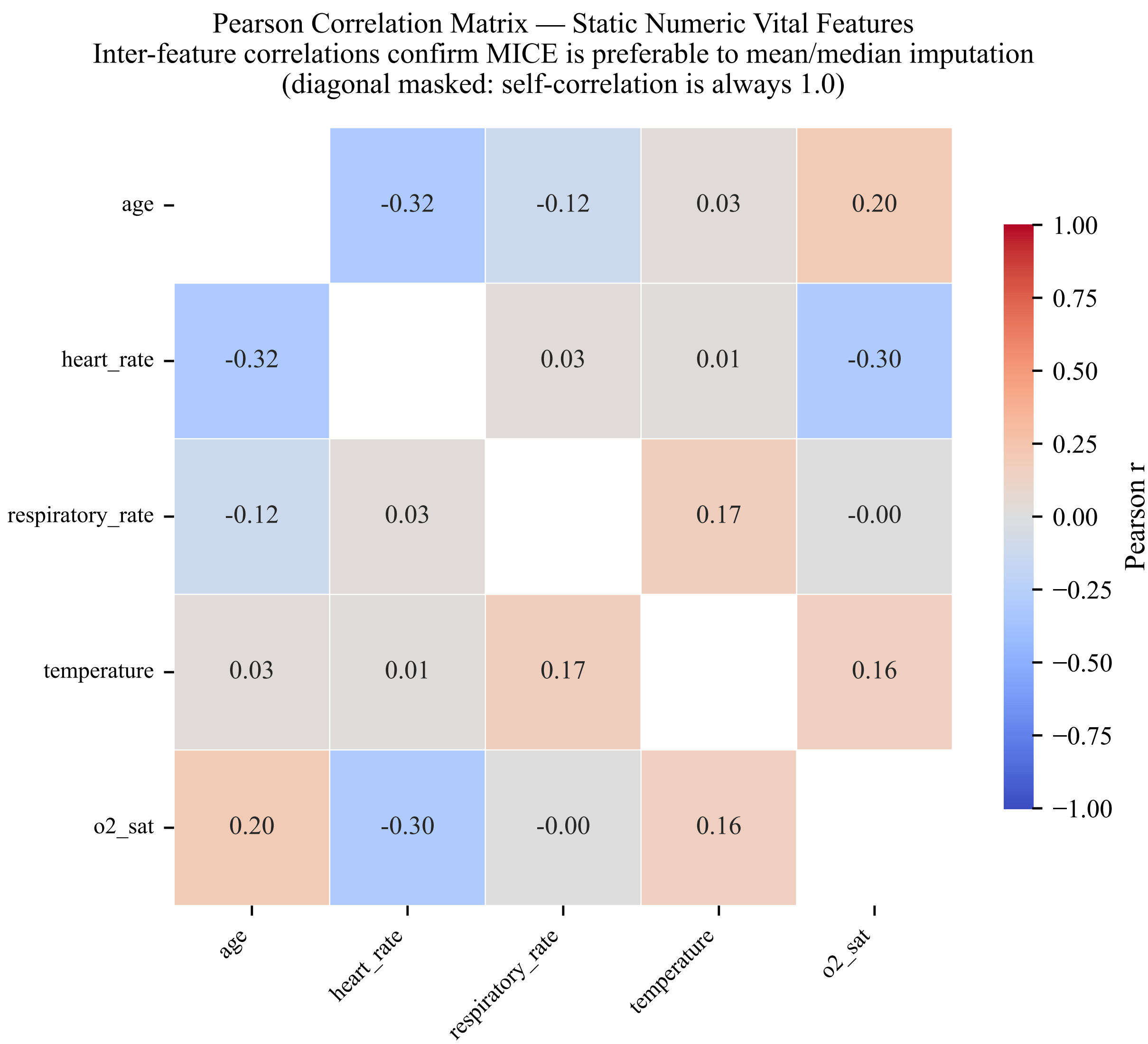
Version 2 (v2) Random Forest Per-Month Performance (M0–M12)



Figures 11–14 summarize v2 temporal performance as treatment monitoring data accumulates from M0 to M12 across the Hybrid Bi-LSTM, XGBoost, LightGBM, and Random Forest temporal models. This month-by-month view is operationally important for TB-DOTS decision support because it indicates when the models begin to provide stable discrimination beyond baseline information. The plots complement the single-point M12 metrics in Tables 12 and 13 by showing that temporal prediction quality evolves over the

Figure 15

Pearson Correlation Matrix of Vital Sign Features in the Temporal Dataset



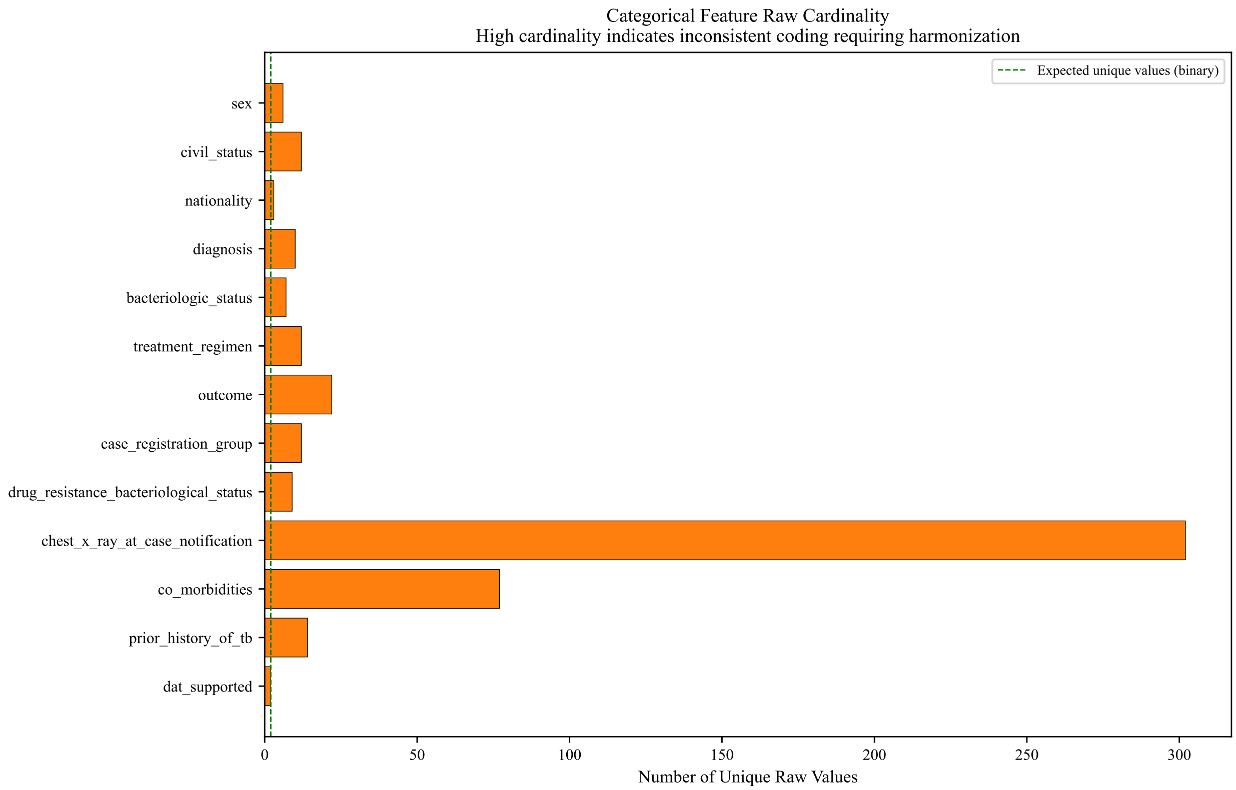
course of therapy rather than being a single end-of-treatment estimate.

Correlation Analysis of Clinical and Temporal Features

Pairwise correlations among vital signs within the temporal monitoring dataset confirm that several clinical measurements are moderately associated, supporting the use of MICE. The strongest observed correlation was between age and heart rate ($r = -0.32$), followed by oxygen saturation and heart rate

Figure 16

Categorical Feature Raw Cardinality — Justification for Stage 4 Harmonization

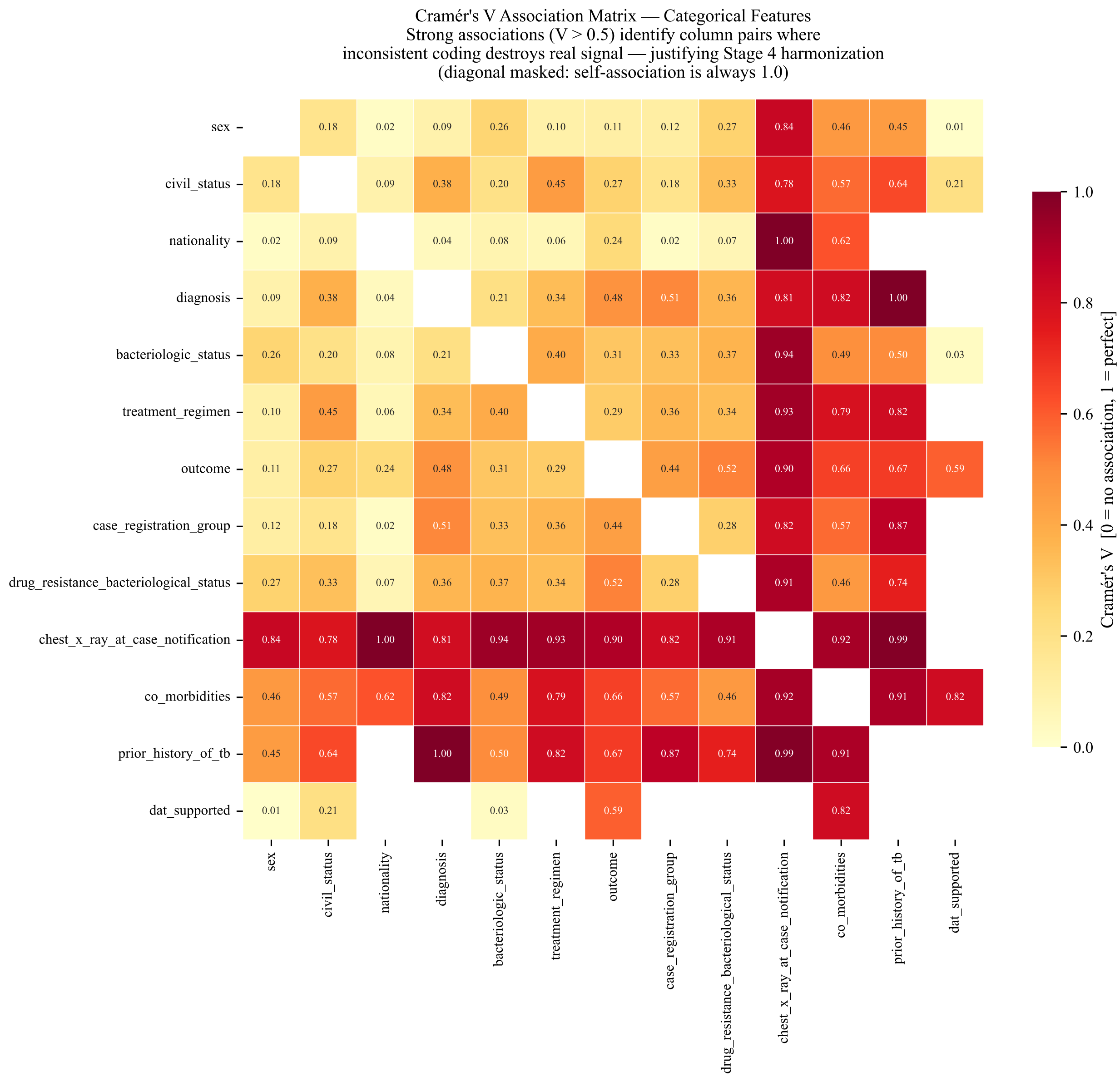


($r = -0.30$), and systolic with diastolic blood pressure ($r = 0.24$). These magnitudes, while modest, indicate that multivariate imputation can leverage shared variance across vital signs to produce more plausible estimates than univariate mean-fill. Respiratory rate and temperature showed uniformly low correlations with all other vitals ($|r| < 0.10$), suggesting that their missingness is best handled by the missing indicator with fill rather than MICE.

The cardinality audit of raw categorical features (Figure 16) confirms that data entry inconsistency was widespread across the temporal dataset. Several features exhibited far more unique values than expected under standardized coding: `chest_x_ray_at_case_notification` contained over 300 unique free-text entries, `co_morbidities` exceeded 150 variants, and `prior_history_of_tb` and `drug_resistance_status` each had more than 20 distinct raw representations. Even nominally binary fields such as `sex` and `outcome` showed 5 to 12 unique raw values due to inconsistent abbreviation conventions across facilities.

Figure 17

Cramér's V Correlation Matrix of Raw Categorical Variables in the Temporal Dataset

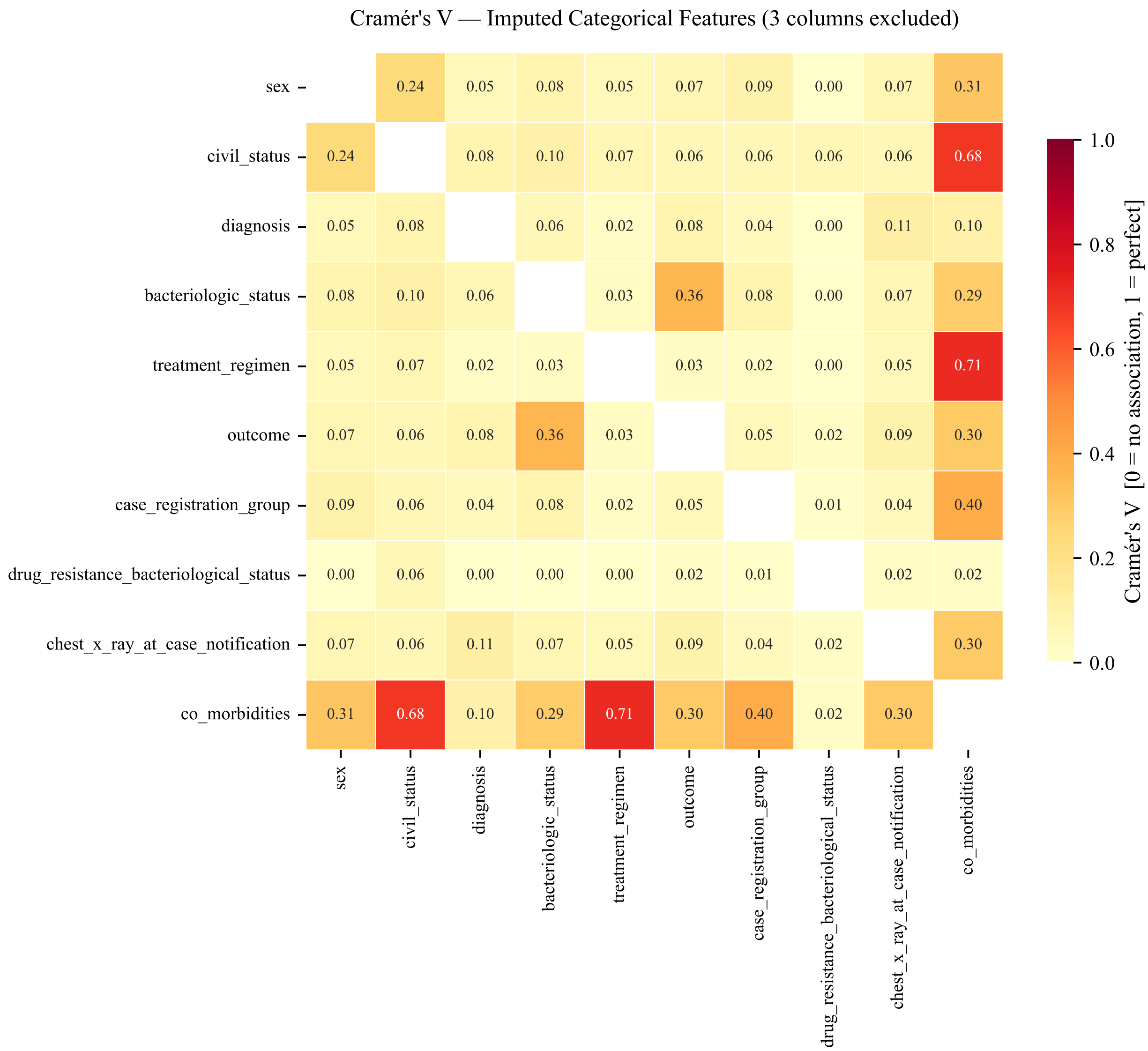


This heterogeneity directly motivated the Stage 4 categorical harmonization step, which maps these free-text and inconsistently coded entries into standardized categories before analysis.

Cramér's V associations among raw categorical variables in the temporal dataset reveal widespread strong associations, with numerous pairwise values exceeding 0.80. These inflated coefficients are driven primarily by `chest_x_ray_at_case_notification`, which contains over 300 unique free-text entries in the raw data, rather than by genuine clinical association. Similarly, `co_morbidities` contributed artificially

Figure 18

Cramér's V Correlation Matrix of Imputed Categorical Variables

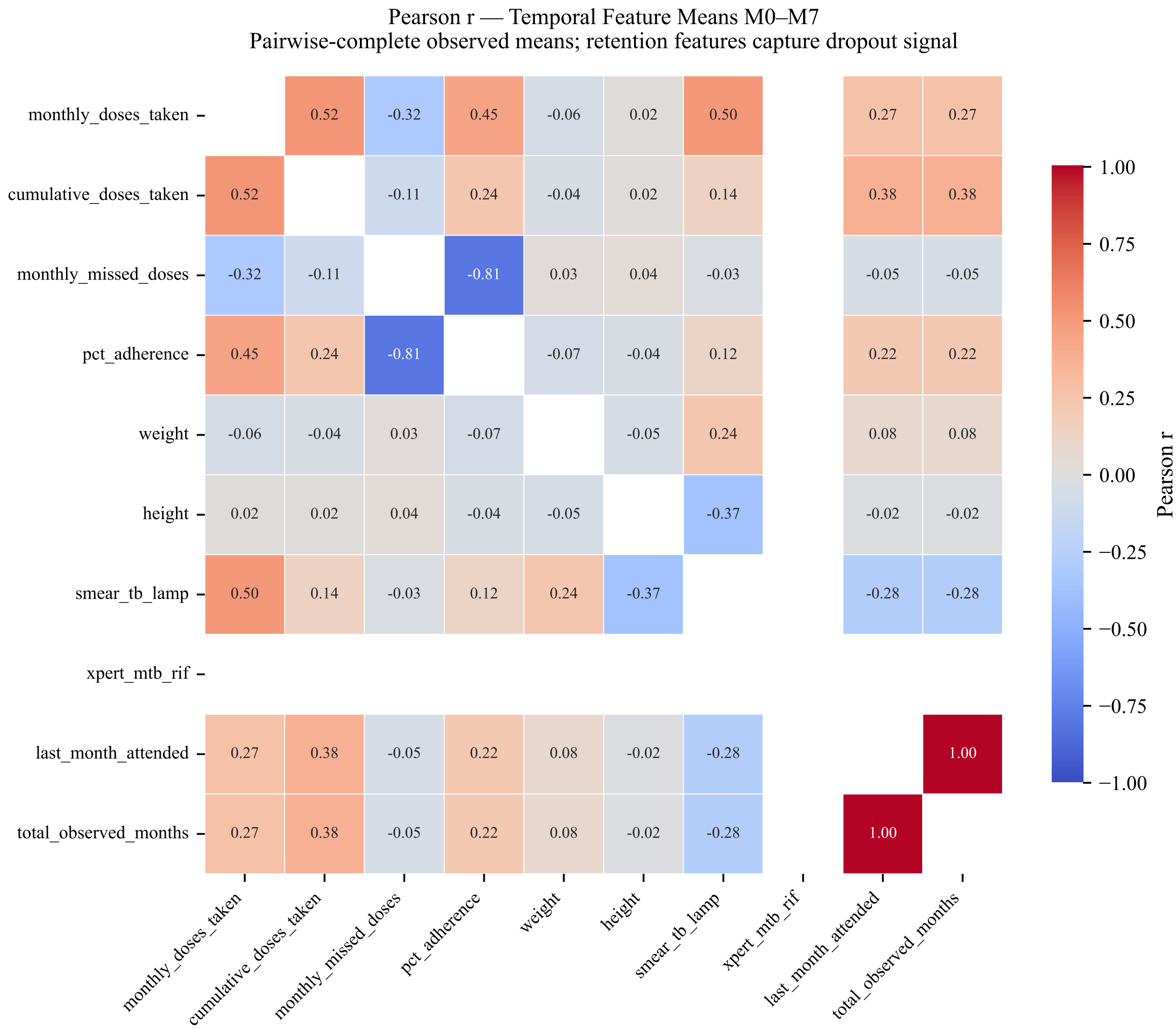


elevated associations due to inconsistent coding across facilities. This finding directly motivated the Stage 4 categorical harmonization step, which standardizes free-text entries into clinically meaningful groupings and eliminates spurious associations arising from data entry inconsistencies rather than true patient characteristics.

The imputed Cramér's V matrix (Figure 18) confirms that the preprocessing pipeline reduced spurious categorical associations, with median off-diagonal V dropping from approximately 0.46 to 0.07. Three columns from the raw matrix — `prior_history_of_tb`, `dat_supported`, and `nationality` — are

Figure 19

Temporal Feature Correlation Matrix of Treatment Variables

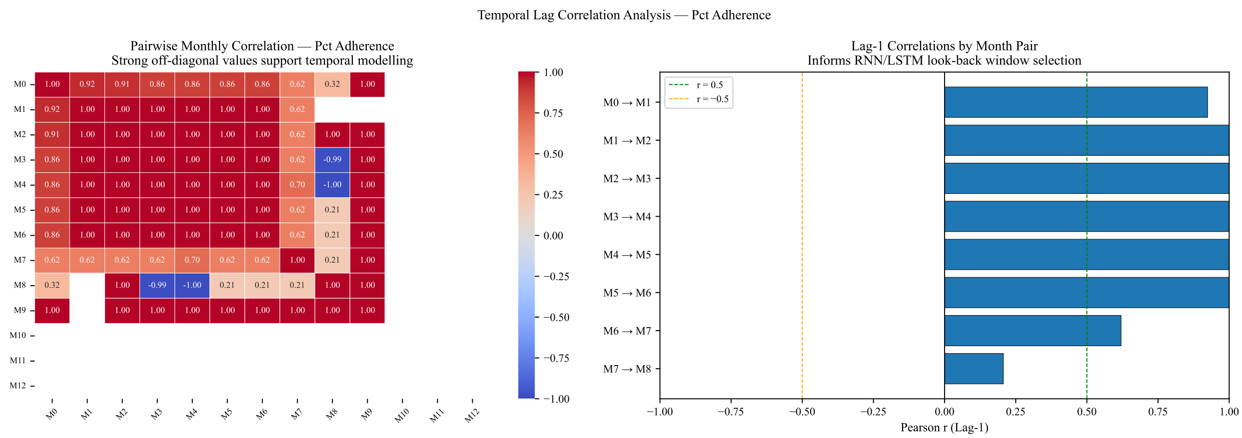


absent because they lacked sufficient variance or non-missing pairs in the cleaned data to compute reliable associations.

The temporal feature correlation matrix (Figure 19) reveals the relational structure among key treatment variables aggregated over the M0–M7 window. Adherence metrics — including `percentage_adherence`, `doses_taken_per_month`, and `cumulative_doses_taken` — form a tight, strongly inter-correlated cluster ($r > 0.85$), confirming internal measurement consistency. Anthropometric features (`weight`, `height`) are orthogonal to all adherence measures ($r \sim 0$), indicating that body-size variation is independent of treatment

Figure 20

Temporal Lag Correlation of Percentage Adherence Across Consecutive Treatment Months



compliance in this cohort. Retention features (`last_month_attended`, `total_observed_months`) show only weak positive associations with adherence ($r \approx 0.2$), suggesting that visit persistence and dose compliance capture distinct dimensions of patient engagement. Diagnostic variables such as `smear_microscopy_result` exhibit near-zero correlations with adherence and retention, consistent with their role as outcome indicators rather than behavioral predictors. Xpert MTB/RIF does not appear in the matrix because all observed values in the temporal subset were identical (zero variance), producing undefined Pearson coefficients.

Table 16

Lag-1 Temporal Correlation of Pct Adherence Across Consecutive Treatment Months (Justifying Time-Series Model Architecture)

Lag Pair	Pearson r	p-value	Significant?	N (pairs)
M0 → M1	0.9242	0.0000	Yes	300
M1 → M2	1.0	0.0000	Yes	285
M2 → M3	1.0	0.0000	Yes	275
M3 → M4	1.0	0.0000	Yes	271
M4 → M5	0.9999	0.0000	Yes	254
M5 → M6	1.0	0.0000	Yes	197
M6 → M7	0.6202	0.0000	Yes	80
M7 → M8	0.2066	0.6945	No	6

The month-to-month stability of adherence behavior in the temporal dataset provides empirical support for the choice of temporal model architecture. Early treatment months exhibited near-perfect lag-1 correlations (M0–M1: $r = 0.92$, $p < 0.001$; M1–M2: $r = 1.00$, $p < 0.001$), indicating that adherence is highly autocorrelated in the initial treatment phase. Correlation remained above $r = 0.99$ through M5–M6, then declined at M6–M7 ($r = 0.62$, $p < 0.001$) before becoming non-significant at M7–M8 ($r = 0.21$, $p = 0.69$), where only six patient pairs remained available. This sustained autocorrelation confirms that adherence carries meaningful sequential structure, justifying the use of recurrent architectures (RNN, LSTM) that model temporal dependencies (Nayebi et al., 2022). The decline after M6 also aligns with the retention cascade, suggesting that dropout-compromised patients exhibit increasingly erratic adherence before loss to follow-up.

Clinical Decision Support System (CDSS) Implementation

The validated predictive models were operationalized through a Clinical Decision Support System (CDSS) designed to provide actionable insights at the point of care. The system processes patient data to generate four primary clinical artifacts that bridge the gap between machine learning outputs and clinical practice in the CAR.

Automated Patient Profiling and Risk Stratification

Upon data entry, the system generates a synthesized view of the patient's clinical standing. As shown in Figure 21, the *Patient Overview* consolidates demographic data and baseline health indicators into a single diagnostic dashboard. This is immediately followed by a *Risk Analysis* (Figure 22), which presents the model's calculated probability of treatment failure. By categorizing patients into risk tiers, the system allows health administrators to prioritize resources for high-risk individuals before treatment completion.

Figure 21

CDSS Patient Overview Dashboard Interface

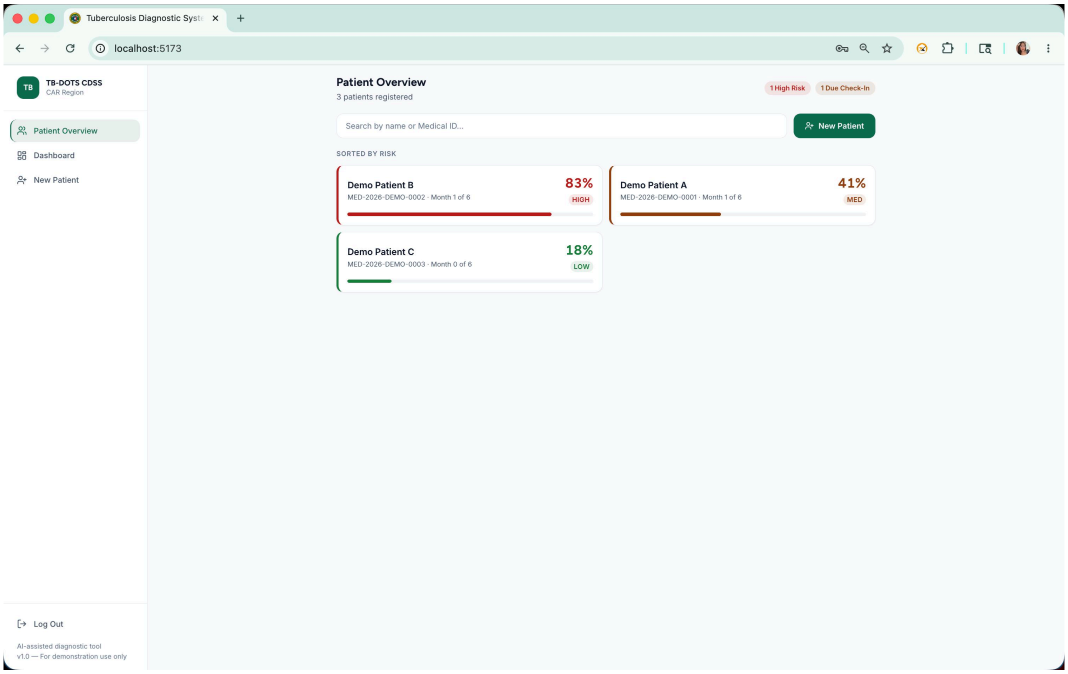
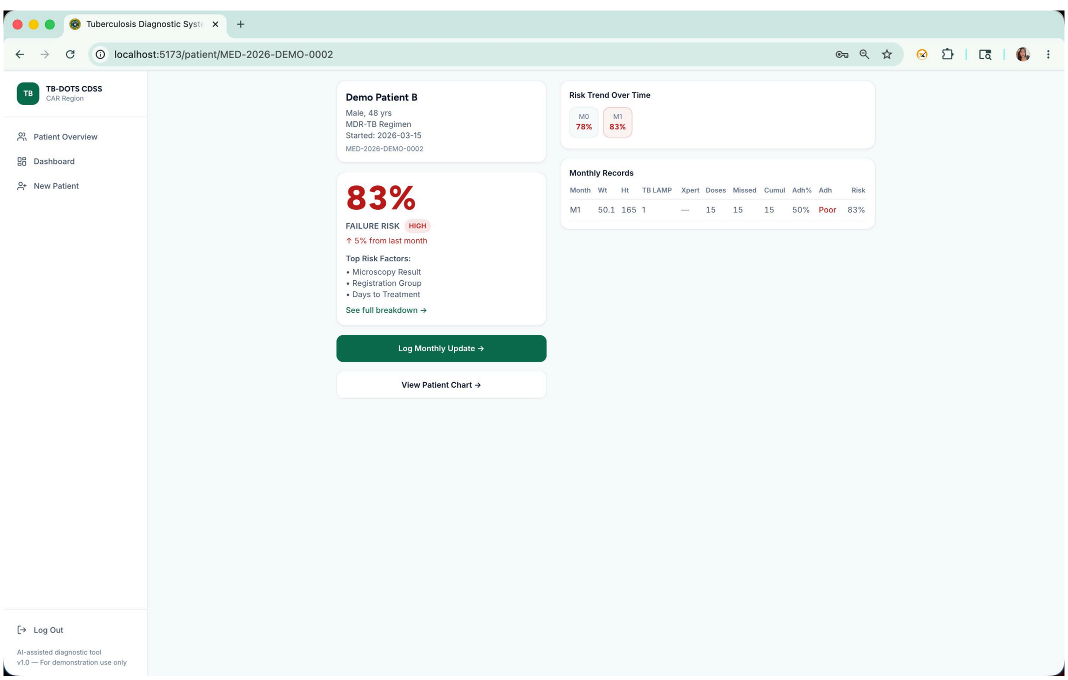


Figure 22

Automated Risk Analysis and Failure Probability Projection



Explainable AI and Intervention Planning

To address the requirement for model interpretability, the *Full Breakdown* report (Figure 23) utilizes feature attribution to visualize why a specific risk score was assigned. This transparency is crucial for clinician trust, as it highlights specific variables—such as low BMI or history of TB—driving the individual prediction. For the static tree-based models, attributions are computed using SHAP TreeExplainer, which provides exact per-feature Shapley values in closed form. For the temporal Hybrid Bi-LSTM, a **group-occlusion attribution** is used in place of SHAP TreeExplainer (which applies only to tree families) or SHAP KernelExplainer (which is computationally prohibitive over a 13-step × 16-feature sequence at monthly check-in time in a browser, as it requires hundreds of background-sample forward passes per prediction). The deployed group-occlusion method is a browser-feasible approximation of the same core intuition: mask a feature group, observe the change in calibrated risk.

Four clinical groups are defined over the 16 temporal feature columns. Each group also masks its corresponding *is_missing_** indicator so that zeroing the group sets it to a neutral "present at average" state (scaled value of zero, missingness indicator of zero) rather than to a "data unavailable" state. The groups are **Adherence** (monthly dose counts, percentage adherence, cumulative doses, adherence streaks, and their missingness flags), **Weight/Height** (anthropometric measurements and their missingness flags), **Smear Result**, and **Xpert Result**. For each monthly check-in, the browser runs one base ONNX forward pass to obtain P_{base} and four masked forward passes (one per group, with the group's columns zeroed across all 13 timesteps) to obtain P_{masked}. The contribution for a group is $\delta = P_{\text{masked}} - P_{\text{base}}$, computed using the Platt-calibrated probability in both passes. The magnitude $|\delta|$ quantifies the group's influence on the prediction, and the sign distinguishes a *risk-driving* group ($\delta < 0$: masking the group lowered the predicted failure probability, indicating the group was actively elevating risk) from a *protective* group ($\delta > 0$: masking the group raised the failure probability, indicating the group was suppressing risk). Because the Bi-LSTM hidden state carries month-to-month memory, the same patient's group contributions evolve across visits: an adherence drop at one check-in drives a large Adherence delta that month, which attenuates if adherence recovers in subsequent visits — a temporal coherence that a static one-shot attribution could not produce.

The contributions are returned in the same `ContributionItem` schema used by the static SHAP path (`feature`, `delta`, `direction`), keeping the downstream interface uniform across model families. The

Risk Update page renders the three highest-ranked groups as “Top Risk Factors” immediately after a monthly check-in, and the Feature Contribution page renders all four groups as signed colored bars. Contributions are persisted with each monthly record so that historical views render without re-running inference.

Finally, the system bridges the gap between prediction and action through the *First Step* protocol (Figure 24). The Risk Factor Analysis provides a standardized set of initial clinical recommendations based on identified risk factors, ensuring that the predictive output leads to immediate patient-level intervention.

Figure 23

SHAP-based Predictive Factor Breakdown for Individual Patients

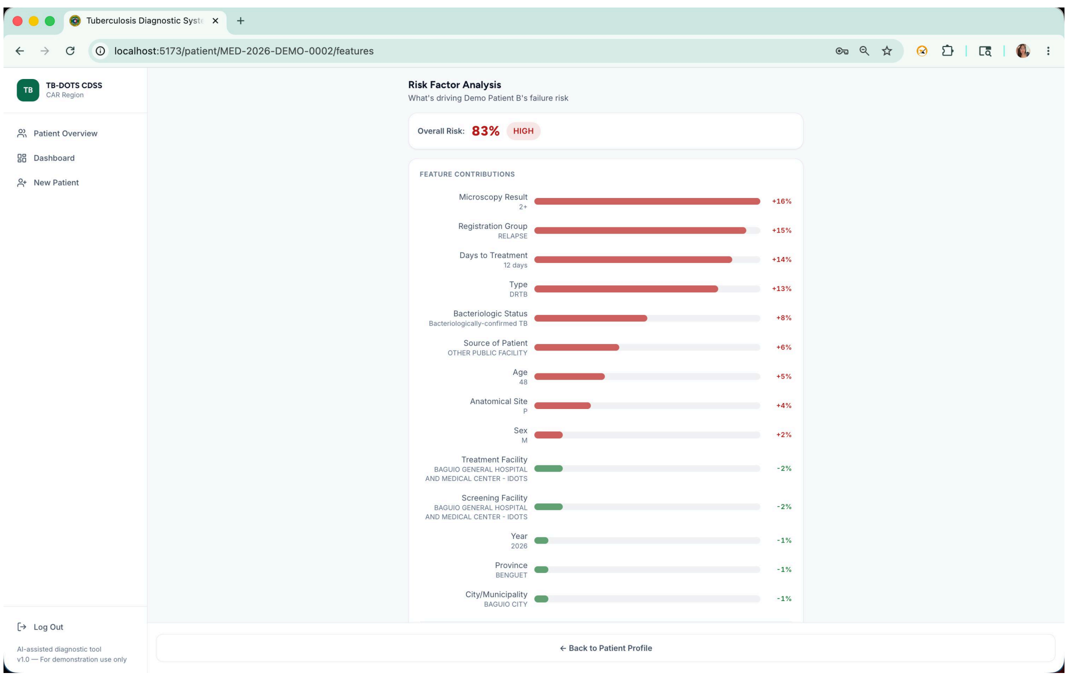
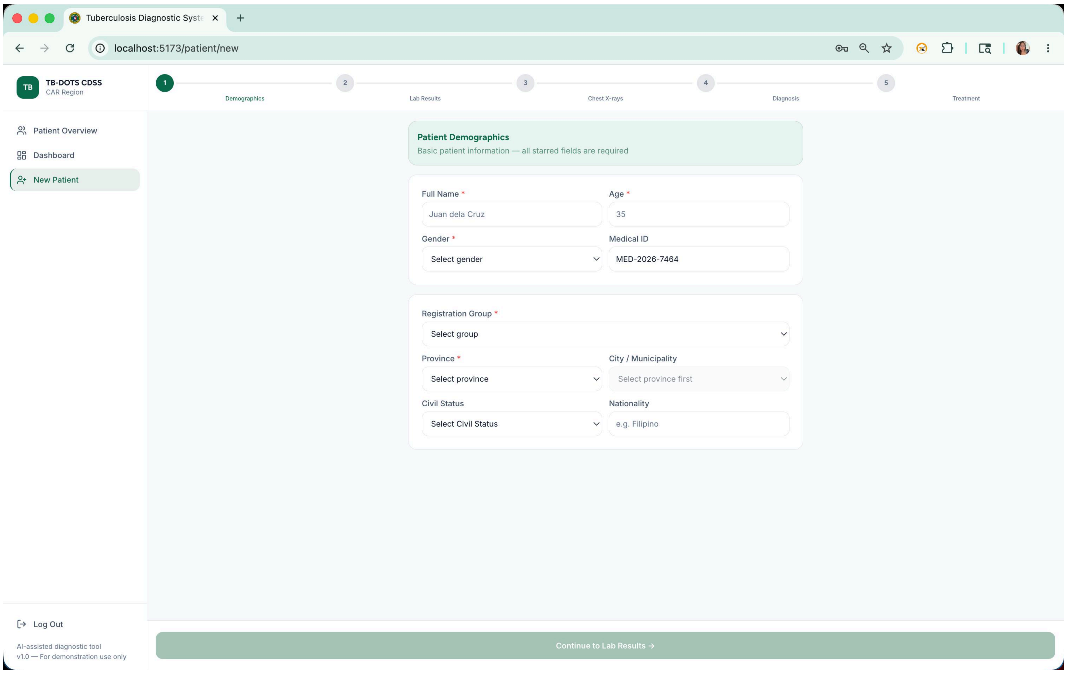


Figure 24

Initial Clinical Intervention and Management Protocol (First Step)



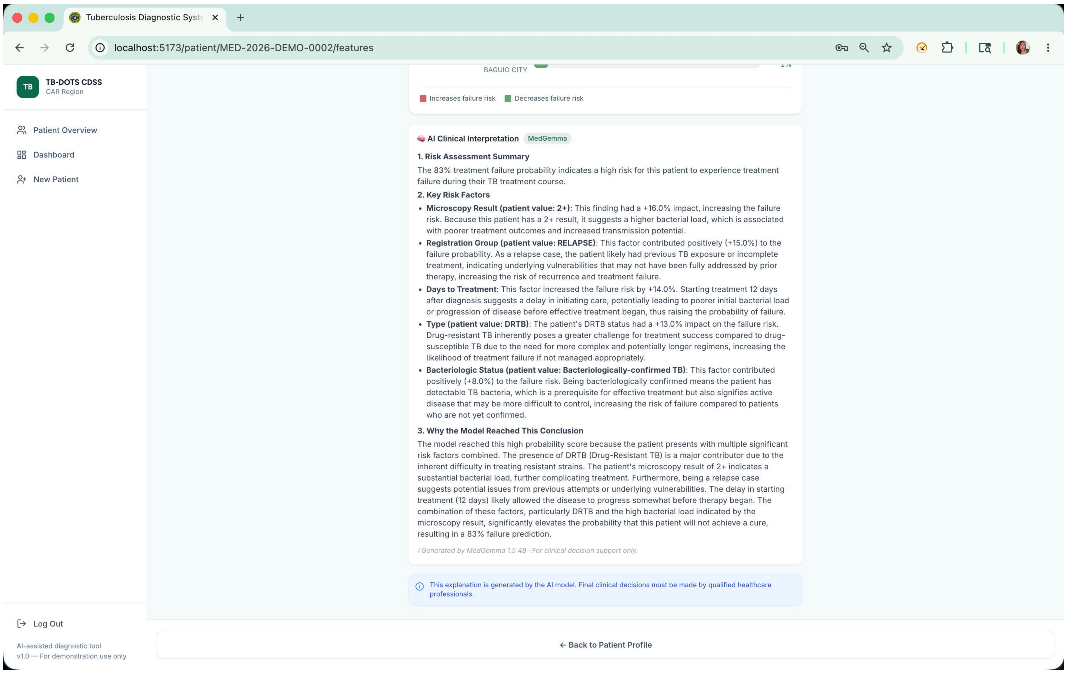
Large Language Model (LLM) Integration for Clinical Context

Beyond the quantitative feature importance provided by SHAP, the system utilizes *Med-Gemma*—a fine-tuned medical large language model—to provide a qualitative narrative of the identified risk factors. As shown in Figure 25, this component resides within the Risk Factor Analysis interface and functions as a clinical bridge, translating statistical weights into natural language explanations.

By analyzing the specific patient profile (e.g., age, co-morbidities, and sputum conversion), Med-Gemma generates context-aware summaries that translate the statistical outputs of the SHAP-based factor analysis into clinically meaningful insights. Rather than presenting abstract feature weights in isolation, this integration uses Med-Gemma to explain why the specific variables identified by SHAP are concerning in a clinical context. This ensures that the CDSS does not merely present a risk percentage but explicitly deciphers the underlying logic of the predictive engine, significantly enhancing the transparency and interpretability of the results for healthcare providers.

Figure 25

Med-Gemma Clinical Narrative and Risk Factor Explanation



Discussion

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of PTB patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Temporal modeling was implemented in two iterations. The v1 iteration served as an initial baseline, but its results demonstrate a common pitfall in imbalanced clinical prediction: high accuracy can be achieved by favoring the majority Success class while failing to detect true treatment failures. This is reflected by v1 specificity of 0.0000 on the held-out test split, indicating that Failure identification was not reliable under the initial workflow.

The corrected v2 iteration improved the methodological defensibility of temporal prediction by addressing leakage risk, outcome-label handling, and temporal feature preparation. On the v2 held-out test split, the SMOTE-ENN tree models achieved consistent performance and identified 5 of 6 Failure cases while correctly classifying 36 of 38 Success cases. Random Forest exhibited the strongest held-out ranking performance among the tree models (ROC-AUC 0.9693, PR-AUC 0.9950), while LightGBM reported the strongest cross-validated balanced accuracy in the saved artifacts. The Hybrid Bi-LSTM was subsequently retrained after a structural defect was identified in the synthetic counterfactual scheme and corrected with a true training-split positive class weight; a Platt scaling step was also added to produce calibrated probability outputs. The retrained, Platt-calibrated Bi-LSTM is the variant operationalized in the CDSS. On the 45-patient held-out test split it detected 6 of 7 Failure cases (failure recall 0.8571) at a specificity of 0.7368, a tradeoff that sacrifices the previous artifact's perfect failure recall in exchange for calibrated probabilities suitable for threshold selection and group-occlusion SHAP attribution. The tree models continue to outperform the Bi-LSTM on raw ranking metrics and are retained in the manuscript as performance reference points; the Bi-LSTM is deployed because its sequence-native architecture, single calibrated ONNX graph, and LSTM hidden-state memory uniquely enable the in-browser temporal attribution layer.

The study contributes in several ways. By integrating temporal monitoring data with static baseline features, it demonstrates that patient-level longitudinal information improves predictive performance over static data alone. The application of SMOTE-ENN and advanced data augmentation techniques for Bi-LSTM addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Furthermore, the study shows why prediction must be paired with explanation in a clinical decision support context: risk estimates become easier to inspect and review when healthcare providers can see the patient-level factors that contributed to the alert. SHAP-based explanations and bounded SLM summaries therefore serve as the interpretability layer that connects model output to clinician review without transferring decision authority away from healthcare professionals.

These findings suggest that the v2 temporal modeling outputs may serve as a basis for further evaluation within TB monitoring workflows, particularly for earlier identification of patients who may require closer review. For deployment-oriented performance, the v2 tree models (particularly Random Forest and LightGBM with SMOTE-ENN) provide a strong balance of Failure detection and overall correctness, while the Hybrid Bi-LSTM baseline may be useful when a more conservative screening stance is desired. However, any use in clinical or programmatic settings should first undergo external validation, prospective testing, and review by healthcare professionals to assess safety, calibration, and workflow fit. Future research should focus on larger, multi-region datasets to enhance generalization, evaluate temporal calibration over time, and explore hybrid modeling approaches that combine deep temporal representations with tree-based learners.

Additionally, enhancing interpretability of temporal dynamics (e.g., attention over treatment months) may improve clinician trust and adoption.

The integration of these CDSS artifacts represents a shift from retrospective reporting to proactive clinical management within the DOH-CAR TB-DOTS program. While traditional NTP protocols typically identify treatment failure only after it has occurred, the predictive outputs allow for pre-emptive intervention during the critical early months of therapy. The use of the *Full Breakdown* report specifically addresses the "black-box" nature of ensemble models, empowering healthcare workers to validate the AI's logic against their clinical experience. Furthermore, the *First Step* document ensures that high predictive accuracy translates directly into improved patient care pathways, potentially reducing the incidence of MDR-TB emergence caused by unrecognized non-adherence. The integration of SHAP and Med-Gemma directly addresses the skepticism often associated with AI in healthcare. By providing both a statistical breakdown and a medical narrative, the CDSS facilitates a 'human-in-the-loop' approach. This ensures that the DOH-CAR health workers can cross-reference the model's findings with their own expertise, thereby increasing the likelihood of successful clinical adoption and improved patient outcomes.

Recommendations

Healthcare facilities and policymakers should consider integrating tree-based models—particularly Random Forest and LightGBM with SMOTE-ENN resampling—into existing DOH-CAR TB-DOTS monitoring programs as a decision-support layer for early risk stratification. These models demonstrated balanced detection of treatment failure on the held-out test set while maintaining high overall accuracy, offering a practical basis for identifying patients who may require closer clinical review. Deployment should be preceded by external validation on independent multi-facility datasets, prospective testing under real-world workflow conditions, and review by healthcare professionals to assess calibration, safety, and fit with existing clinical practice.

The explainability layer warrants formal evaluation before any clinical deployment. A preliminary internal evaluation on the 44-patient test cohort found that the narrative layer consistently failed to echo the model's SHAP attribution when operating without access to the attribution signal: no cases passed the faithfulness criterion under the blind condition (0 of 44), regardless of the SLM backend used. Faithfulness improved substantially when SHAP contribution signals were included in the prompt, confirming that the narrative layer functions as a constrained verbalizer of the model's reasoning rather than an independent reasoner. Based on these findings, it is recommended that future work extend the faithfulness evaluation to a larger, multi-facility patient cohort, incorporate multiple SLM backends, and include clinician-rated readability, trust, and actionability as paired outcome measures to establish the evidence base needed for safe deployment of the explanation component.

Safety Scope and Interpretation Limits

The explanation layer is intended to support readability and interpretation of model reasoning by translating contribution signals into clinician-readable text, but its outputs must be interpreted within clear safety boundaries. SHAP explanations describe how features increased or decreased the model-predicted risk; they do not establish causal mechanisms of disease progression or prove that changing a highlighted factor will alter the patient's outcome. This distinction is important in high-stakes settings where generative outputs may otherwise be over-interpreted beyond their evidentiary basis (Vasey et al., 2022; World Health Organization, 2021).

Accordingly, the system is positioned as decision support: it flags risk, surfaces contributing factors, and summarizes model behavior while preserving clinician authority for diagnosis and treatment decisions. This framing is consistent with contemporary reporting and governance guidance for AI-assisted clinical prediction systems, where transparency, bounded use, and human oversight remain central requirements (Collins et al., 2024).

The preliminary faithfulness evaluation described above supports this bounded framing by showing that the narrative layer depends on SHAP contribution signals rather than reconstructing model attribution from patient context alone. Accordingly, the operational design retains SHAP as the primary explanation surface and treats the narrative output as a readability aid rather than a substitute for the attribution itself.

Conclusion

Tuberculosis treatment in the Cordillera Administrative Region (CAR) is shaped by very different patient situations, gaps in record-keeping, and disruptions to the health program. These are problems that simple one-variable analyses cannot fully explain. To turn this raw data into something useful for prediction, the study first built a clear and repeatable cleaning pipeline for two related datasets: a 7,389-patient DOH-CAR registry and a 599-patient month-by-month monitoring set. The pipeline standardized formats, removed patient identifiers, and handled missing values by first checking why each variable was missing (MCAR, MAR, or MNAR) before applying one of four matching strategies. Because the cleaned data was trustworthy rather than blindly filled in, the descriptive and exploratory analyses could clearly show outcome trends over the years, the rise in deaths and lost follow-ups during the COVID-19 period, and the patient-level, time-based, and operational patterns that later guided model design.

These cleaned signals fed directly into how well the models performed. On the static dataset, XGBoost with SMOTE-ENN gave the strongest screening-focused setup (composite score $S = 0.699$; failure recall 72.9%), while XGBoost without resampling produced the best-calibrated risk ranking (test AUC 0.712; MCC 0.258), offering two different setups depending on how the tool will be used. The longitudinal models pushed performance further: after the v2 pipeline fixed the data leakage and label-handling issues that had limited v1, the SMOTE-ENN tree models (Random Forest, LightGBM, and XGBoost) reached 0.9318 accuracy and 0.8333 specificity, correctly catching five of six failure cases in the held-out test set, while the retrained, Platt-calibrated Hybrid Bi-LSTM detected 6 of 7 failure cases with a failure recall of 0.8571 and a specificity of 0.7368. The Bi-LSTM is the deployed temporal model in the CDSS because its sequence-native architecture and calibrated probability output support the in-browser temporal attribution layer. Building on these predictions, feature attribution (SHAP for tree models; group-occlusion for the Bi-LSTM) and a bounded SLM narrative layer were combined into a Clinical Decision Support System (CDSS) that shows each patient's risk level, the factors driving that risk, and updated estimates as new monthly data come in.

These results are strong enough to be clinically promising, but not yet ready for actual clinical use. Since the test cases came from only one regional registry, and since the narrative layer's faithfulness was only checked on a small group of 44 patients, the system must first go through external validation using data from other facilities, prospective testing in real clinical workflows, and review by healthcare professionals to confirm safety, calibration, and fit. The narrative explanation layer will also need a larger faithfulness study before it can be safely used in clinical practice. Once these conditions are met, the framework and the published pseudonymized version of the DOH-CAR PTB Treatment Outcome Dataset may only then be able to support both regional TB control efforts and future research on explainable clinical prediction.

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Appendix A: CDSS Web Application Walkthrough

This appendix presents a sequential walkthrough of the deployed CDSS web application, following the order in which a clinical user would traverse the system: authentication, patient intake, profile review, longitudinal monitoring, and risk explanation. Several interfaces shown here also appear in the body of the paper; they are reproduced in this appendix for continuity of the walkthrough.

B.1 Authentication and Dashboard

Figure 26

CDSS login screen.

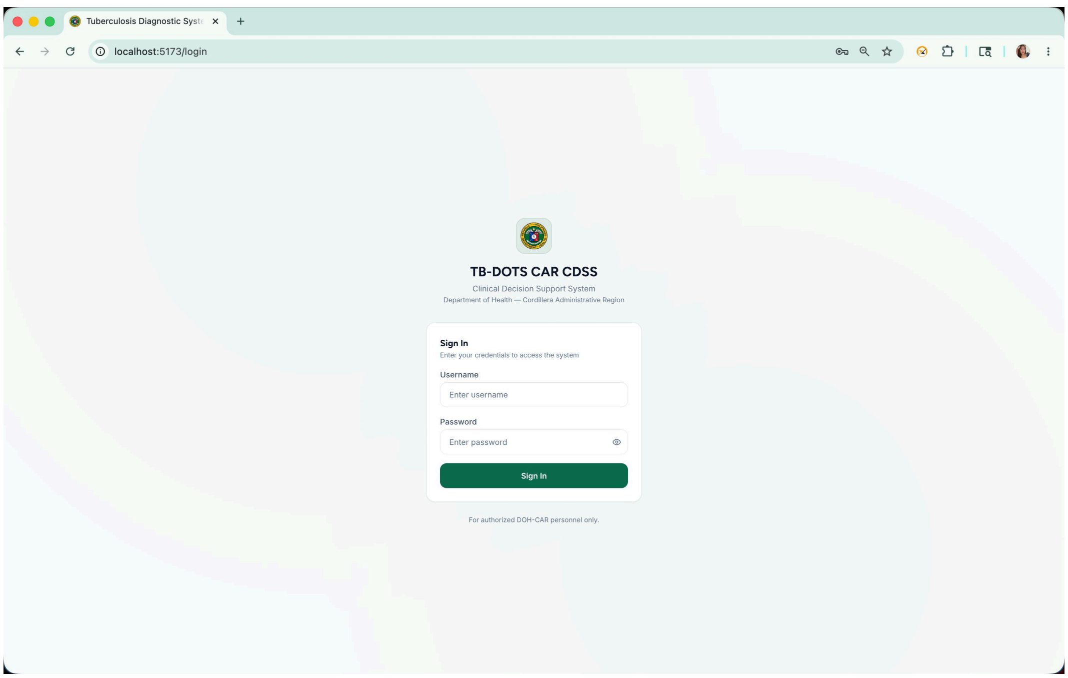
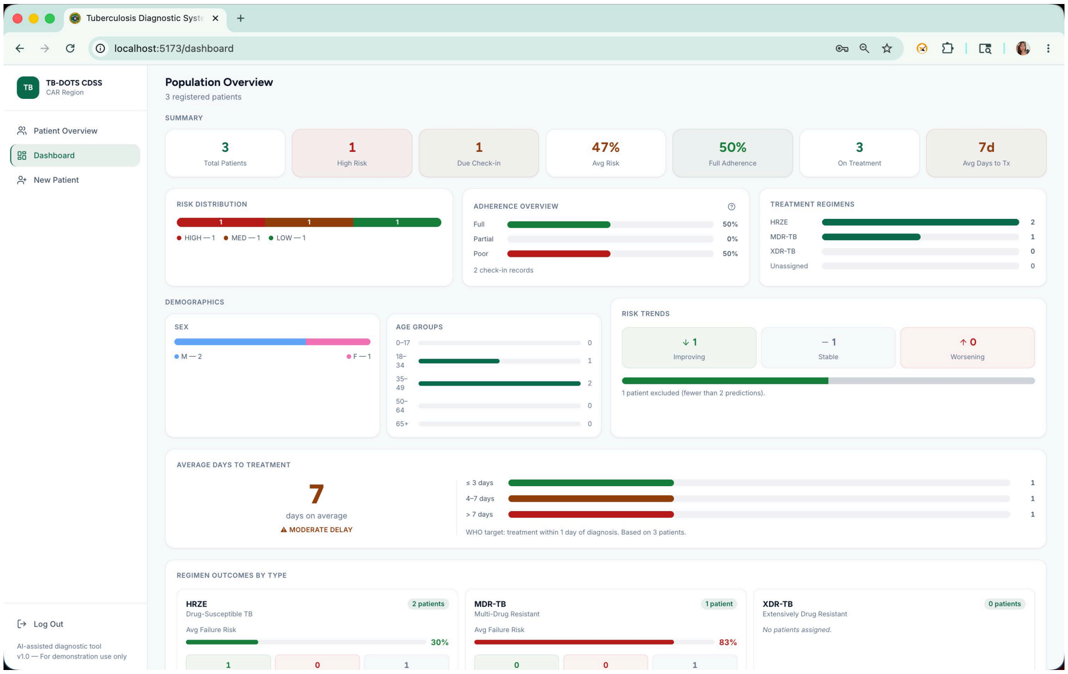


Figure 27

Main dashboard listing enrolled patients.



B.2 Patient Intake Workflow

The intake workflow consists of five sequential steps. The Lab Results step (Step 2) is omitted here as it was not captured in the available screenshot set.

Figure 28

Patient intake Step 1: Demographics.

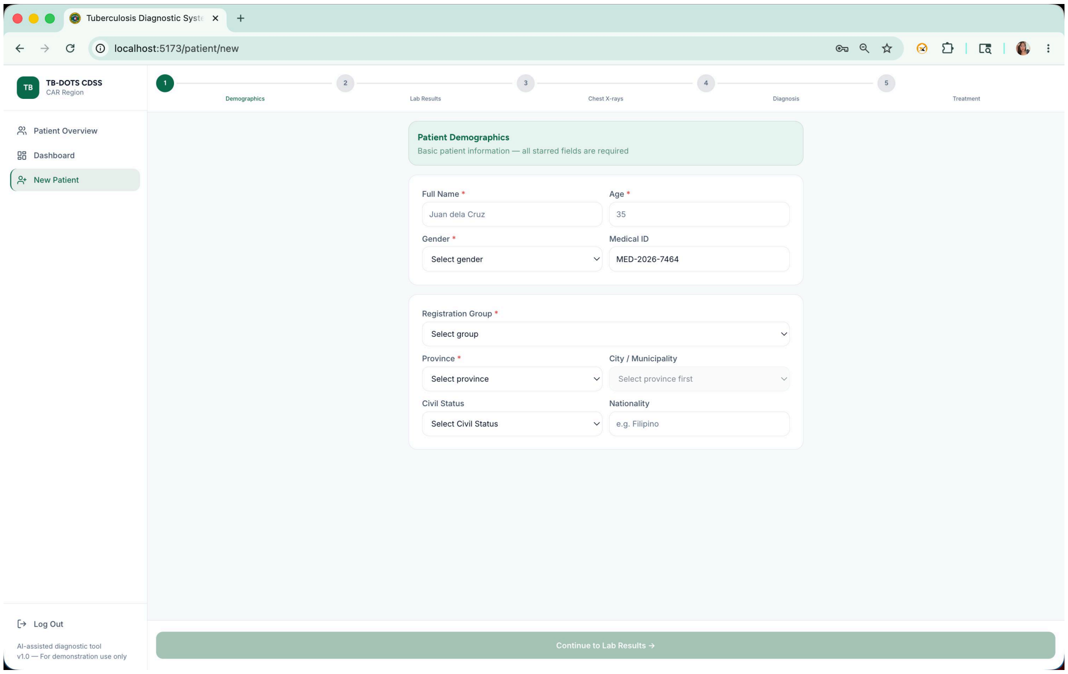


Figure 29

Patient intake Step 3: Chest X-ray upload.

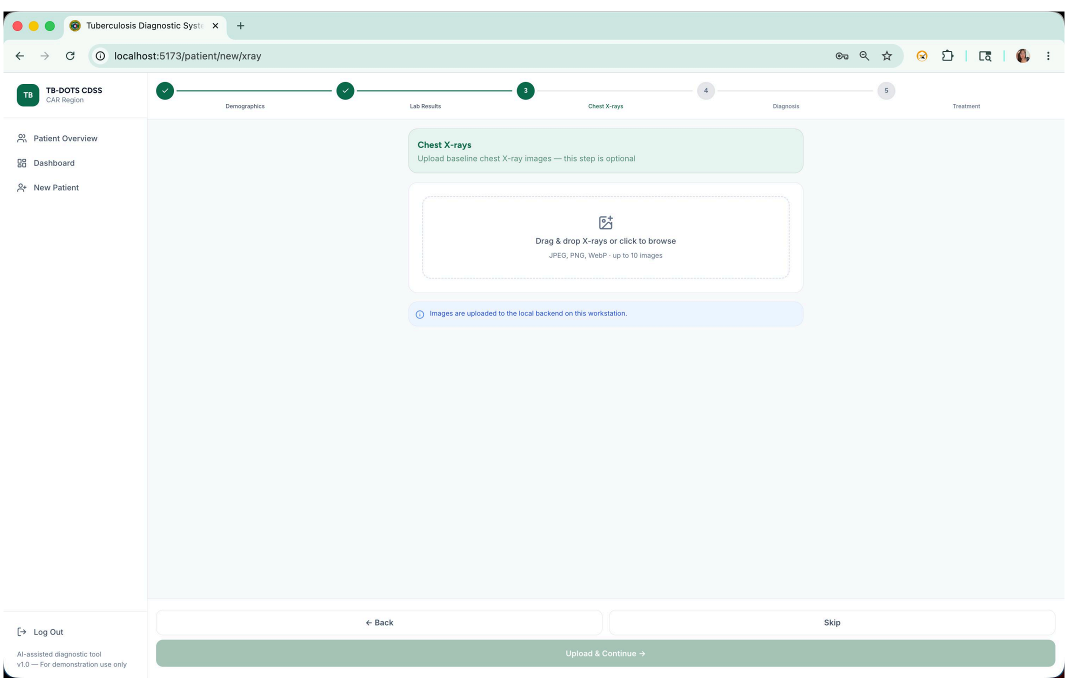


Figure 30

Patient intake Step 4: Diagnostic result.

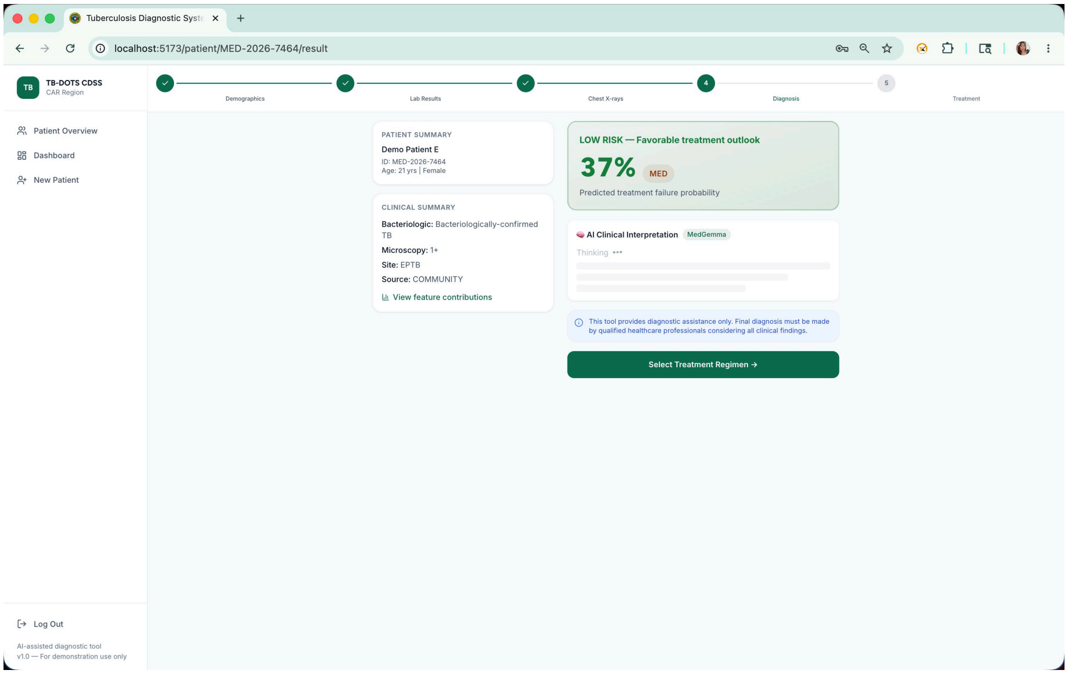
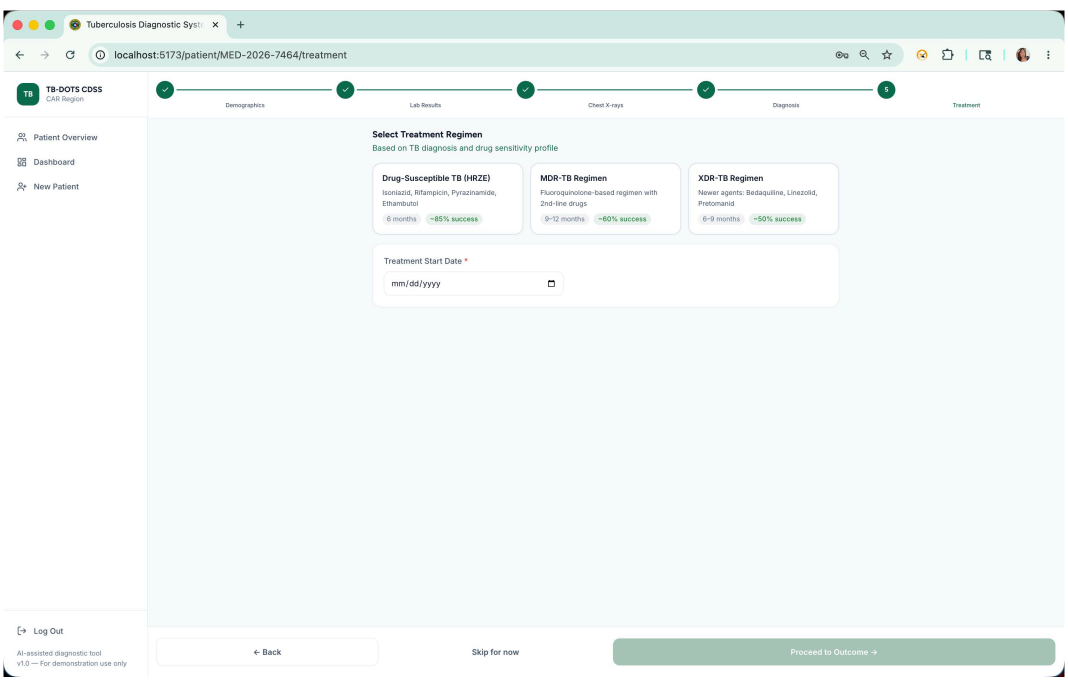


Figure 31

Patient intake Step 5: Treatment regimen selection.



B.3 Patient Profile and Records

Figure 32

Patient profile overview tab.

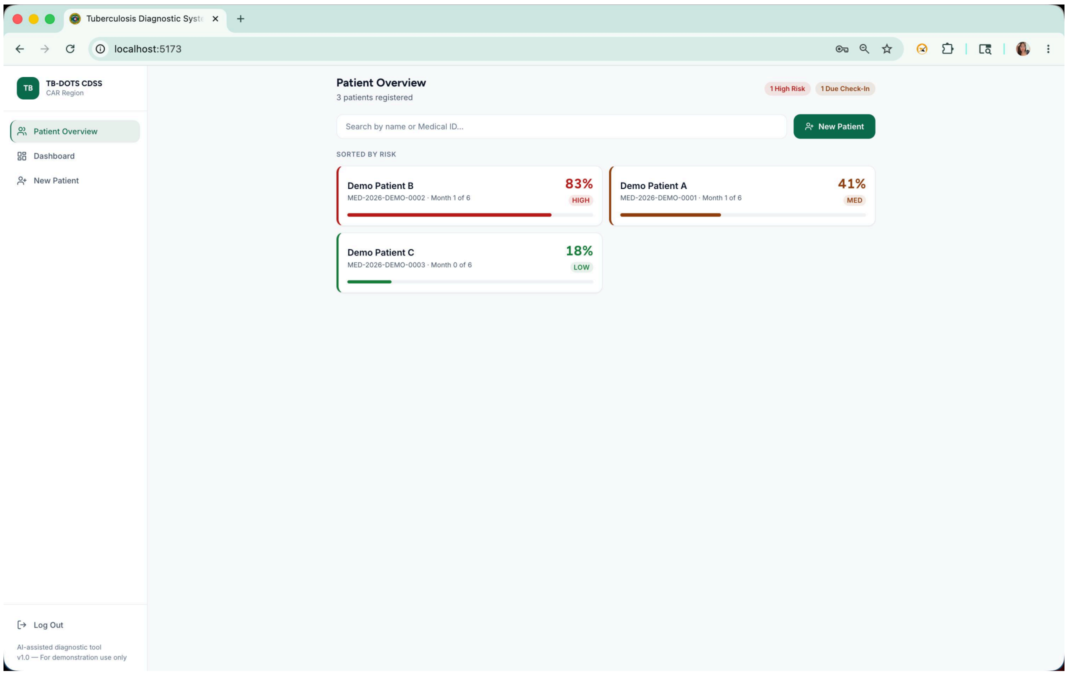
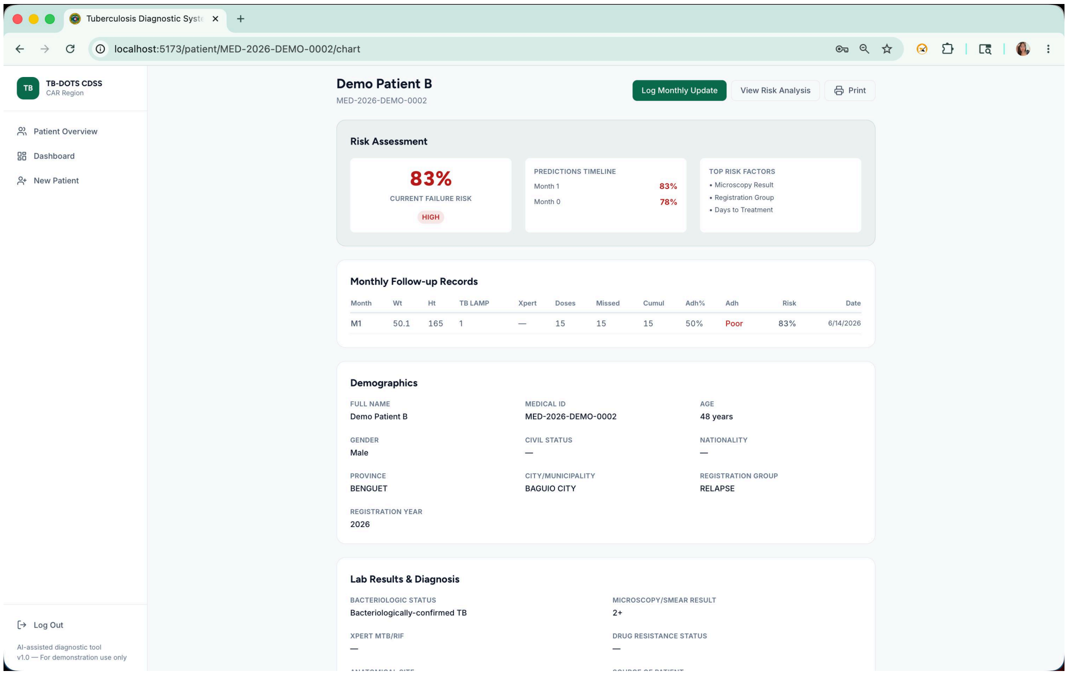


Figure 33

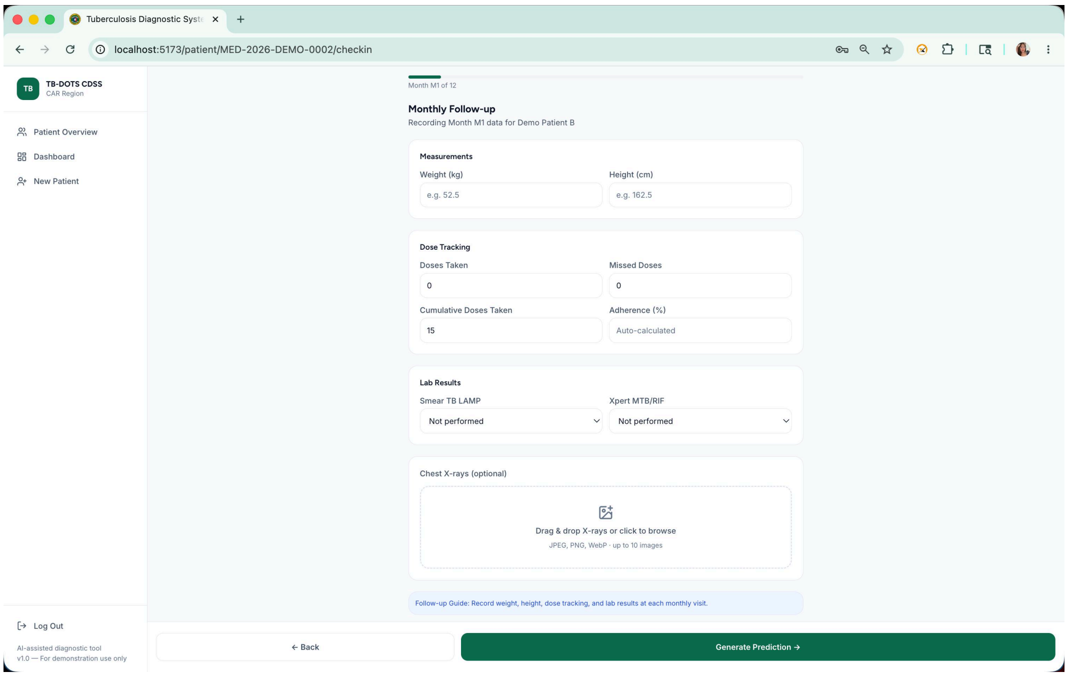
Complete patient chart with longitudinal record.



B.4 Longitudinal Monitoring

Figure 34

Monthly check-in log for adherence and follow-up imaging.



B.5 Risk Prediction and Explainability

Figure 35

Updated risk analysis after monthly follow-up.

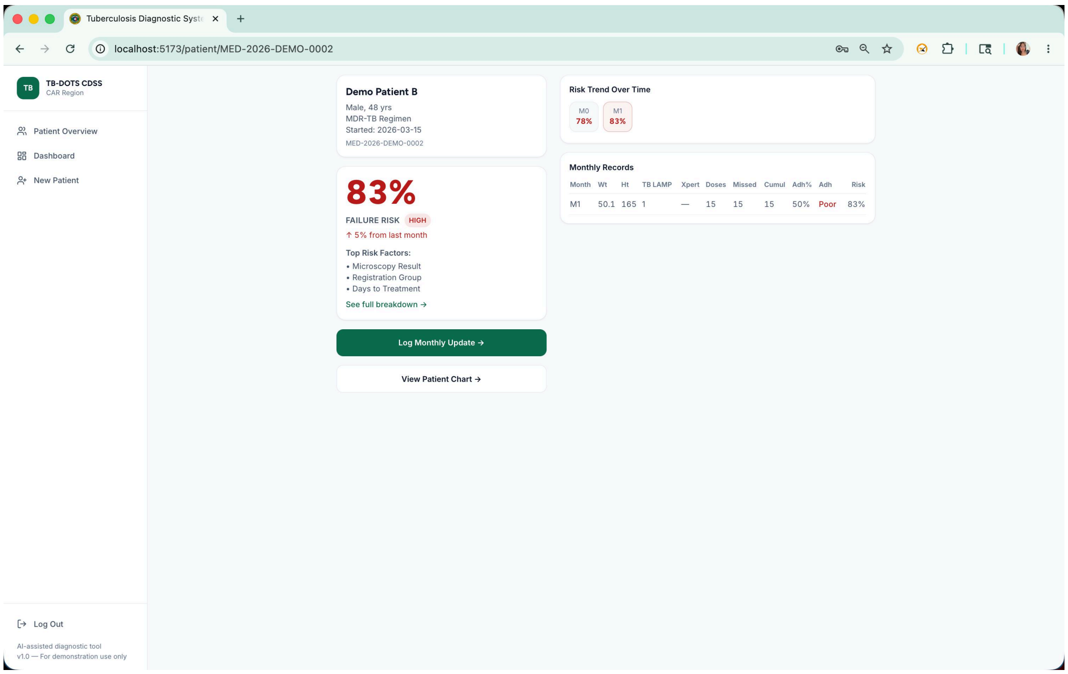


Figure 36

SHAP-based full breakdown of contributing factors.

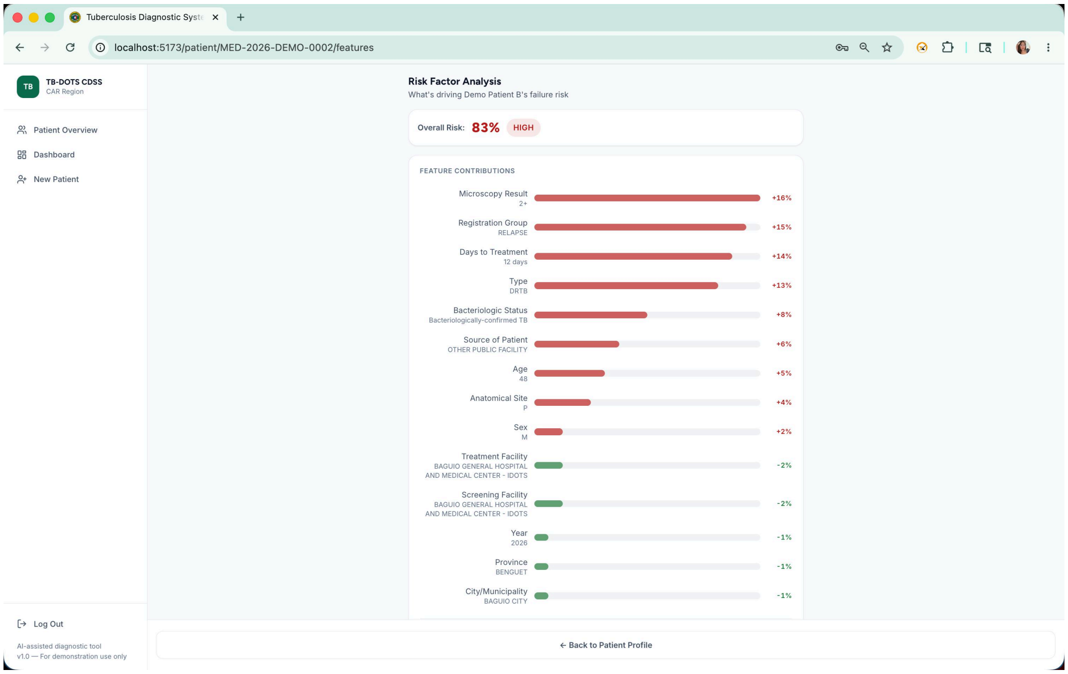
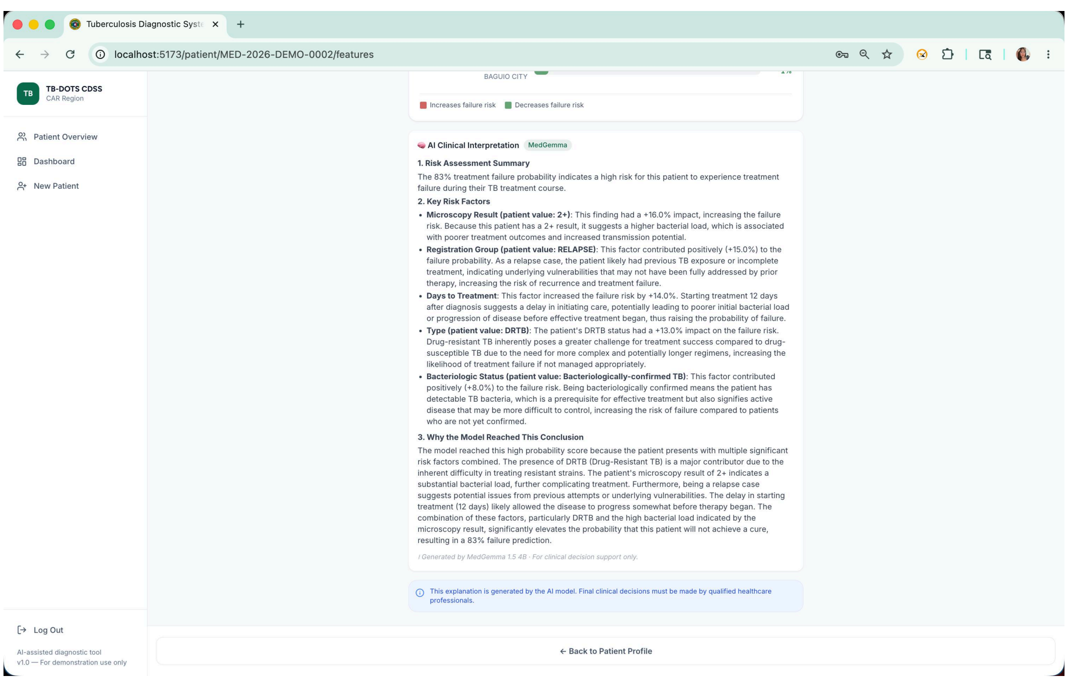


Figure 37

Med-Gemma clinical narrative explaining risk factors.



Appendix B: iCreate Supplementary Material

The following pages reproduce the original iCreate 2026 supplementary material in its source format for archival completeness.

Treatment Success and Failure Prediction using Machine Learning Models for Pulmonary Tuberculosis Patients under the DOH-CAR TB-DOTS Program

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Abstract

This study proposes a multi-layer analytical framework using machine learning to predict pulmonary tuberculosis (PTB) treatment success or failure in the Cordillera Administrative Region (CAR), Philippines. Utilizing demographic, clinical, and diagnostic data from 2015 to 2025, supervised models (Logistic Regression, Support Vector Machines, and Extreme Gradient Boosting) and temporal models (Recurrent Neural Networks and Long Short-Term Memory) are trained to identify risk factors. Shapley Additive Explanations (SHAP) ensure interpretability. The models will be integrated into a clinical decision support tool for early intervention. Aligning with the Sustainable Development Goals for good health and well-being, this predictive tool aims to enhance TB control efforts, optimize resource allocation, and improve patient management.

Keywords

Clinical decision support, machine learning, predictive modeling, pulmonary tuberculosis, treatment outcomes

1. Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease that continues to challenge public health due to its gradual progression and risk of treatment failure (Tobin & Tristram, 2024). In the Philippines, TB remains a major public health concern, with the country being one of the top contributors to the global TB burden (World Health Organization, 2022). In the Cordillera Administrative Region (CAR), the Department of Health (DOH) reported 437 confirmed TB cases in Baguio City alone in 2022, alongside an estimated 4,405 missing or unreported cases (Department of Health Republic of the Philippines, n.d.).

While multiple studies have examined risk factors such as sociodemographic characteristics, comorbidities, and medical history, findings remain fragmented as traditional statistical analyses often assume linear relationships and examine only a few factors simultaneously (Amante &

Abdosh, 2015). To address this gap, this study adopts an integrative machine learning framework to combine multiple dimensions of data in analyzing interrelated variables.

The specific research objectives of this study are as follows:

1. **Data Pre-processing & Feature Engineering**: To clean, pre-process, and digitalize PTB patient records, creating engineered features that ensure the dataset is optimized for statistical analysis and machine learning applications.
2. **Statistical Analysis & Profiling**: To perform descriptive and inferential statistics to profile the demographic, clinical, and treatment characteristics of the study population, identifying baseline variables that significantly influence treatment outcomes.
3. **AI Modeling & Comparative Analysis**: To develop, validate, and evaluate both non-temporal and temporal machine learning models for predicting PTB treatment outcomes, comparing the predictive performance of static baseline data against dynamic, time-dependent patient patterns throughout the course of therapy.
4. **Data Visualization & Clinical Interpretation**: To visualize patient risk levels, statistical trends, and predictive outputs from the AI models, providing clear insights to support clinical interpretation and the early identification of individuals at risk of poor outcomes.

2. Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering the period from 2015 to 2025, consolidated by the DOH-CAR. Data undergoes a standardized preprocessing pipeline, including schema harmonization, error correction, and missing value imputation via Multivariate Imputation by Chained Equations, or MICE (Alruhaymi & Kim, 2021).

To predict treatment outcomes defined by the National Tuberculosis Program criteria, supervised models including Logistic Regression, Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost) are utilized. Furthermore, temporal models using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) are employed to capture longitudinal treatment dynamics (Lu et al., 2022; Nayebi et al., 2022). Model performance is assessed using standard classification metrics, and Shapley Additive Explanations (SHAP) are applied to ensure clinical interpretability (Peng et al., 2024).

3. Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static (Random Forest, Gradient Boosting, LightGBM, XGBoost, Logistic Regression) and temporal models are compared, and feature importance analyses provide insights into the key factors influencing treatment outcomes. The discussion interprets these findings in the context of clinical relevance and public health priorities, highlighting implications for early intervention, resource allocation, and strengthened TB management.

3.1 Dataset Characteristics and Class Distribution

The final dataset exhibited significant class imbalance, with 86.2% of cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, including four newly introduced geographic and facility-level variables and one derived clinical delay variable (Diagnosis_to_Treatment_days). Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio. This approach mitigated class bias while preserving realistic evaluation conditions in the test and validation sets.

3.2 Validation Set Performance and Generalization

Table 1: Test Set Model Performance

Model	AUC (Area Under the Curve)	Recall (Failure)	F1 (Failure)
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Among all evaluated models, Random Forest achieved the highest overall performance on the test set, with an AUC of 0.7154 and an F1-score of 0.3539 for the failure class. It also demonstrated balanced recall (57.98%) without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (~0.71), though with slightly lower recall for treatment failure. Logistic Regression demonstrated the highest recall (77.13%); however, its precision was notably low (19.6%), resulting in over-alerting and reduced overall accuracy (~51%). These findings suggest that while Logistic Regression aggressively identifies high-risk patients, it lacks specificity, potentially burdening healthcare systems with excessive false alarms.

On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure. Notably, all models demonstrated positive AUC gains from test to validation sets, indicating strong generalization and absence of overfitting. The consistency of Random Forest across both evaluation phases confirms its robustness and suitability for deployment within a clinical decision support framework.

3.3 Validation Set Performance and Generalization

Table 2: Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 1 presents the performance metrics of all temporal models evaluated on the held-out test set. Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. Precision values were similarly high (0.9048), indicating few false positives for the majority class. However, recall for the minority Failure class reached 1.0 for these models, while specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Tree-based models augmented with SMOTE-ENN, particularly Random Forest, showed a more balanced performance. Random Forest achieved slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. ROC-AUC scores further illustrate differences in probability calibration, with XGBoost Baseline achieving the highest ROC-AUC (0.5789), while the Bi-LSTM Augmented variant performed lower (0.3684), likely due to overfitting on limited Failure samples. Overall, these results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

4. Conclusions and Recommendations

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Both temporal models, specifically the Bi-LSTM Baseline and Augmented variants, and tree-based models, including XGBoost, LightGBM, and Random Forest with SMOTE-ENN, were evaluated on a stratified 70/20/10 patient-level split, incorporating progressive temporal samples from treatment months M0 through M12. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited, as indicated by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes. These results highlight that tree-based models augmented with SMOTE-ENN are particularly effective at handling severe class imbalance, while temporal models provide strong predictive accuracy for the majority outcome.

The study contributes in several ways. By integrating temporal monitoring data with static baseline features, it demonstrates that patient-level longitudinal information improves predictive performance over static data alone. The application of SMOTE-ENN and advanced data augmentation techniques for Bi-LSTM effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Furthermore, the findings suggest that predictive models, particularly Random Forest with SMOTE-ENN, have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly.

Based on these findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance generalization, explore hybrid modeling approaches combining deep learning and tree-based methods, and investigate additional features such as social determinants of health and geographic risk factors. Additionally, enhancing model interpretability, particularly of Bi-LSTM attention outputs, could improve clinician trust and adoption.

Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes. By accurately identifying patients at risk of treatment failure, these predictive models have the potential to reduce TB-related morbidity and mortality, inform targeted public health interventions, and contribute to more efficient and effective TB care within the Cordillera Administrative Region.

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